QUERCUSTOA-APP

(Quercus Taxonomy-oriented Annotation)

v0.23

A software package to integrate functional annotation and comparative genomics across oak lineages

GI en Especies Leñosas (WooSp)
Dpto. Sistemas y Recursos Naturales
ETSI Montes, Forestal y del Medio Natural
Universidad Politécnica de Madrid

https://github.com/ggfhf/

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Disclaimer

The software package QUERCUSTOA-APP (Quercus Taxonomy-oriented Annotation) is available for free download from the GitHub repository:

https://github.com/GGFHF/quercusTOA-app

under GNU General Public License v3.0.

How to cite

If you are using QUERCUSTOA-APP or QUERCUSTOA database, you should cite the following paper:

Fernando Mora-Márquez, Mikel Hurtado & Unai López de Heredia (2026). quercusTOA: integrating functional annotations and comparative genomics across oak lineages. *Frontiers in Bioinformatics*, **Volume 6, 1821531**. DOI: https://doi.org/10.3389/fbinf.2026.1821531

Introduction

The genus *Quercus* comprises 400–500 species of immense ecological and economic importance across the Northern Hemisphere. Given the taxonomic complexity arising from frequent hybridization, the integration of genomic and transcriptomic data has become essential. This molecular approach is not only clarifying the evolutionary and diversification history of oaks but is also proving indispensable for advancing modern forest breeding and biotechnology, particularly in understanding resilience and stress responses.

Comparative genomics is based on the premise that all living organisms descend from a common ancestor, and it studies the differences and similarities between the genomes of different species, more or less closely related in evolutionary terms. Homology search is a technique used in comparative genomics for the identification of similar sequences (DNA, RNA or proteins) to find homologous genes or proteins. This technique involves comparing a query sequence against a database of known sequences to find matches with significant similarity.

The QUERCUSTOA-APP is a taxonomy-aware functional annotation and comparative genomics tool designed to leverage resources from the QUERCUSTOA database. As an integrative resource for oaks, this database serves as a public reference to facilitate accurate functional annotation, enrichment analysis, and comparative genomics reports for *Quercus* species' genomics and transcriptomics experiments. QUERCUSTOA-APP is a desktop application designed with a user-friendly GUI to easily perform:

(1) comparative genome analyses in oaks:
   - protein/transcript sequence homology searches
   - protein and gene ortholog identification through protein IDs
   - ortholog protein and gene alignments and tree drawing

(2) functional annotation and enrichment analysis:
   - protein /transcript homology searches on an optimized genome-assembly based protein set in oaks
   - Retrieval of functional annotation features: (GO terms, KEGG, MetaCyc, EC, etc.
   - Retrieval of orthologs in *Arabidopsis thaliana* (TAIR10).

Furthermore, QUERCUSTOA-APP can also run on Linux servers using the BASH scripts included in this software (see the chapter “Functional annotation, enrichment analysis and homology search on Linux servers”).

For technical validation of QUERCUSTOA, refer to the manuscript:

Fernando Mora-Márquez, Mikel Hurtado & Unai López de Heredia (2026). quercusTOA: integrating functional annotations and comparative genomics across oak lineages. *Frontiers in Bioinformatics*, **Volume 6, 1821531**. DOI: https://doi.org/10.3389/fbinf.2026.1821531

The QuercusTOA framework

The whole QUERCUSTOA framework is designed in two separate parts with different functions:

- QUERCUSTOA is a database that includes curated records of *Quercus* proteins and related taxonomic, functional and structural information.
- QUERCUSTOA-APP is a desktop application that allows us to perform the functional annotation, enrichment analysis and comparative genomics of transcriptomes and proteins yielded in *Quercus* experiments exploiting the cross-references uploaded in the QUERCUSTOA database.

The **QUERCUSTOA database** consists of three data sets:

- **The sequences data set** (Figure 1):

  There are four sources of sequences:

  - the Genome database of NCBI (https://www.ncbi.nlm.nih.gov/protein)
  - the Plants database of the Genome Warehouse of the CNCB-NGDC (https://download.cncb.ac.cn/gwh/Plants/)
  - TAIR10 database (https://www.arabidopsis.org/)
  - CANTATA lncRNA database (http://cantata.amu.edu.pl/)

  Genome and protein FASTA sequences, along with their corresponding GFF annotation files, are retrieved for oak species with publicly available assembly records. The most recent assemblies for *Q. suber*, *Q. robur*, *Q. lobata*, and *Q. rubra* were obtained from the NCBI Genomes database. Similarly, equivalent genomic and proteomic data for *Q. variabilis*, *Q. dentata*, *Q. gilva*, *Q. longispica*, and Q. *acutissima* were downloaded from the Plant Genome Warehouse of the CNCB-NGDC. The protein and gene sequences are loaded in a SQLite database file using the utility load-species-seqs.py of **NGShelper** (https://github.com/GGFHF/NGShelper).

  The FASTA file of protein sequences of *Arabidopsis thaliana* is downloaded from the TAIR10 database. The FASTA file of lncRNA sequences (*A. thaliana*, *Q. lobata* and *Q. suber*) is downloaded from the CANTATA database.

- **The comparative genomics data set** (Figure 2):

  The program `extract-gff-feature-seqs.py` (**NGShelper**) obtains gene, CDS and concatenated CDS sequences from FASTA files of genomes and their GFFs. Next, the concatenated CDS sequences are clustered by **MMseqs2** and the cluster sequences are loaded by `split-mmseqs2-concds-clusters.py` (**NGShelper**) and `load-mmseqs2-concds-clusters.py` (**NGShelper**).

  The GFF file is processed by `remove-gff-redundancies.py` (**NGShelper**) to obtain a GFF without redundancies (mRNAs with the same concatenated CDS sequence and their corresponding features in a gene).

  FASTA files of genomes and their GFFs without redundancies included in the sequence data set are processed by **Liftoff** to obtain CDS and gene relationships between the species. These relationships are loaded by programs `get-liftoff-gff-data.py` (**NGShelper**), and `load-liftoff-gff-data.py` (**NGShelper**) into the comparative genomics SQLite database.

Finally, `get-liftoff-genome-relationships.py` (**NGShelper**) gets the relationships between each Quercus species genome and genomes of other species, and `get-liftoff-homologous-proteins.py` (**NGShelper**) obtain the homologous proteins.

- **The functional annotation data set** (Figure 3):

  First, redundant protein sequences included in the sequence data set are removed using **MMseqs2** (https://github.com/soedinglab/MMseqs2) to produce clusters of sequences that are aligned with **MAFFT** (https://mafft.cbrc.jp/alignment/software/) to produce a consensus clustered sequence with **EMBOSS** (http://emboss.open-bio.org/). The overall quality of the set consensus sequences performed by **BUSCO** (https://busco.ezlab.org/).

  Then, annotation records corresponding to the consensus sequences are searched using **InterProScan** and **eggnog-mapper**. Both consensus sequences and associated records are uploaded into an SQLite database (see appendix A) with `load-interproscan-annotations.py` (**NGShelper**) and `load-emapper-annotations.py` (**NGShelper**). **BLAST+** and **DIAMOND** databases are built using `blastp` and `diamond blastp` respectively.

  TAIR10 protein sequences are processed by the program `makeblastdb` to get a **BLAST+** database, to find the *Arabidopsis thaliana* orthologs of consensus sequences loaded into the SQLite database by the program `load-tair10-orthologs.py` (**NGShelper**) with a `blastx` homology search,.

  On the other hand, the consensus sequences and lncRNA sequences (downloaded from CANTATA lncRNA database) are processed by **makeblastdb** to build BLAST+ databases.

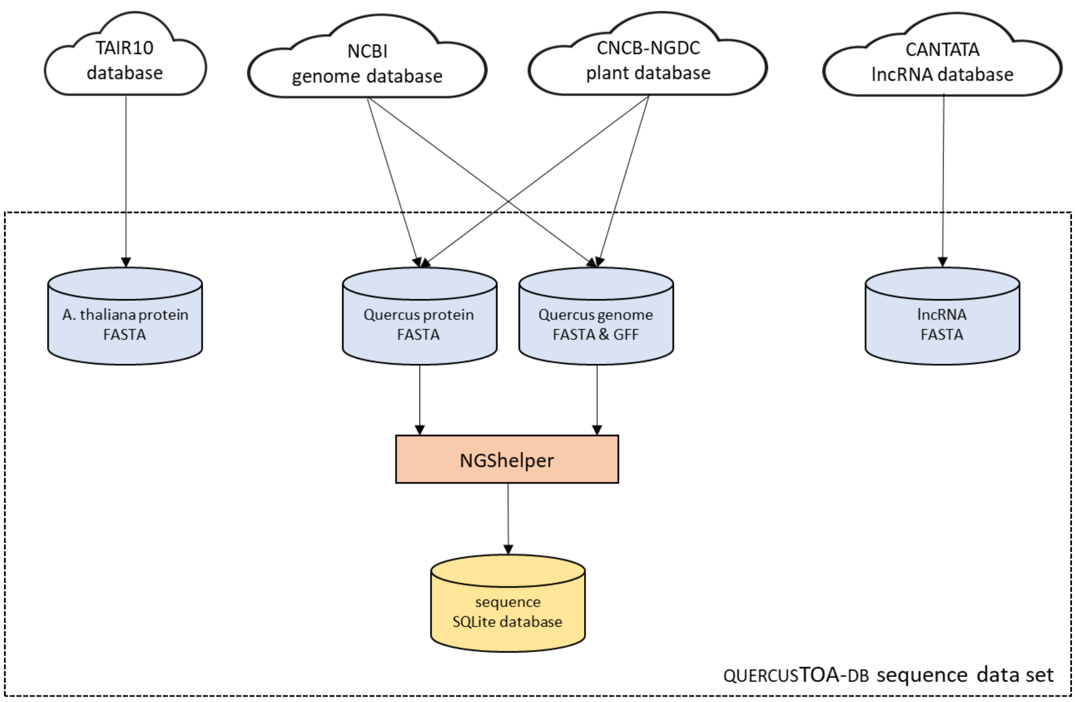


Figure 1. Flow-chart of the creation of the QUERCUSTOA database sequence data set.

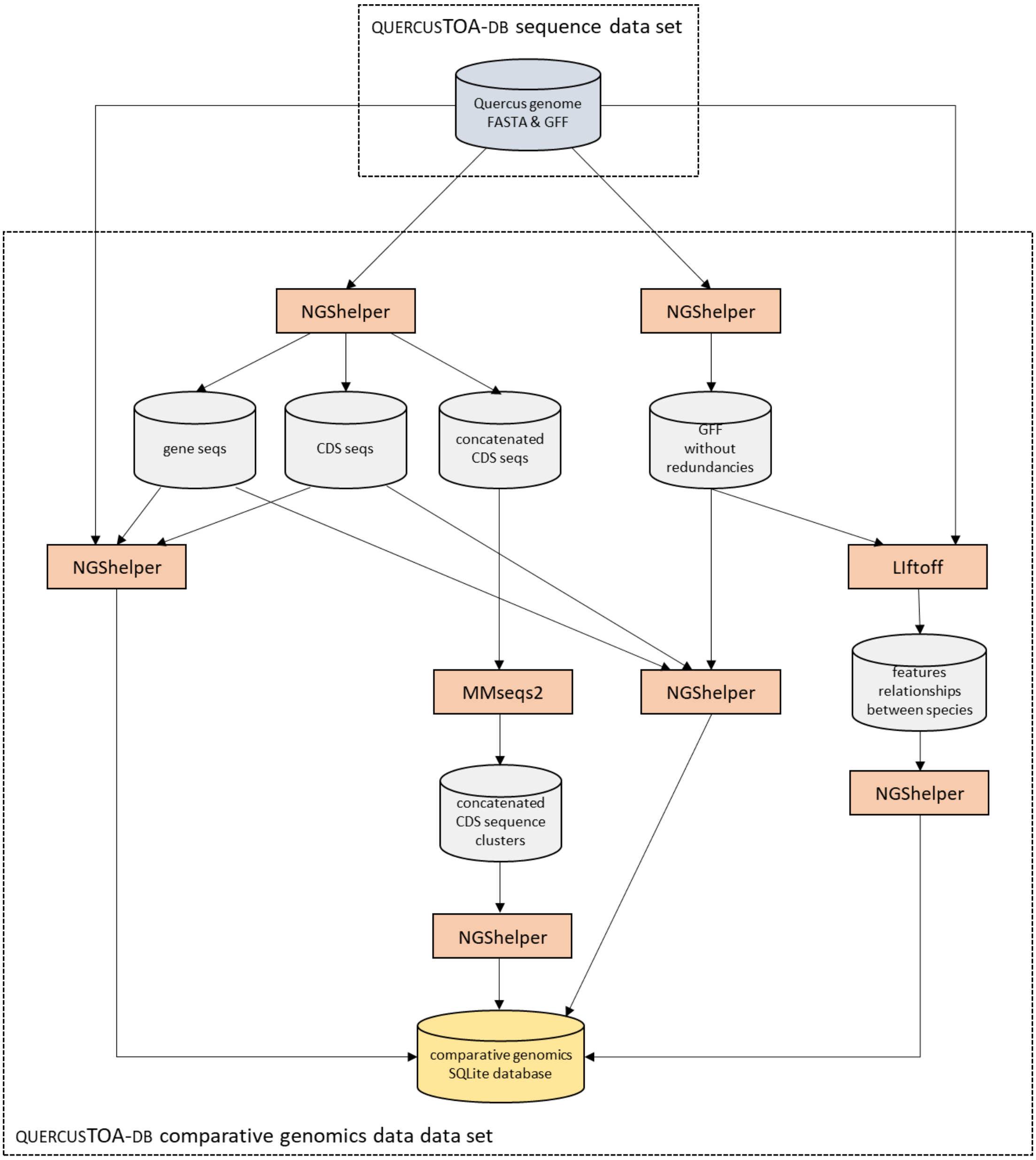


Figure 2. Flow-chart of the creation of the QUERCUSTOA database comparative genomics data set.

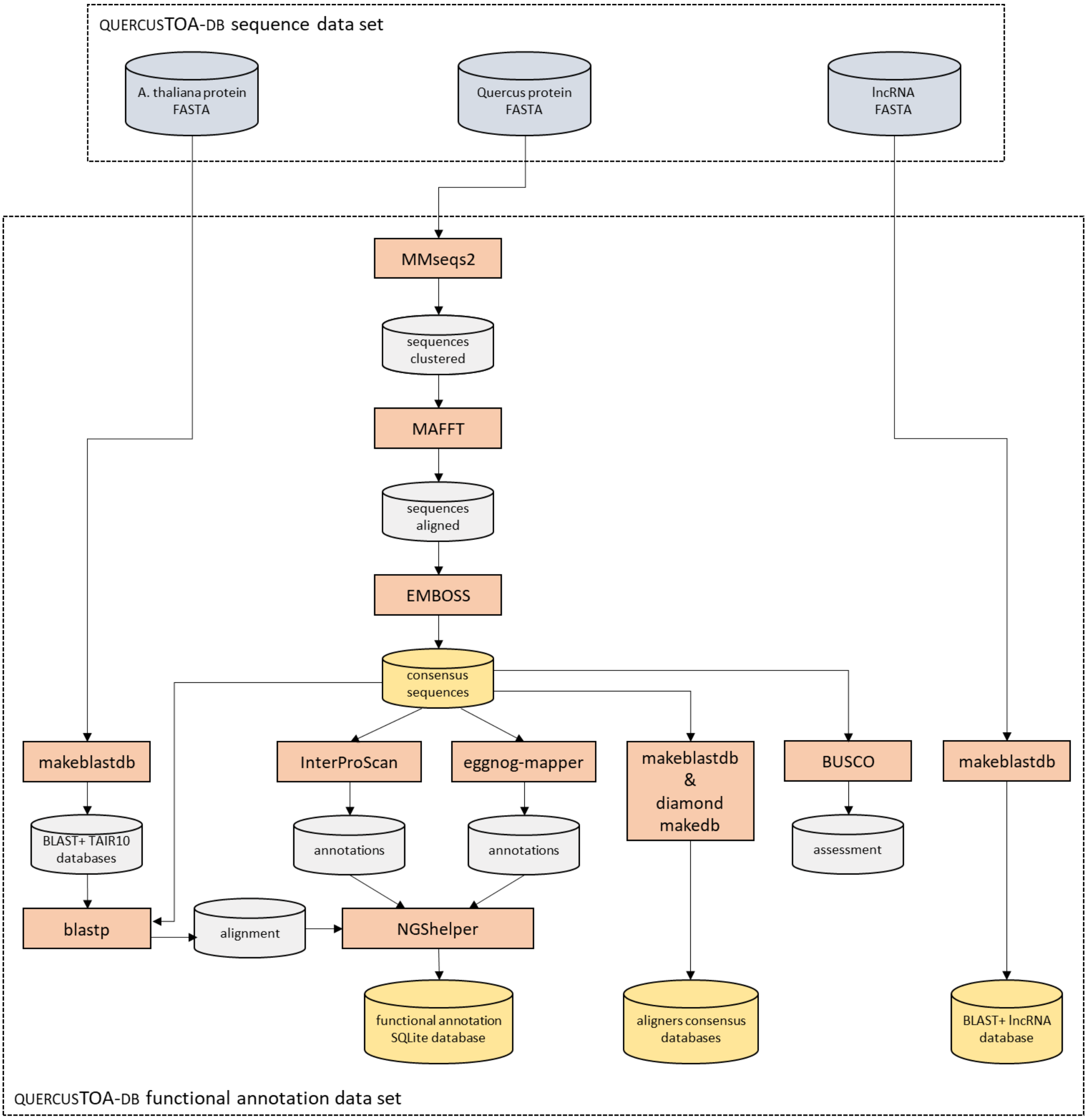


Figure 3. Flow-chart of the creation of the QUERCUSTOA database functional annotation data set.

The QUERCUSTOA database is hosted at the CESVIMA server of the Universidad Politécnica de Madrid and can be downloaded directly (https://blogs.upm.es/quercustoa/) or (preferably) using the QUERCUSTOA-APP (https://github.com/GGFHF/quercusTOA-app). The database is periodically updated by the development team, and statistics of the database are provided at the web page.

Once the database is downloaded by the **QUERCUSTOA-APP**, transcripts or proteins can be annotated (Figure 4). The transcript sequences of a genomics or transcriptomics experiment are processed by **CodAn** (https://github.com/pedronachtigall/CodAn/) to get predicted ORFs. These sequences are aligned to **BLAST+** consensus database by `blastp` (if **BLAST+** aligner is used) or `diamond blastp` (for alignments performed with **DIAMOND**). Then, transcripts sequences are aligned to **BLAST+** or **DIAMOND** consensus database by `blastx` or `diamond blastx`, respectively. And finally, transcript sequences are aligned to **BLAST+** lncRNA datatabase using `blastn`. All alignments are concatenated to their functional annotations by the program `concat-functional-annotations.py` included in QUERCUSTOA-APP. This program yields two annotation files: a file containing all annotations yielded by **BLAST+** programs for every transcript sequence, and the other one contains the annotations of the subject sequence identification with best hit yielded by blast programs for every transcript sequence.

For a transcript sequence, the `blastp/diamond blastp` annotations are chosen. If these annotations do not exist, the `blastx/diamond blastx` annotations are taken. And finally, if they do not exist either, the `blastn` annotation is selected.

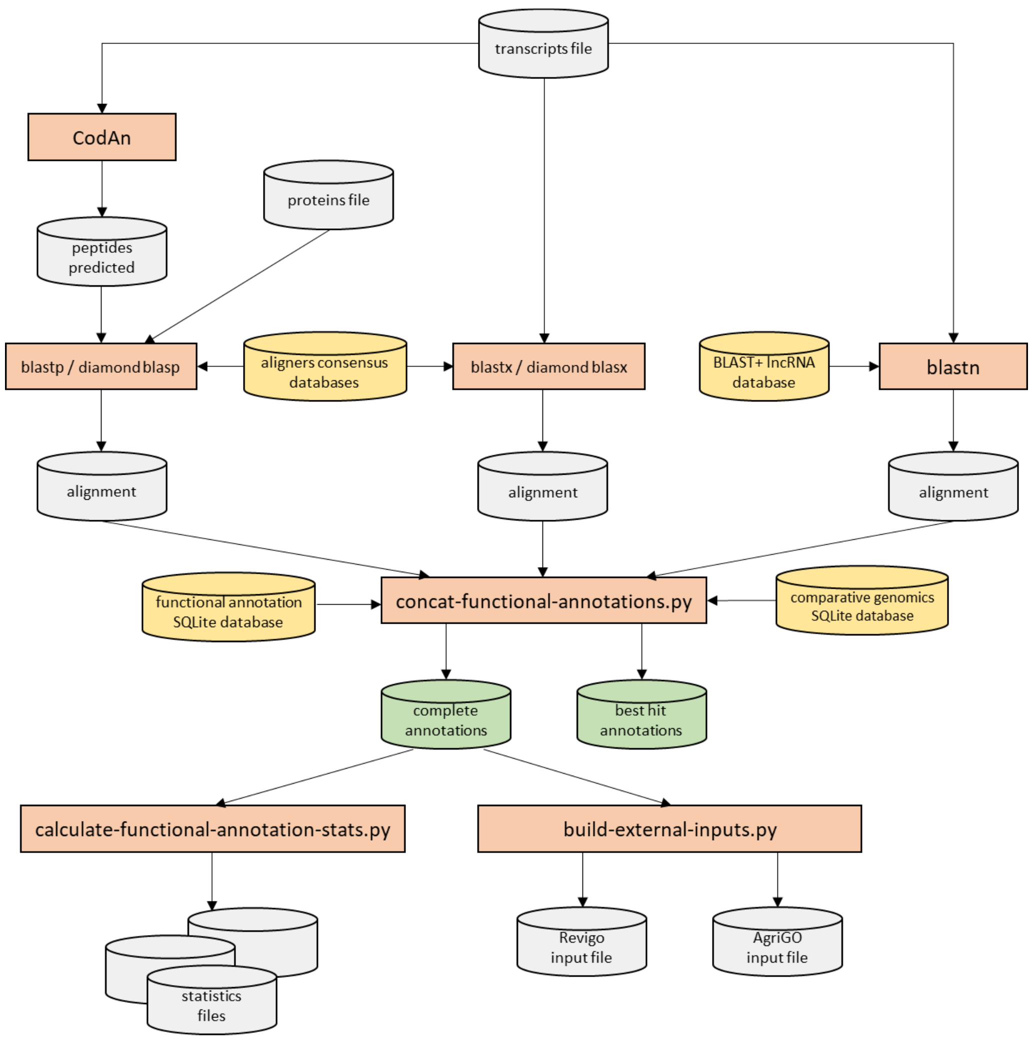


Figure 4. Flow-chart of the process of functional annotation in the QUERCUSTOA-APP desktop application.

The input file of a functional annotation process can also be a file with protein sequences (Figure 2). In this case the corresponding steps with `blastx/diamond blastx` and `blastn` do not run.

The annotation files contain the following data:

- qseqid: query (transcriptome) sequence identification
- sseqid: subject (**BLAST+** databases of QUERCUSTOA) sequence identification
- pident: percentage of identical positions
- length: alignment length
- mismatch: number of mismatches
- gapopen: number of gap openings
- qstart: start of alignment in query
- qend: end of alignment in query
- sstart: start of alignment in subject
- send: end of alignment in subject
- evalue: expect value
- bitscore: bit score
- aligner: BLAST+ program that yielded the annotation
- protein_description: description from the protein sequence
- protein_species: species from the protein sequence
- tair10_ortholog_seq_id: ortholog sequence identification from TAR10
- tair10_description: description of the ortholog sequence identification from TAR10
- qacutissima_homology: homology data with *Quercus acutissima*
- qdentata_homology: homology data with *Quercus dentata*
- qgilva_homology: homology data with *Quercus gilva*
- qlobata_homology: homology data with *Quercus lobata*
- qlongispica_homology: homology data with *Quercus longispica*
- qrobur_homology: homology data with *Quercus robur*
- qrubra_homology: homology data with *Quercus rubra*
- qsuber_homology: homology data with *Quercus suber*
- qvariabilis_homology: homology data with *Quercus variabilis*
- interpro_goterms: concatenated list of GO terms from InterPro
- panther_goterms: concatenated list of GO terms from Panther
- metacyc_pathways: concatenated list of pathway identifications from MetaCyc
- eggnog_ortholog_seq_id: ortholog sequence identification from eggNOG
- eggnog_ortholog_species: species from eggNOG
- eggnog_ogs: OGs (Orthologous Groups) of proteins from eggNOG
- cog_category: COG (Cluster of Orthologous Genes) from eggNOG
- eggnog_description: description from eggNOG
- eggnog_goterms: concatenated list of GO terms from eggNOG
- ec: concatenated list of EC (Enzyme Commission) numbers
- kegg_kos: concatenated list of KO from KEGG
- kegg_pathways: concatenated list of pathway identifications from KEGG
- kegg_modules: concatenated list of module identifications from KEGG
- kegg_reactions: concatenated list of chemical reactions identifications from KEGG
- kegg_rclasses: concatenated list of reactions classification identifications from KEGG

- brite: functional hierarchy of OGs assigned to the sequence
- kegg_tc: T cell receptor (TCR) signaling pathway
- cazy: concatenated list of Carbohydrate-Active Enzymes (CAZymes)
- pfams: concatenated list of protein families from Pfam

The program `calculate-functional-stats.py` (QUERCUSTOA-APP) yields functional annotations statistics and `build-external-inputs.py` (QUERCUSTOA-APP) buids specific inputs for **AgriGO** and **REVIGO** servers.

From the annotation files, both complete annotations as the best hit annotations, QUERCUSTOA-APP can run an enrichment analysis using as background the registries from the QUERCUSTOA database (Figure 5).

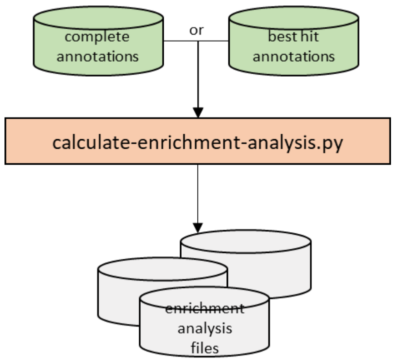


Figure 5. Flow-chart of the process of enrichement analysis in the QUERCUSTOA-APP desktop application.

The program `calculate-enrichment-analysis.py` included in QUERCUSTOA-APP yields files with the output of enrichment analysis of GO terms, MetaCyc pathways, KEGG KOs, and KEGG pathways.

These files contain the following data:

- Identification: GO term, MetaCyc pathway, KEGG KO or KEGG pathway
- Description (only for GO terms)
- Namespace (only for GO terms)
- Sequence number with the identification in transcript annotations
- Sequence number with identifications in transcript annotations
- Sequence number with the identification in species or *Quercus* annotations
- Sequence number with identifications in species or *Quercus* annotations
- Enrichment
- p-value
- FDR (False Discovery Rate)

To search the homology (Figure 6) corresponding to query sequences of proteins or predicted peptides by **CodAn** from a transcriptome, these sequences are aligned using `diamond blastp` to the **DIAMOND** consensus database built in the functional annotation data set. The alignments yielded by this run are processed by `get-cluster-homology-relationships.py` (quercusTOA-app) using the functional annotation and comparative genomics SQLite databases and the result is a set of homology relationships of each query sequence. Homology relationships are processed by `get-protein-fasta-files.py` (quercusTOA-app) and `align-fasta-seqs.py` (QUERCUSTOA-APP) to get the sequences of homologous proteins and genes and the alignments for each query sequence. Alignment plots and phylogenetic plots are drawn for orthologous genes and proteins.

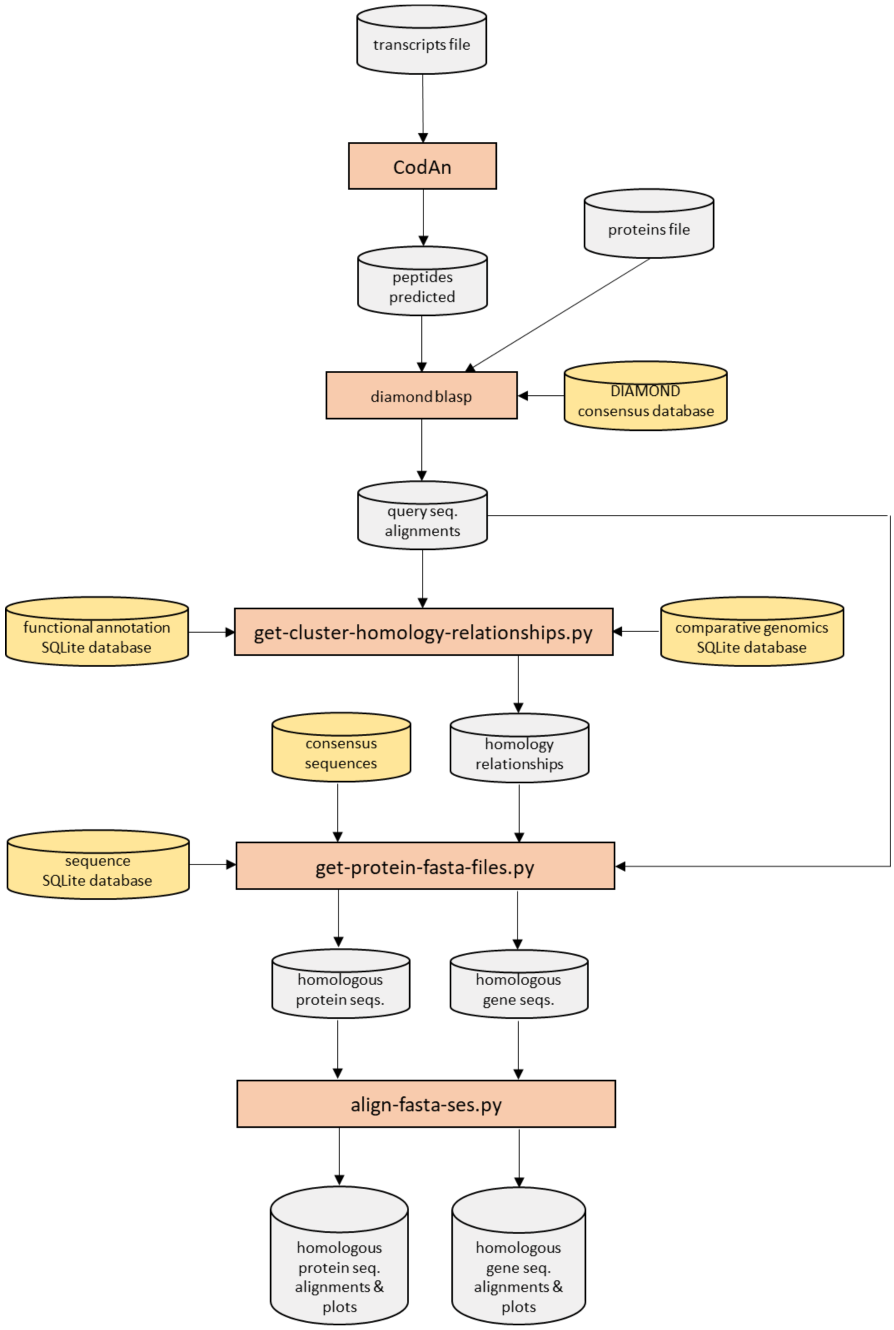


Figure 6. Flow-chart of the process of homology search in the QUERCUSTOA-APP desktop application.

Notice: All files with the ".csv" extension yielded by QUERCUSTOA-APP are delimited by the semicolon character.

Installation in GUI environments

QUERCUSTOA-APP installation

The source code for QUERCUSTOA-APP, including documentation and installation instructions, is hosted on GitHub (https://github.com/GGFHF/quercusTOA-app) and has been archived in the Zenodo repository (https://doi.org/10.5281/zenodo.18740268).

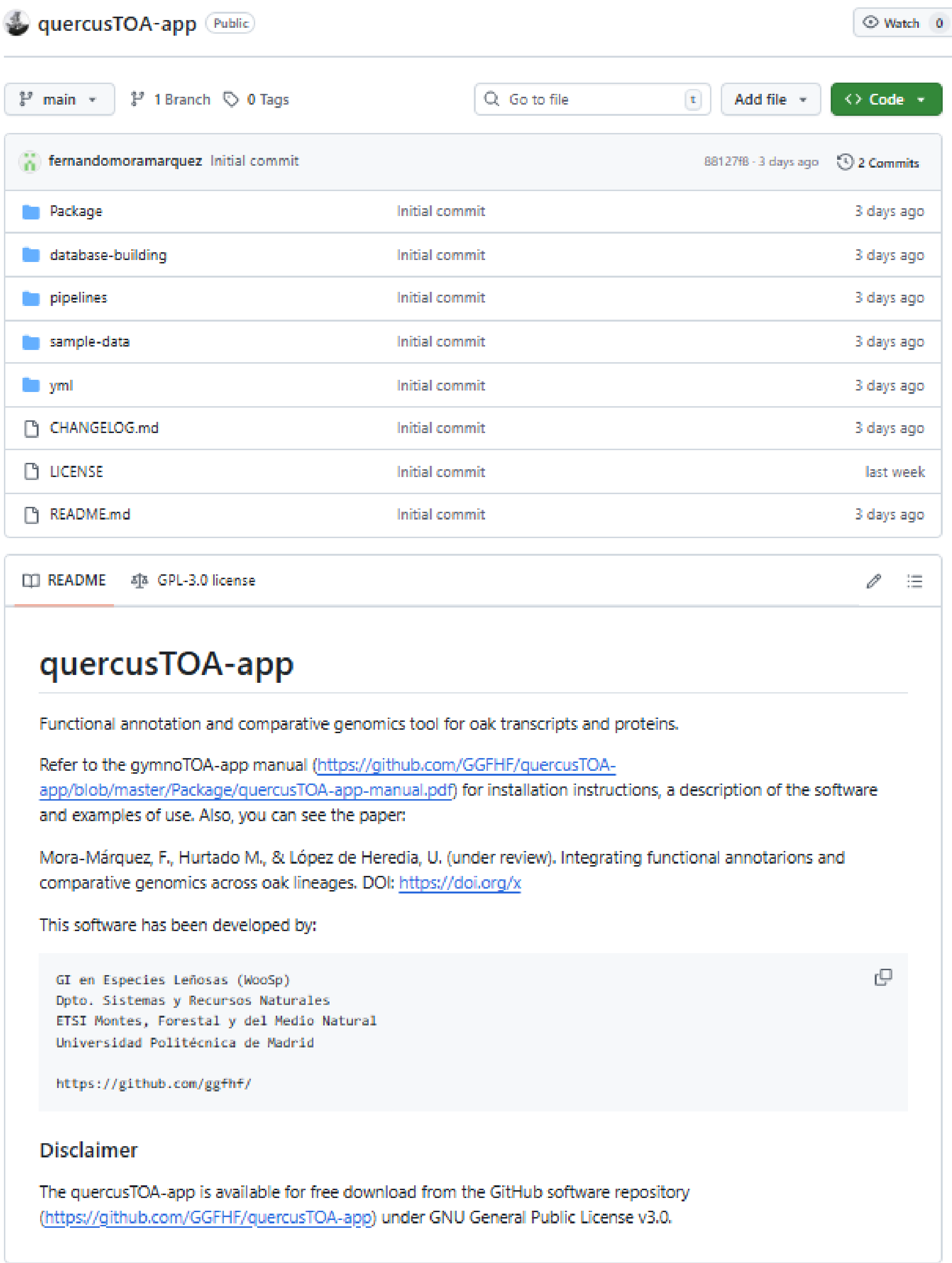


Figure 7. QUERCUSTOA-APP home at GitHub software repository.

NOTE: In this manual, red typeface is used to highlight file paths and environment names that require user modification.

LINUX / macOS

Download the QUERCUSTOA-APP software in a directory, e.g., $HOME/Apps/quercusTOA-app, and decompress the ZIP file using the following command:

```
$ mkdir -p $HOME/Apps

$ cd $HOME/Apps

$ wget --output-document main.zip https://github.com/GGFHF/quercusTOA-app/archive/refs/heads/main.zip

$ unzip main.zip

$ mv quercusTOA-app-main quercusTOA-app

$ rm main.zip
```

The QUERCUSTOA-APP directory will be hosted in $HOME/Apps. Then, set the execution permissions of the programs using these commands:

```
$ find $HOME/Apps/quercusTOA-app -type f \( -name "*.py" -o -name "*.sh" \) -exec chmod +x {} \;
```

Windows

Go to the GitHub repository https://github.com/GGFHF/quercusTOA-app/ (Figure 7), click *Code* and, in the pop-up window, click in *Download ZIP*. Decompress the downloaded ZIP file using "Extract All.." of the File Explorer on *quercusTOA-app-master.zip* extracting the files in the directory, e.g., %USERPROFILE%\Apps\quercusTOA-app.

NOTE: In this manual we will refer to the QUERCUSTOA-APP directory as $HOME/Apps/quercusTOA-app (Linux or macOS) or %USERPROFILE%\Apps\quercusTOA-app (Windows), but you can install QUERCUSTOA-APP in the directory you think is appropriate.

Conda and additional infrastructure software installation

Python 3 (https://www.python.org/), version 3.12 or higher, is required by QUERCUSTOA-APP. Also, this application needs the following **Python** modules:

- **PyQt5** (https://www.riverbankcomputing.com/static/Docs/PyQt5/), a **Python** interface for QT software package.
- **Pandas** (https://pandas.pydata.org/), a **Python** library for data analysis and manipulation tool
- **Matplotlib** (https://matplotlib.org/), a software for creating static, animated, and interactive visualizations in **Python**.

- **NumPy** (https://numpy.org/), a **Python** library that supports for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays.
- **SciPy** (https://scipy.org/), a **Python** library used for scientific computing and technical computing.
- **Plotnine** (https://plotnine.readthedocs.io/en/stable/), an implementation of a grammar of graphics in Python based on ggplot2

Conda is an open-source package management and environment management tool that allows you to install, manage, and configure software packages and dependencies, particularly for **Python** and data science projects. We will use **Miniforge3** (https://github.com/conda-forge/miniforge) that is a minimal installer for Conda that uses the `conda-forge` channel by default (https://conda-forge.org/), but you could use other **Conda** installer.

Next, we present examples of how to install Python, **Conda** software and this additional software in Linux Ubuntu, macOS and Windows using **WSL**.

Installation on Ubuntu 22.04 LTS or Ubuntu 24.04 LTS

The commands to install **Miniforge3** in the directory, e.g. $HOME/Apps/Miniforge3, if there is no previous installation:

```
$ mkdir -p $HOME/Apps

$ cd $HOME/Apps

$ wget "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-$(uname)-$(uname -m).sh"

$ bash Miniforge3-$(uname)-$(uname -m).sh -b -p $HOME/Apps/Miniforge3

$ rm Miniforge3-$(uname)-$(uname -m).sh

$ $HOME/Apps/Miniforge3/condabin/conda init bash
```

Close the terminal and open a new terminal. The **Conda** environment base is now active. Then type the following commands to configure the download environment:

```
$ conda config --add channels bioconda

$ conda config --add channels conda-forge

$ conda config --set channel_priority strict
```

Finally, we will create the environment *quercustoa* used to run QUERCUSTOA-APP programs typing:

```
$ conda env create -f $HOME/Apps/quercusTOA-app/yml/quercustoa.yml
```

Installation on macOS 15.0.1

First, install **Homebrew** and **wget** command, if necessary, typing in a terminal window:

```
$ /bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

$ brew install wget
```

Now install **Miniforge3** in the directory, e.g. $HOME/Apps/Miniforge3, if there is no previous installation:

```
$ mkdir -p $HOME/Apps

$ cd $HOME/Apps

$ wget "https://github.com/conda-
forge/miniforge/releases/latest/download/Miniforge3-$(uname)-$(uname -
m).sh"

$ bash Miniforge3-$(uname)-$(uname -m).sh -b -p $HOME/Apps/Miniforge3

$ rm Miniforge3-$(uname)-$(uname -m).sh

$ $HOME/Apps/Miniforge3/condabin/conda init zsh
```

Then close the terminal and open a new terminal and type the following commands to configure the download environment:

```
$ conda config --add channels bioconda

$ conda config --add channels conda-forge

$ conda config --set channel_priority strict
```

Finally, we will create the environment *quercustoa* used to run QUERCUSTOA-APP programs typing:

```
$ conda env create -f $HOME/Apps/quercusTOA-app/yml/quercustoa.yml
```

Installation on Microsoft Windows 10 (64 bits) and Windows 11 using WSL

QUERCUSTOA-APP uses the Windows Subsystem for Linux (**WSL**) and Ubuntu to run some scripts coded in Bash. **WSL** has to be installed before using QUERCUSTOA-APP. For further clarification about **WSL**, you can see the URL https://learn.microsoft.com/en-us/windows/wsl. In order to install **WSL** and Ubuntu 24.04, open a command prompt as administrator and type the following command:

```
> wsl --install --distribution Ubuntu-24.04
```

Restart Windows. Then a window will appear installing Ubuntu. This process may take a few minutes. You will have to enter a username and its password. When the process ends, close this window.

To install **Miniforge3**, e.g. in the user root directory (%USERPROFILE%), open a command prompt and type the following commands:

```
> cd %USERPROFILE%

> curl -O "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-Windows-x86_64.exe"

> Miniforge3-Windows-x86_64.exe /InstallationType=JustMe /AddToPath=1 /RegisterPython=1 /S /D=%USERPROFILE%\Miniforge3

> del Miniforge3-Windows-x86_64.exe
```

Then close the terminal and open a new terminal and type the following commands to configure the download environment:

```
> conda config --add channels conda-forge

> conda config --set channel_priority strict
```

Finally, we will create the environment *quercustoa* used to run QUERCUSTOA-APP programs typing:

```
$ conda env create -f %USERPROFILE%\Apps\quercusTOA-app/yml/quercustoa-win.yml
```

Starting QUERCUSTOA-APP

QUERCUSTOA-APP runs in graphical mode using the graphical user interface (GUI).

If your O.S. is Ubuntu or macOS, start QUERCUSTOA-APP typing the following commands in a terminal window:

```
$ cd $HOME/Apps/quercusTOA-app/Package

$ conda activate quercustoa

$ ./quercusTOA.py
```

If your O.S. is Windows, type the command:

```
> cd %USERPROFILE%\Apps\quercusTOA-app\Package

> conda activate quercustoa

> python quercusTOA.py
```

The initial appearance of QUERCUSTOA-APP at application startup in GUI mode is shown in Figure 8.

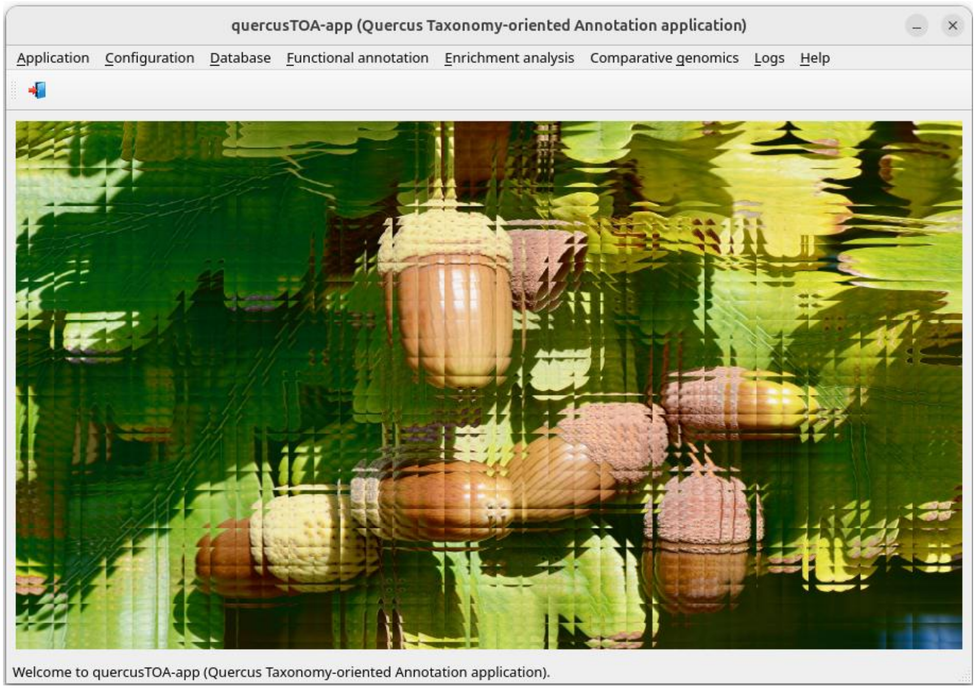


Figure 8. Front-end of the QUERCUSTOA-APP application interface at startup.

First steps in GUI environments

QUERCUSTOA-APP menus

QUERCUSTOA-APP is structured in several menus:

Application

Just to exit the application.

Configuration

This menu contains all the items related to:

- Recreate QUERCUSTOA-APP config file
- View QUERCUSTOA-APP config file
- Install Bioinfo software
    - Miniforge3 and additional infrastructure software
    - BLAST+
    - CodAn
    - DIAMOND
    - Liftoff
    - LiftoffTools
    - MAFFT

Database

This menu contains all the items related to:

- Download QUERCUSTOA database
- Statistics

Functional annotation

This menu contains all the items related to:

- Run pipeline
- Restart pipeline
- Browse results
- Statistics
    - Summary report
    - Species
        - Frequency distribution data
        - Frequency distribution plot
    - Gene Ontology

        - Frequency distribution per GO term data
        - Frequency distribution per GO term plot
        - Frequency distribution per namespace data
        - Frequency distribution per namespace plot
        - Sequences # per GO terms # data
        - Sequences # per GO terms # plot

Enrichment analysis

This menu contains all the items related to:

- Run analysis
- Restart analysis
- Browse results
    - GO enrichment analysis
    - Metacyc pathway enrichment analysis
    - KEGG KO enrichment analysis
    - KEGG pathway enrichment analysis

Comparative genomics

This menu contains all the items related to:

- Search homology of sequences
- Restart a sequences homology search
- Browse results
- Get protein ids homology
- Create a GFF file
- Restart a GFF file creation

Logs menu

This menu allows access to the application logs:

- Submitting logs
- Result logs

Help menu

It contains the documentation of the application.

Configuring the QUERCUSTOA-APP environment

When QUERCUSTOA-APP starts for the first time it is required to configure the QUERCUSTOA-APP environment. To do so, we select the menu item with the following path:

Main menu > Configuration > Recreate quercusTOA-app config file

Figure 9 shows the window corresponding to this menu item. Default values are presented for *Database directory* and *Result directory*. If necessary, modify them and press the button *[Execute]*.

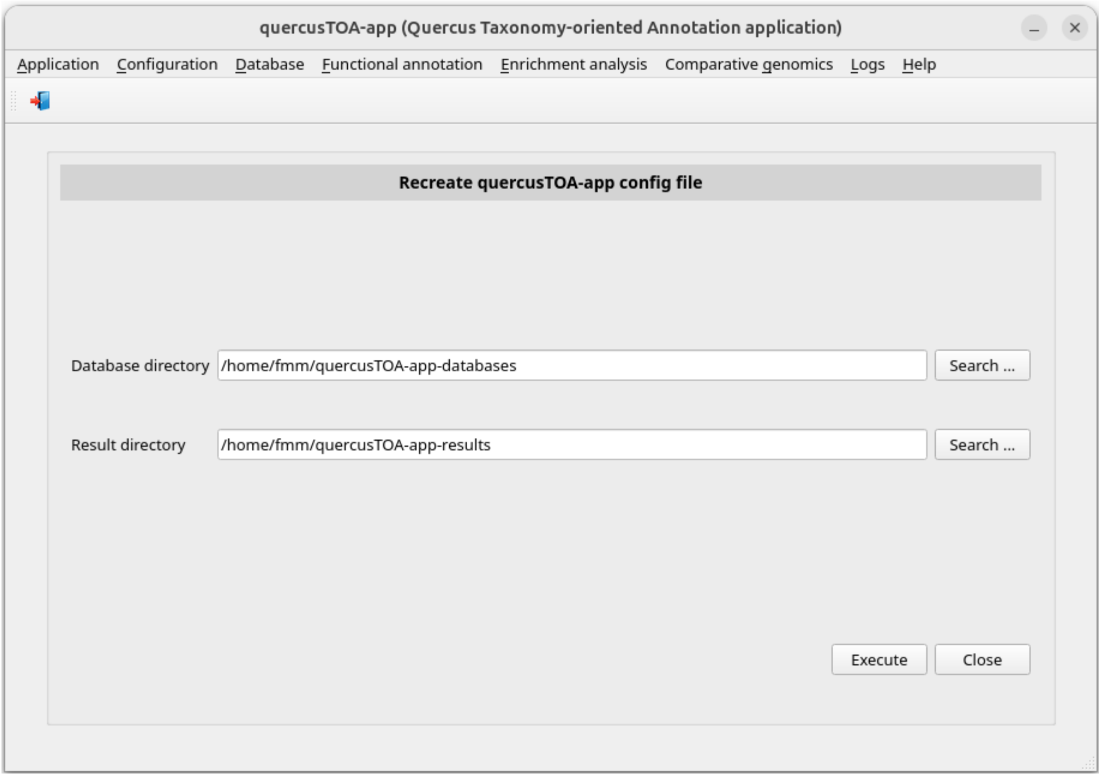


Figure 9. Window *Recreate quercusTOA-app config file*.

Installing infrastructure software on WSL (Windows only)

As stated in chapter "Installation in GUI environments", GYMNOTOA-APP uses the Windows Subsystem for Linux (**WSL**) and Ubuntu to run some scripts coded in Bash. Therefore, it is necessary to install **Miniforge3** and the environment *quercustoa* on **WSL**.

First, install Miniforge3·(Conda infrastructure) selecting the menu item with this path:

> *Main menu > Configuration > Bioinfo software installation > Miniforge3 (Conda infrastructure) on WSL [Execute]*

Press the bottom *[Execute]*. A pop-up window will display the submission log (Figure 10).

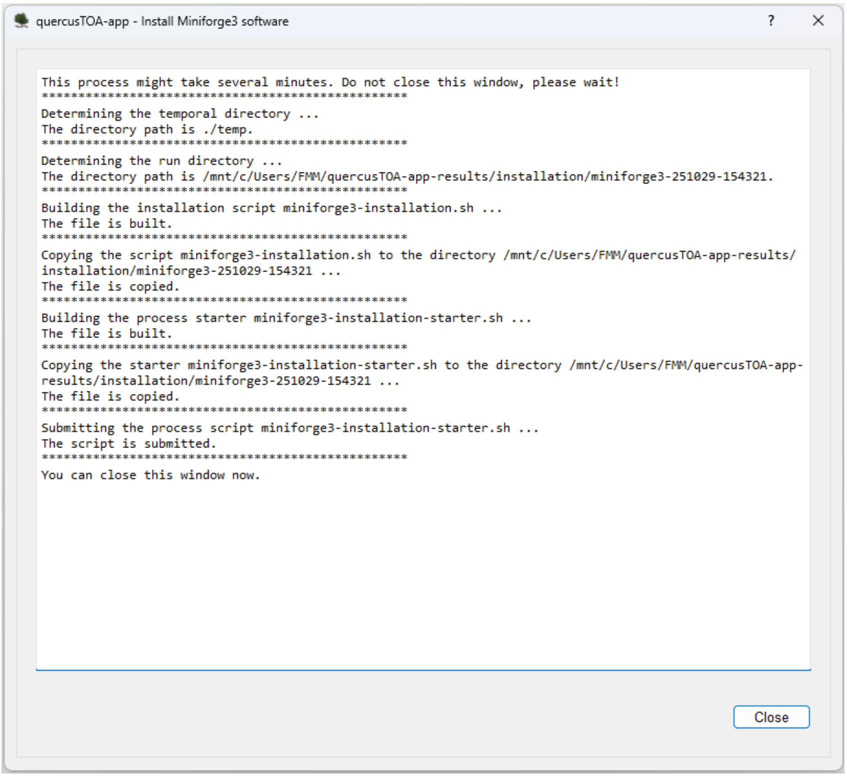


Figure 10. Submitting log of a Miniforge3 installation process on a computer with Windows.

To view the process log during and after the run, select the menu item with this path:

> *Main menu > Logs > Result logs*

Figure 11 shows the window *Browse results logs*.

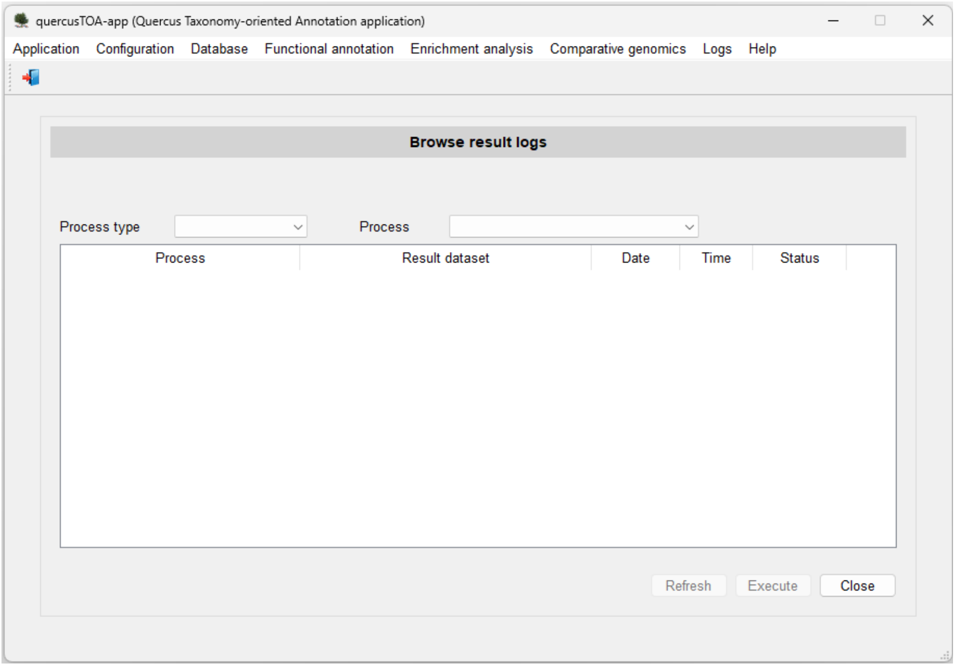


Figure 11. Initial appearance of the window *Browse results logs* on a computer with Windows.

Select **installation** in the *Process type* combo-box. Then the window is updated (Figure 12).

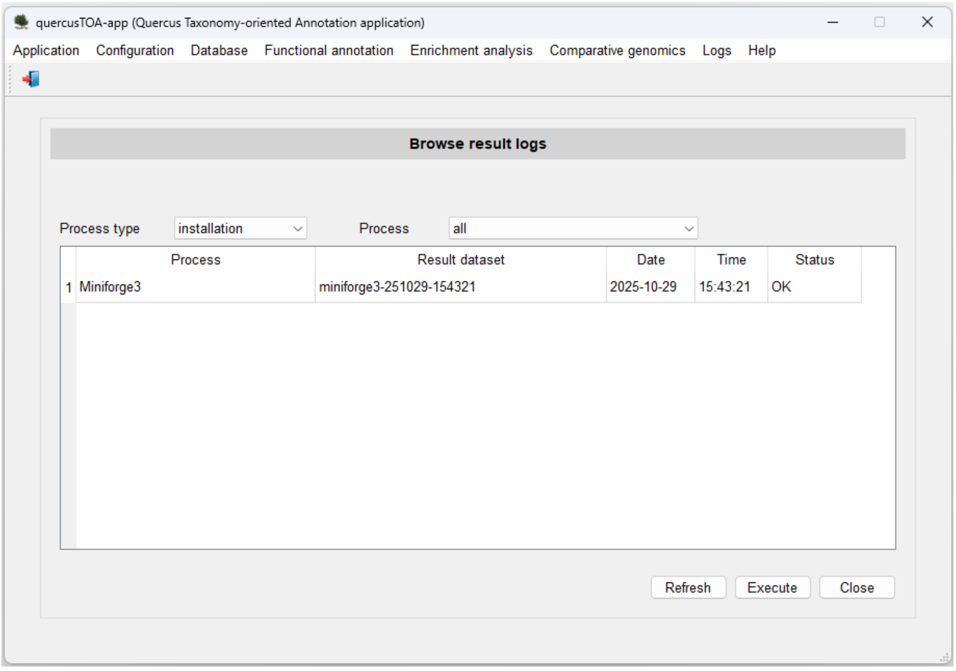
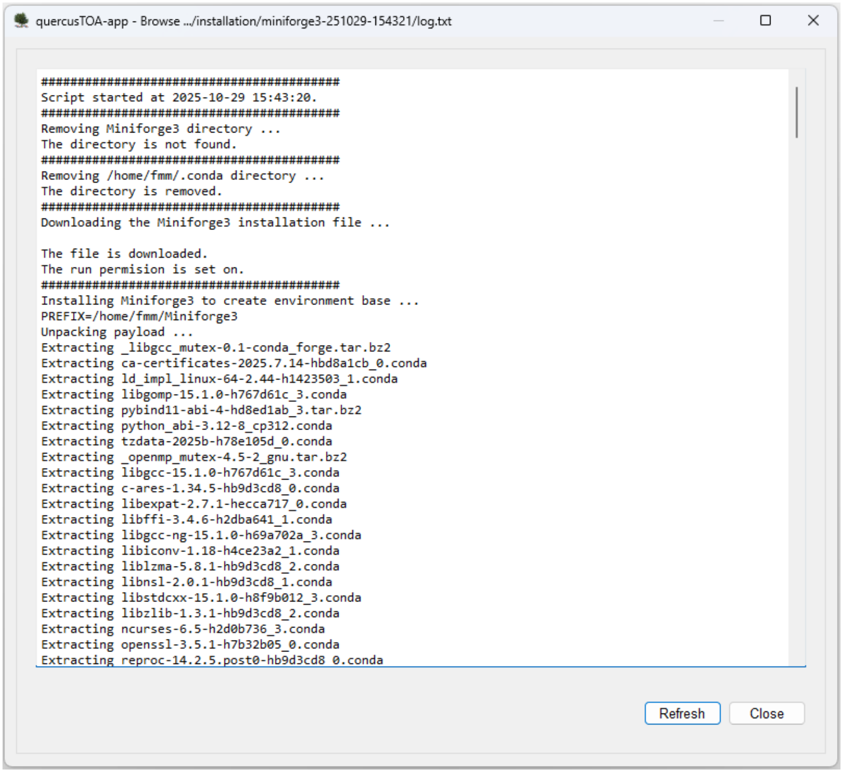


Figure 12. Window *Browse results logs* showing the Miniforge3 installation process finished on a computer with Windows.

So far, we have only performed a single installation process: the process corresponding to the last (and unique) **Miniforge3** installation. By double-clicking on it or selecting its row with a click and pressing the Execute button, a new pop-up window appears with its corresponding log (Figure 13).



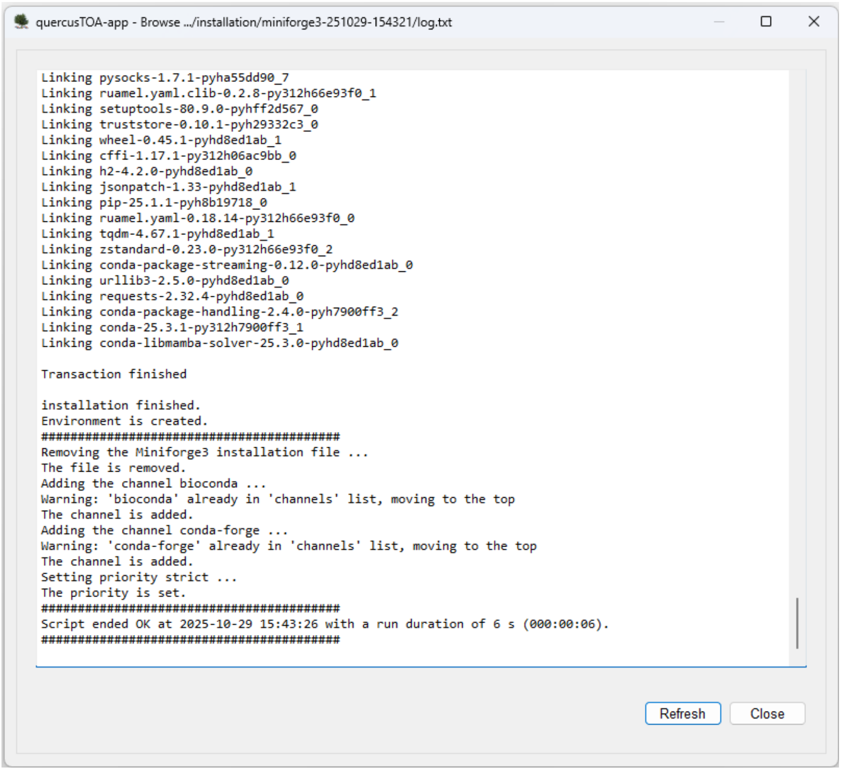


Figure 13. Begin and end of a log of Miniforge3 installation process on a computer with Windows.

There is a button to refresh the run status. Clicking it, the log will be updated.

All the process logs have:

- A header with the time when it started.
- At the bottom, a summary with the status (OK, if all the programs have ended without errors; WRONG, otherwise), the end time, and the duration of the script run.

Once **Miniforge3** software is installed on **WSL**, install the environment *quercusTOA* clicking the following menu item:

> *Main menu > Configuration > Bioinfo software installation > quercusTOA environment on WSL [Execute]*

Installing bioinformatic software

QUERCUSTOA-APP's dependencies are the following:

- **BLAST+** (https://blast.ncbi.nlm.nih.gov/). It is used to find local similarity between transcripts or predicted peptides and sequences of genomic databases.
- **CodAn** (https://github.com/pedronachtigall/CodAn/). It is a software package to characterize the CDS and UTR regions on transcripts from any Eukaryote species.
- **DIAMOND** (https://github.com/bbuchfink/diamond/) is a faster BLAST alternative (100x-10,000x) to align protein and predicted peptides though it reports fewer matches.
- **Liftoff** (https://github.com/agshumate/Liftoff/).It is a tool that accurately maps annotations in GFF or GTF between assemblies of the same, or closely-related species
- **LiftoffTools** (https://github.com/agshumate/LiftoffTools/). It is a toolkit to compare genes lifted between genome assemblies.
- **MAFFT** (https://mafft.cbrc.jp/). It is a multiple alignment program for amino acid or nucleotide sequences.

The installation of these packages is automatic by using Bioconda (https://bioconda.github.io/) which holds many bioinformatics software packages.

We install the necessary software selecting the menu items with these paths:

> *Main menu > Configuration > Bioinfo software installation > BLAST+ [Execute]*
>
> *Main menu > Configuration > Bioinfo software installation > CodAn [Execute]*
>
> *Main menu > Configuration > Bioinfo software installation > DIAMOND [Execute]*
>
> *Main menu > Configuration > Bioinfo software installation > Liftoff [Execute]*
>
> *Main menu > Configuration > Bioinfo software installation > LiftoffTools [Execute]*
>
> *Main menu > Configuration > Bioinfo software installation > MAFFT [Execute]*

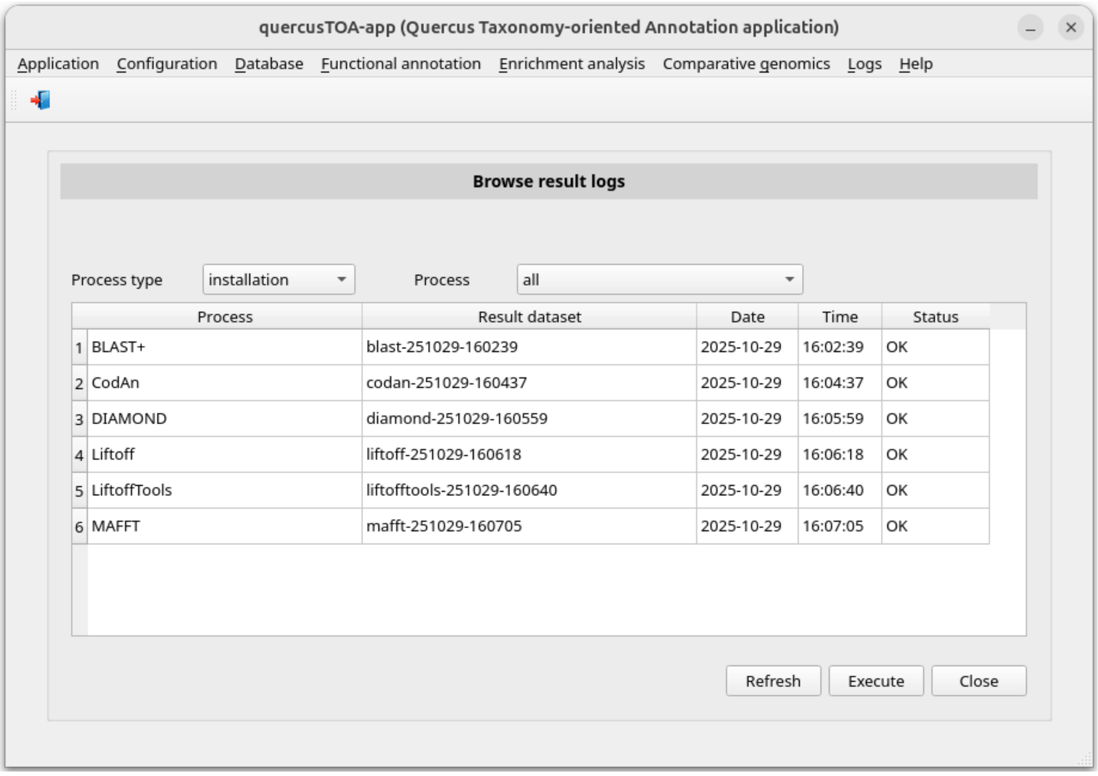


Figure 14. Window *Browse results logs* once all installation processes are finished on a computer with Ubuntu.

Consulting submitted processes and troubleshooting

The correct operability of the submitted processes is controlled by logs like the one described in the section “Installing infrastructure software on WSL (Windows only)” (Figure 13). Doing so, the user can monitor the performance of the process at any time and detect problems. Please, confirm that each process is finished before submitting another one.

A step-by-step example in GUI environments

The following step-by-step examples demonstrate how to use the QUERCUSTOA-APP for functional annotation and comparative genomics:

- QUERCUSTOA database
    - Download QUERCUSTOA database
    - View statistics of QUERCUSTOA database
    - Run a functional annotation pipeline
- Functional annotation
    - Browse result of the functional annotation
    - View statistics of the functional annotation
- Enrichment analysis
    - Run an enrichment analysis
    - Browse results of the enrichment analysis
- Comparative genomics
    - Search of sequence homology
    - Browse results of the homology search

The QUERCUSTOA database

Download the QUERCUSTOA database

NOTE: The QUERCUSTOA database must be downloaded before using the QUERCUSTOA-APP. Subsequent downloads are only necessary when a database update is available on the server

The QUERCUSTOA database is downloaded selecting the menu item with this path:

Main menu > Database > Download quercusTOA-db [Execute]

Then a process is submitted which will access the server, download the latest version of the database and decompress it. You can check when the process has been completed by reviewing its log. To do so, click the following menu item:

Main menu > Logs > Result logs

Select **database** in the *Process type* combo-box. Then the window will update (Figure 15).

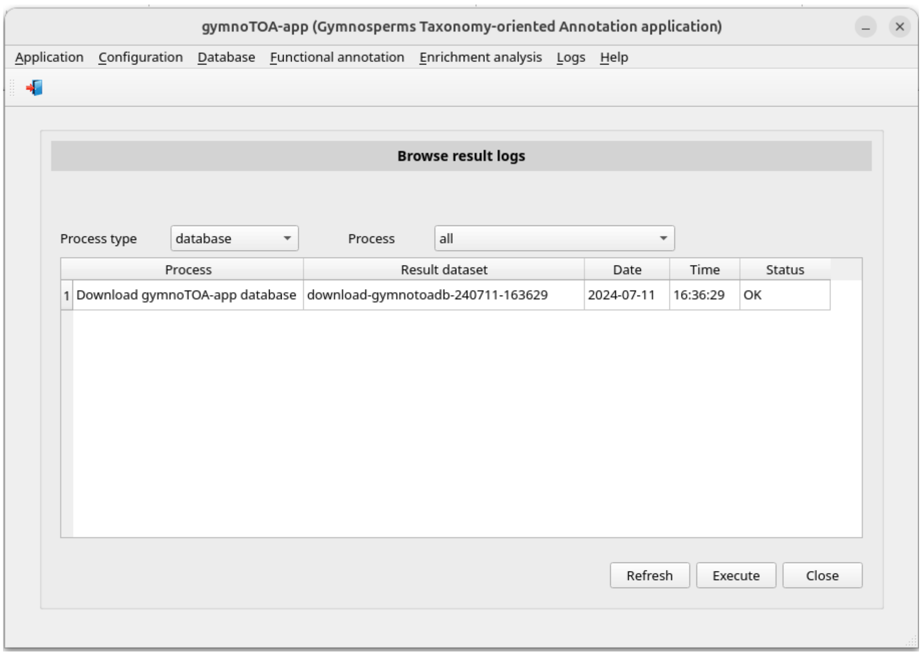


Figure 15. Window *Browse results logs* showing the database process finished.

View statistics of QUERCUSTOA database

The statistics of QUERCUSTOA database can be consulted in the menu item with this path:

Main menu > Database > Statistics

The window *View quercusTOA-db statistics* will appear (Figure 16).

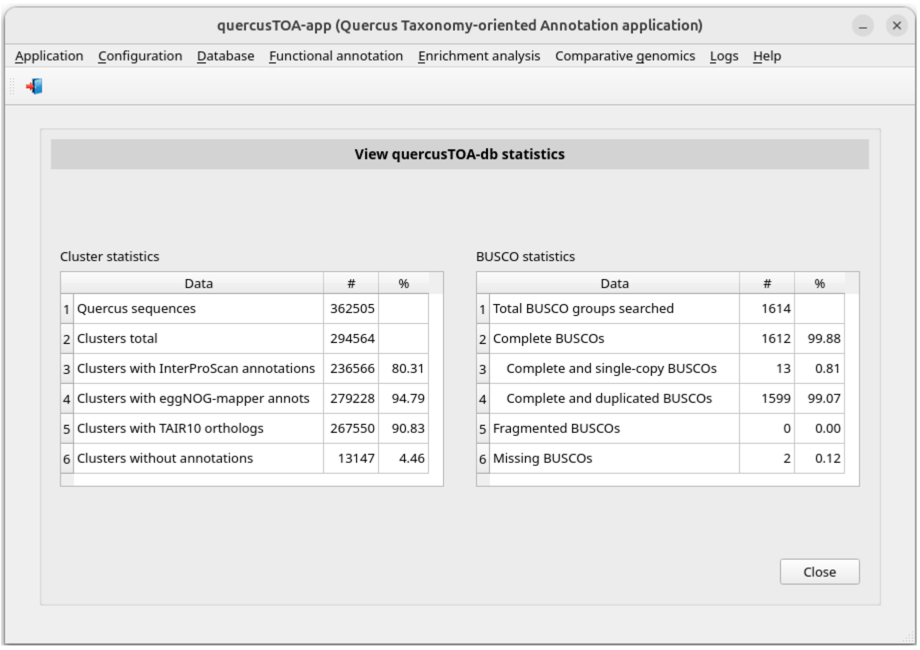


Figure 16. Window *View quercusTOA-db statistics* with data of July 2025 version data.

Functional annotation

Run a functional annotation pipeline

TREEGENES is a forest tree genome database that holds genome and transcriptome information of many forest tree. Their *Quercus suber* transcriptome is located in the URL https://treegenesdb.org/FTP/Transcriptome/TSA/Qusu/Qusu_TSA.fasta. In this example, we will use a subset of 1000 sequences, file *Qusu_TSA-1000seqs.fasta*, included in the subdirectory *sample-data* of QUERCUSTOA-APP.

To perform an annotation process (Figure 4) we select the menu item with this path:

Main menu > Functional annotation > Run pipeline

The window *Run an annotation pipeline* appears (Figure 17).

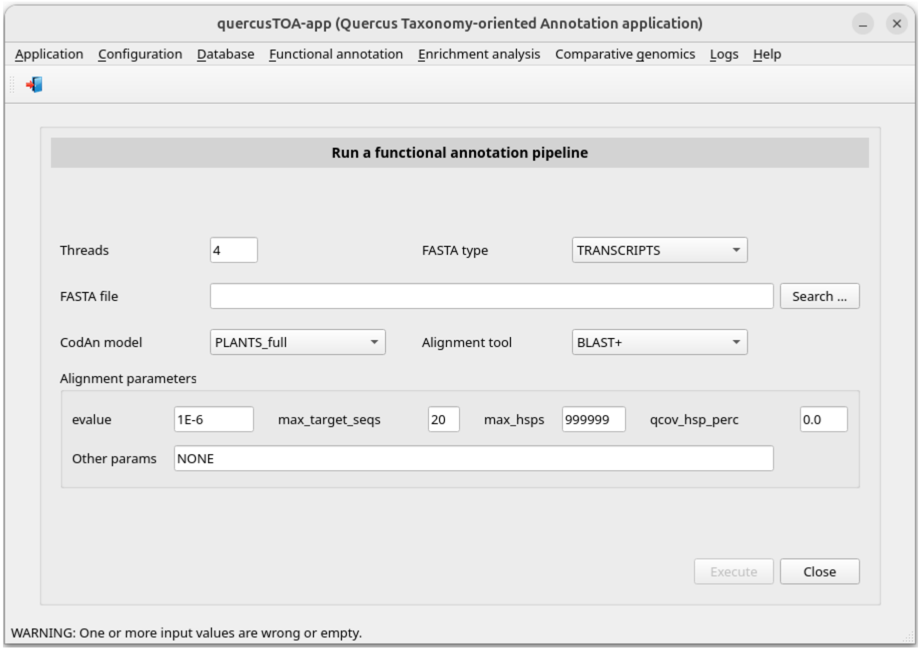


Figure 17. Initial appearance of the window *Run annotation pipeline*.

Select the FASTA type option (TRANSCRIPTS in our case). Then type the path of the FASTA file or select it using the button Search... (in this example we use the file transcripts file *Qusu_TSA-1000seqs.fasta* located in the subdirectory *sample-data* of QUERCUSTOA-APP). Then the Execute button will be available (Figure 18). Before running the process, you can also modify the default value of:

- threads number.
- **CodAn** model: **PLANTS_full** or **PLANTS_partial.**
- alignment software (**BLAST+** or **DIAMOND**) and its parameters: e_value (number of expected hits of similar quality -score- that could be found just by chance), the

number of aligned sequences to keep, the number of HSPs (high-scoring segment pairs) to keep and the query coverage per HSP (**0.0** if **DIAMOND**).

- Other parameters: additional parameters and their corresponding values.

Once the parameters are updated, click the Execute button.

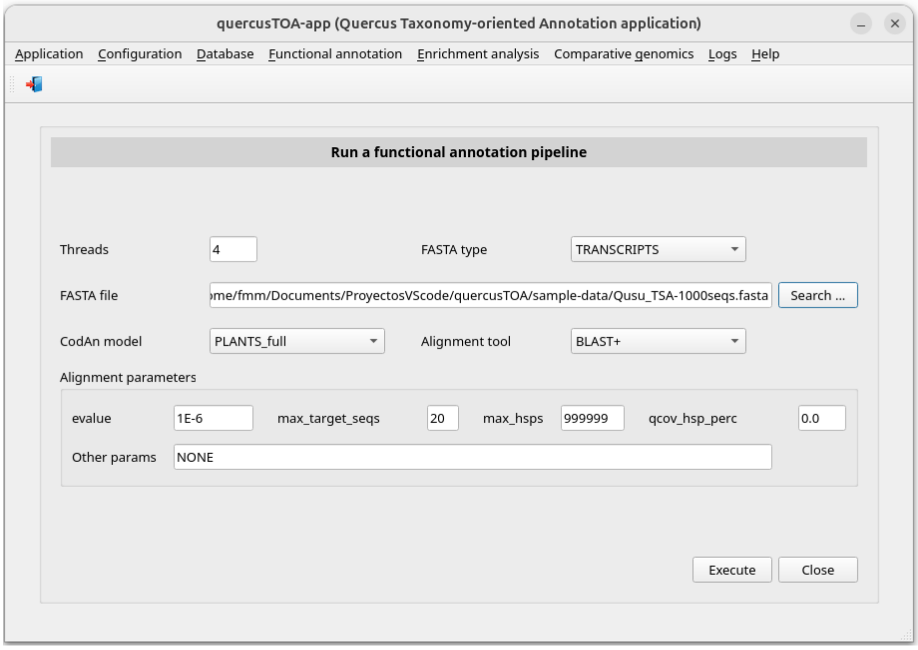


Figure 18. Window *Run annotation pipeline* once the transcriptome file has been typed.

You can check the process status and when it is completed reviewing the log. to do so, click the following menu item:

Main menu > Logs > Result logs

Select **run** in the *Process type* combo-box. Then the window is updated (Figure 19).

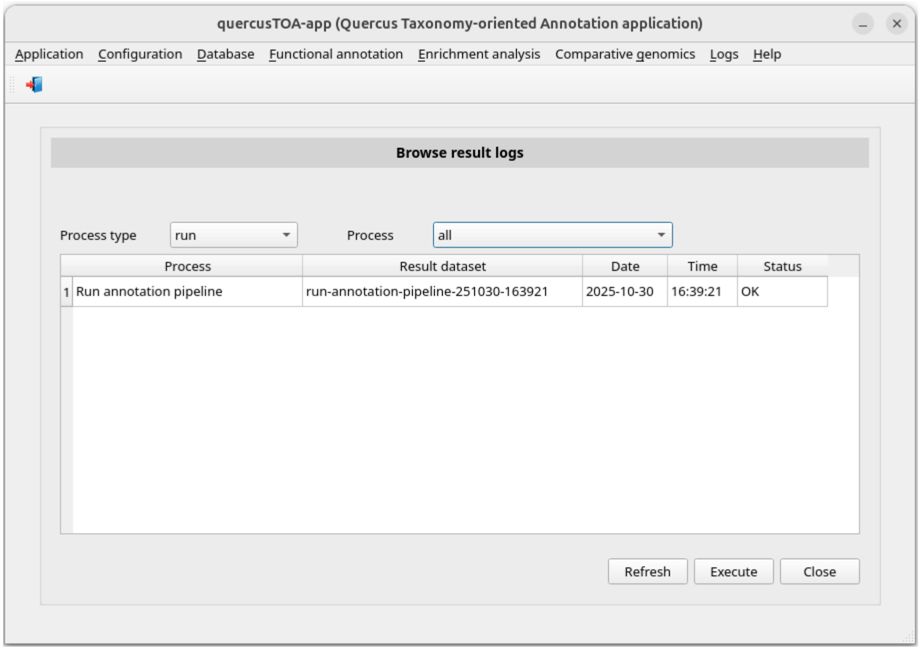


Figure 19. Window *Browse results logs* showing annotation pipeline finished.

When the process has finished, we could go the subdirectory of *...\quercusTOA-app-results\run* corresponding to the script run, e.g. *run-annotation-pipeline-251030-163921*, and consult the generated files (Figure 20). Some of these files are:

- Annotation:
    - *functional-annotations-complete.csv*: It contains all annotations yielded by blast programs for every query sequence identification.
    - *functional-annotations-besthit.csv*: It contains the annotations of the subject sequence identification with best hit yielded by blast programs for every query sequence identification.
- Statistics:
    - *stats-goterms.csv*: It contains
    - *stats-namespaces.csv*: It contains
    - *stats-seq-per-goterm.csv*: It contains
    - *stats-species.csv*: It contains
- Other applications inputs:
    - *agrigo-input-file.txt*: It contains data to be used as **agriGO** input.
    - *revigo-input-file.txt*: It contains data to be used as **REVIGO** input.

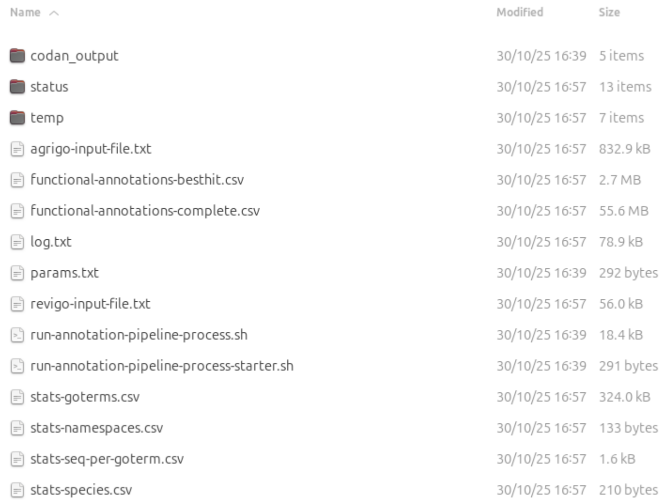


Figure 20. Folders and files in *…/quercusTOA-app-results/run/run-annotation-pipeline-251030-163921* yielded by the annotation pipeline.

Browse result of the functional annotation

If you edit the files *functional-annotations-complete.csv* or *functional-annotations-besthit.csv*, you will see the result of the function annotation You can browse it using QUERCUSTOA-APP in the following menu item:

Main menu > Functional annotation > Browse results of an annotation Pipeline

The window (Figure 21) allows you to choose **best hit per sequence** (*functional-annotations-besthit.csv*) or **all hits per sequence** (*functional-annotations-complete.csv*) in the combo-box *Result type*.

Click on the annotation pipeline that you are interested in consulting, in our example *run-annotation-pipeline-251030-163921*. The window will be updated showing the parameters of the process (Figure 22).

Then click push button Execute and a pop-up window will appear showing the functional annotation data (Figure 23).

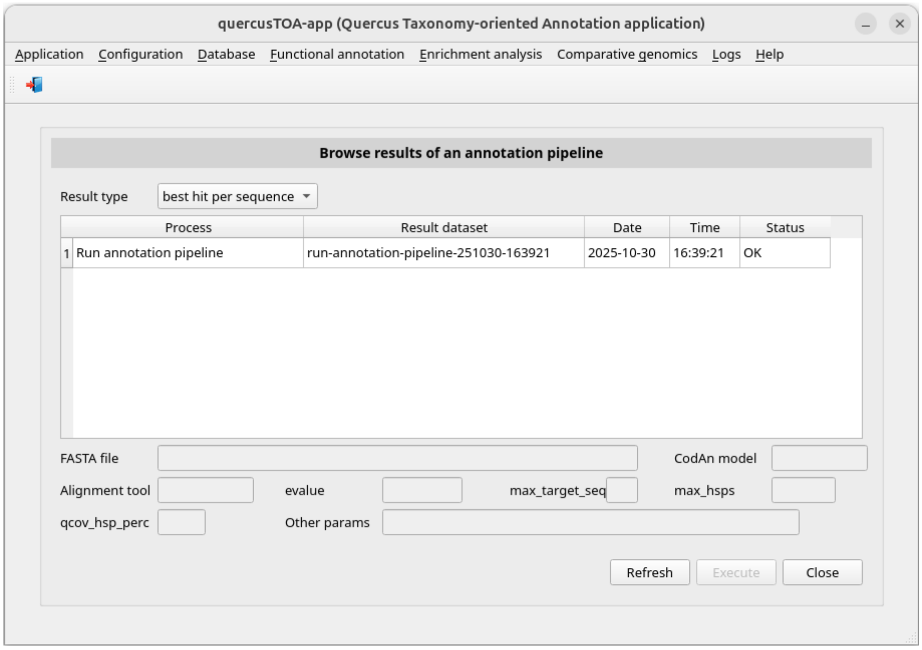


Figure 21. Window *Browse results of an annotation pipelines* showing annotation pipelines finished.

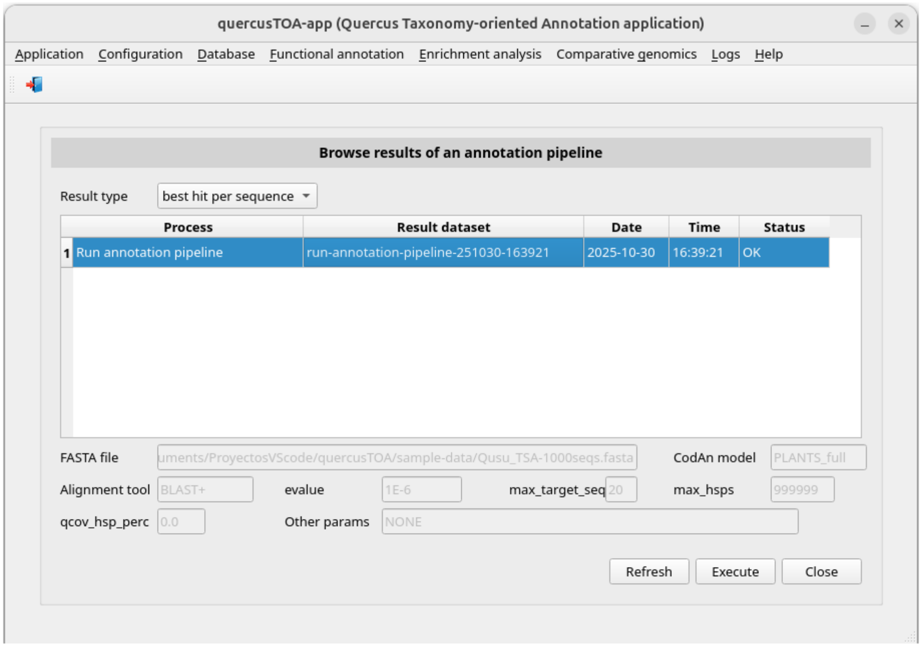
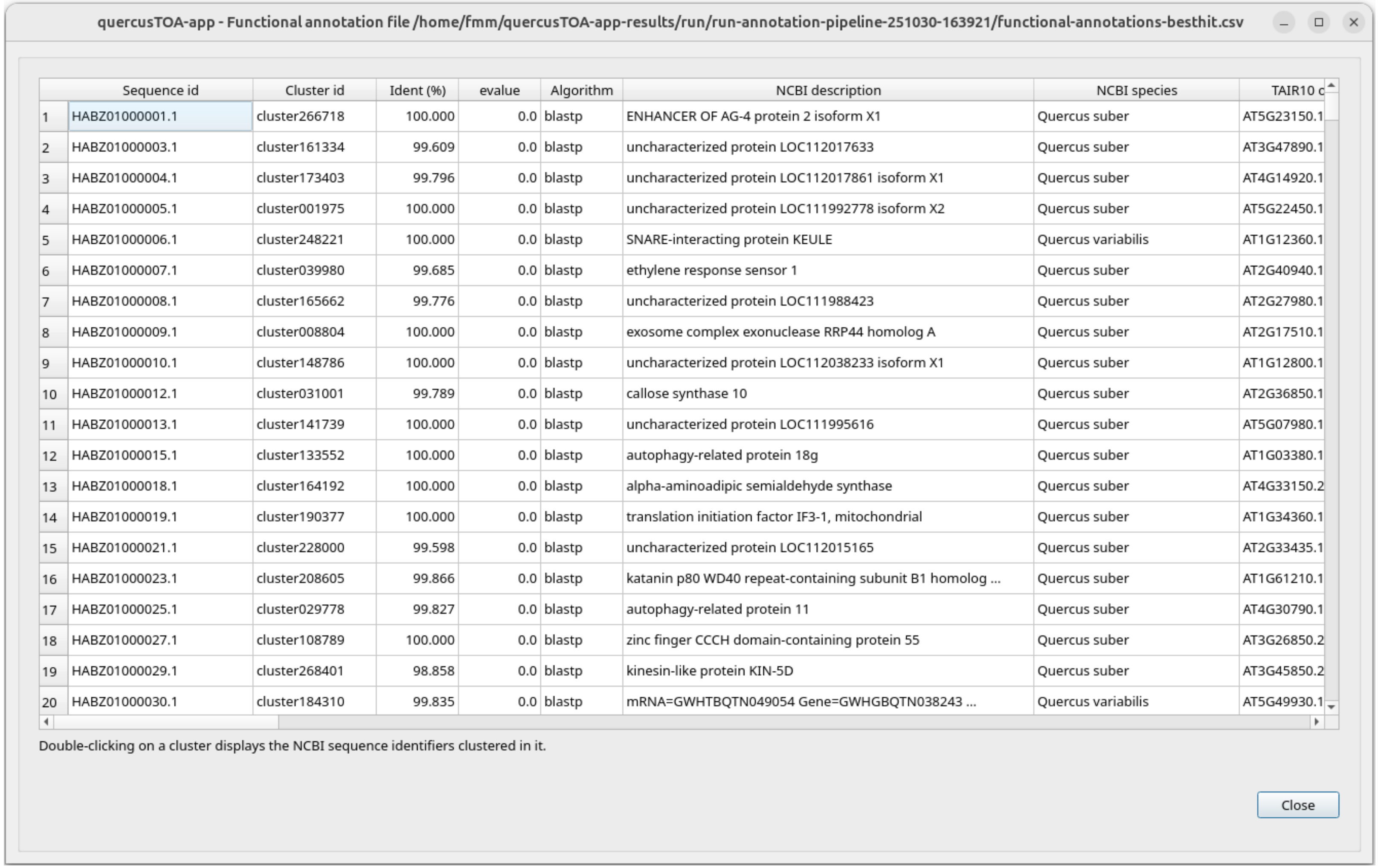


Figure 22. Window *Browse results of an annotation pipelines* after *run-annotation-pipeline-251030-163921* was selected.



quercusTOA-app - Functional annotation file /home/fmm/quercusTOA-app-results/run/run-annotation-pipeline-251030-163921/functional-annotations-besthit.csv

	Sequence id	Cluster id	Ident (%)	evalue	Algorithm	NCBI description	NCBI species	TAIR10 c
1	HABZ01000001.1	cluster266718	100.000	0.0	blastp	ENHANCER OF AG-4 protein 2 isoform X1	Quercus suber	AT5G23150.1
2	HABZ01000003.1	cluster161334	99.609	0.0	blastp	uncharacterized protein LOC112017633	Quercus suber	AT3G47890.1
3	HABZ01000004.1	cluster173403	99.796	0.0	blastp	uncharacterized protein LOC112017861 isoform X1	Quercus suber	AT4G14920.1
4	HABZ01000005.1	cluster001975	100.000	0.0	blastp	uncharacterized protein LOC111992778 isoform X2	Quercus suber	AT5G22450.1
5	HABZ01000006.1	cluster248221	100.000	0.0	blastp	SNARE-interacting protein KEULE	Quercus variabilis	AT1G12360.1
6	HABZ01000007.1	cluster039980	99.685	0.0	blastp	ethylene response sensor 1	Quercus suber	AT2G40940.1
7	HABZ01000008.1	cluster165662	99.776	0.0	blastp	uncharacterized protein LOC111988423	Quercus suber	AT2G27980.1
8	HABZ01000009.1	cluster008804	100.000	0.0	blastp	exosome complex exonuclease RRP44 homolog A	Quercus suber	AT2G17510.1
9	HABZ01000010.1	cluster148786	100.000	0.0	blastp	uncharacterized protein LOC112038233 isoform X1	Quercus suber	AT1G12800.1
10	HABZ01000012.1	cluster031001	99.789	0.0	blastp	callose synthase 10	Quercus suber	AT2G36850.1
11	HABZ01000013.1	cluster141739	100.000	0.0	blastp	uncharacterized protein LOC111995616	Quercus suber	AT5G07980.1
12	HABZ01000015.1	cluster133552	100.000	0.0	blastp	autophagy-related protein 18g	Quercus suber	AT1G03380.1
13	HABZ01000018.1	cluster164192	100.000	0.0	blastp	alpha-aminoadipic semialdehyde synthase	Quercus suber	AT4G33150.2
14	HABZ01000019.1	cluster190377	100.000	0.0	blastp	translation initiation factor IF3-1, mitochondrial	Quercus suber	AT1G34360.1
15	HABZ01000021.1	cluster228000	99.598	0.0	blastp	uncharacterized protein LOC112015165	Quercus suber	AT2G33435.1
16	HABZ01000023.1	cluster208605	99.866	0.0	blastp	katanin p80 WD40 repeat-containing subunit B1 homolog ...	Quercus suber	AT1G61210.1
17	HABZ01000025.1	cluster029778	99.827	0.0	blastp	autophagy-related protein 11	Quercus suber	AT4G30790.1
18	HABZ01000027.1	cluster108789	100.000	0.0	blastp	zinc finger CCCH domain-containing protein 55	Quercus suber	AT3G26850.2
19	HABZ01000029.1	cluster268401	98.858	0.0	blastp	kinesin-like protein KIN-5D	Quercus suber	AT3G45850.2
20	HABZ01000030.1	cluster184310	99.835	0.0	blastp	mRNA=GWHTBQTN049054 Gene=GWHGBQTN038243 ...	Quercus variabilis	AT5G49930.1

Double-clicking on a cluster displays the NCBI sequence identifiers clustered in it.

Close

Figure 23. Pop-up window showing the annotation performed by the process *run-annotation-pipeline-251030-163921*.

If you double-click on a cluster identification, e.g. *cluster266718*, other pop-up will appear with the NCBI or CNCB-NGDC sequences that form the cluster (Figure 24).

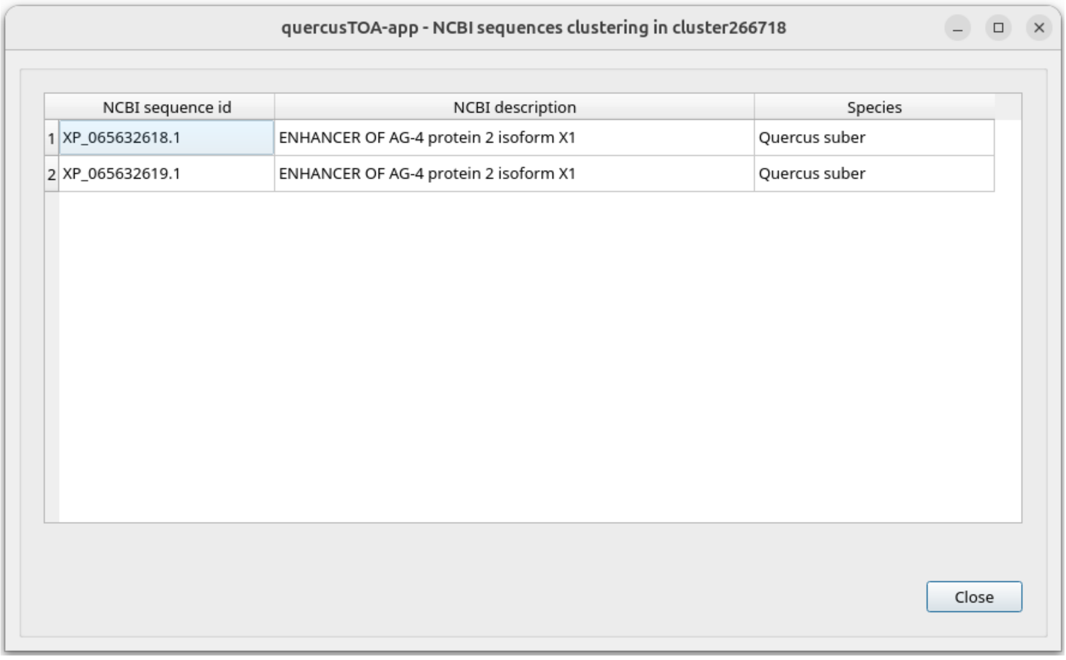


Figure 24. Pop-up window showing the sequence composition of the cluster *cluster266718*.

View statistics of the functional annotation

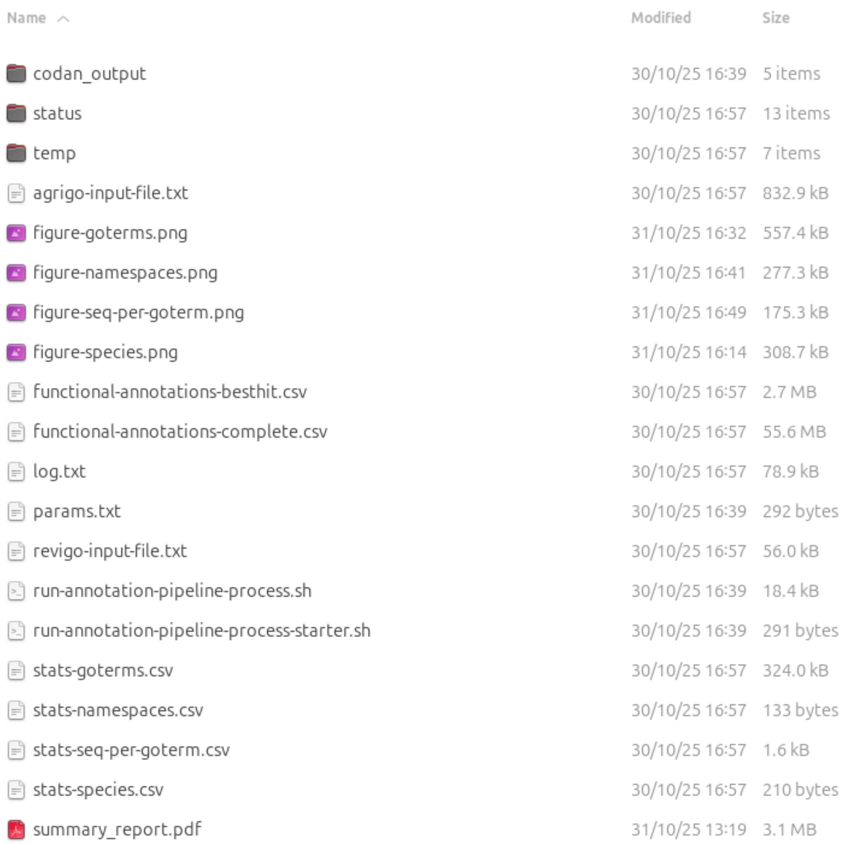
Annotation pipelines generate some statistics: frequency distribution of species, frequency distribution per GO term, frequency distribution per namespace and sequence number per GO term number.

First, we build a summary report using the menu item with this path:

Main menu > Functional annotation > Statistics > Summary report [Execute]

A PDF document will be shown with plots of above statistics. If you go the subdirectory of *…\quercusTOA-app-results\run* corresponding to the script run and consult the files (Figure 25), you will see these new files:

- *summary_report.pdf*: report including the following PNG files
- *figure-species.png*: frequency distribution of species
- *figure-goterms.png*: frequency distribution per GO term
- *figure-namespaces.png*: frequency distribution per namespace
- *figure-seq-per-goterm.png*: sequence number per GO term number



Name	Modified	Size
codan_output	30/10/25 16:39	5 items
status	30/10/25 16:57	13 items
temp	30/10/25 16:57	7 items
agrigo-input-file.txt	30/10/25 16:57	832.9 kB
figure-goterms.png	31/10/25 16:32	557.4 kB
figure-namespaces.png	31/10/25 16:41	277.3 kB
figure-seq-per-goterm.png	31/10/25 16:49	175.3 kB
figure-species.png	31/10/25 16:14	308.7 kB
functional-annotations-besthit.csv	30/10/25 16:57	2.7 MB
functional-annotations-complete.csv	30/10/25 16:57	55.6 MB
log.txt	30/10/25 16:57	78.9 kB
params.txt	30/10/25 16:39	292 bytes
revigo-input-file.txt	30/10/25 16:57	56.0 kB
run-annotation-pipeline-process.sh	30/10/25 16:39	18.4 kB
run-annotation-pipeline-process-starter.sh	30/10/25 16:39	291 bytes
stats-goterms.csv	30/10/25 16:57	324.0 kB
stats-namespaces.csv	30/10/25 16:57	133 bytes
stats-seq-per-goterm.csv	30/10/25 16:57	1.6 kB
stats-species.csv	30/10/25 16:57	210 bytes
summary_report.pdf	31/10/25 13:19	3.1 MB

Figure 25. Folders and files in *.../quercusTOA-app-results/run/run-annotation-pipeline-251030-163921* after summary report was built.

You also can browse statistics data or make plots with different formats and resolutions using QUERCUSTOA-APP. First, we consult data of frequency distribution of species in the following menu item:

Main menu > Functional annotation > Statistics > Species > Frequency distribution data

In the new window (Figure 26), double-click on the annotation process or select with a click on it and press the *Execute* button. A pop-up window will appear with data of frequency distribution of species (Figure 27).

We view the plot corresponding to the top ten data selecting the menu item

Main menu > Functional annotation > Statistics > Species > Frequency distribution data

In the window shown below (Figure 28), you can modify the default values of the file name and its format and resolution. After, double-click on the annotation process or select with a click on it and press the *Execute* button. A pop-up window will appear with the plot of frequency distribution of species (Figure 29). The corresponding file is saved in the subdirectory of *...\quercusTOA-results\run* corresponding to the script run.

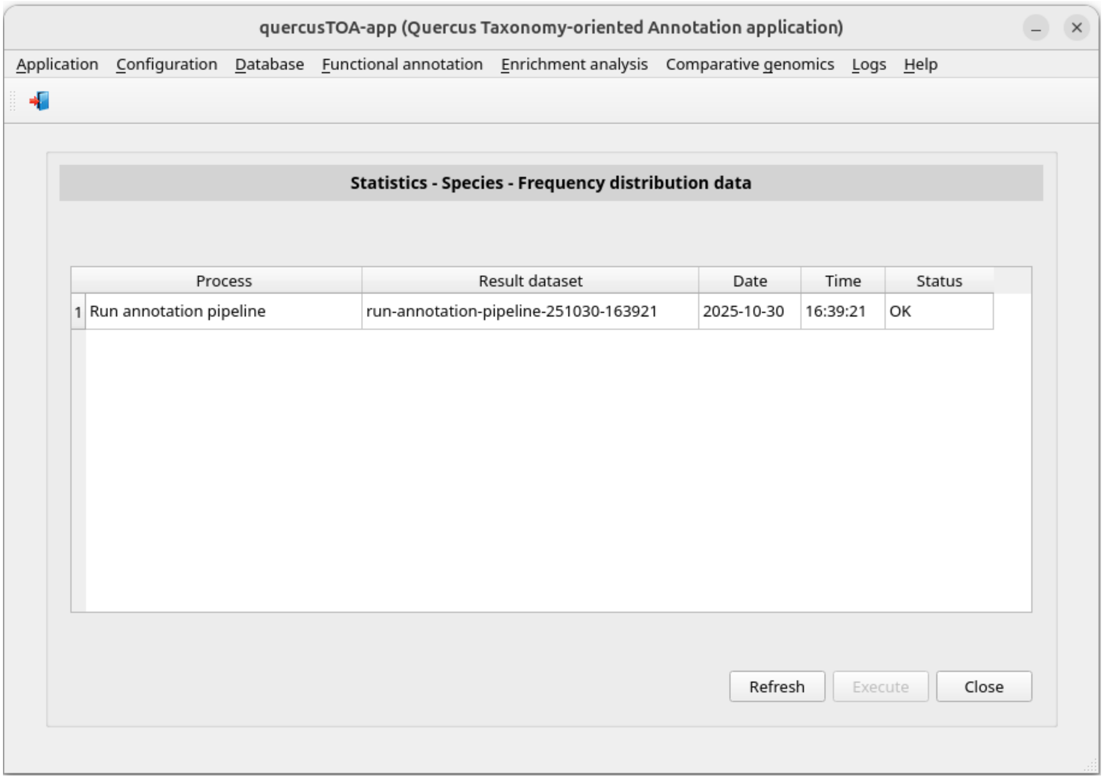


Figure 26. Window *Statistics - Species - Frequency distribution data* showing annotation pipelines finished.

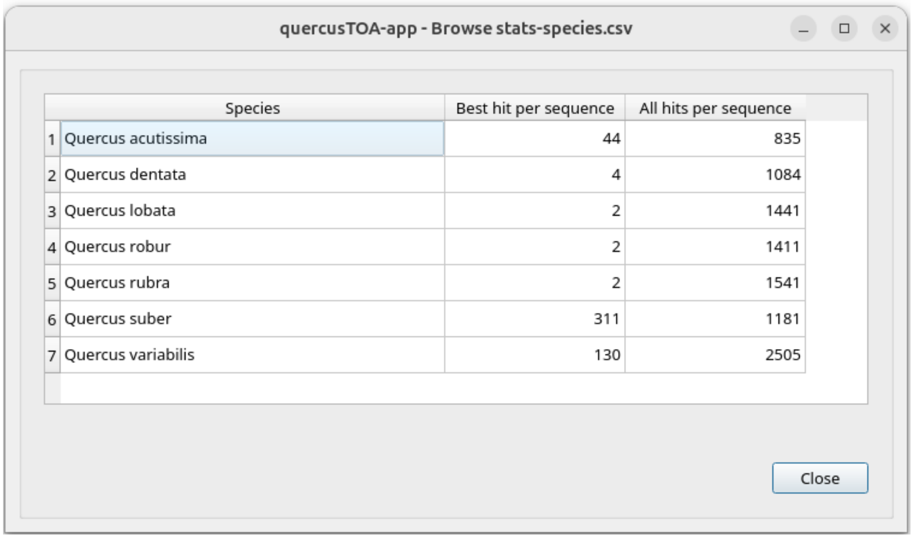


Figure 27. Pop-up window showing the complete list of frequency distribution performed by the process *run-annotation-pipeline-251030-163921*.

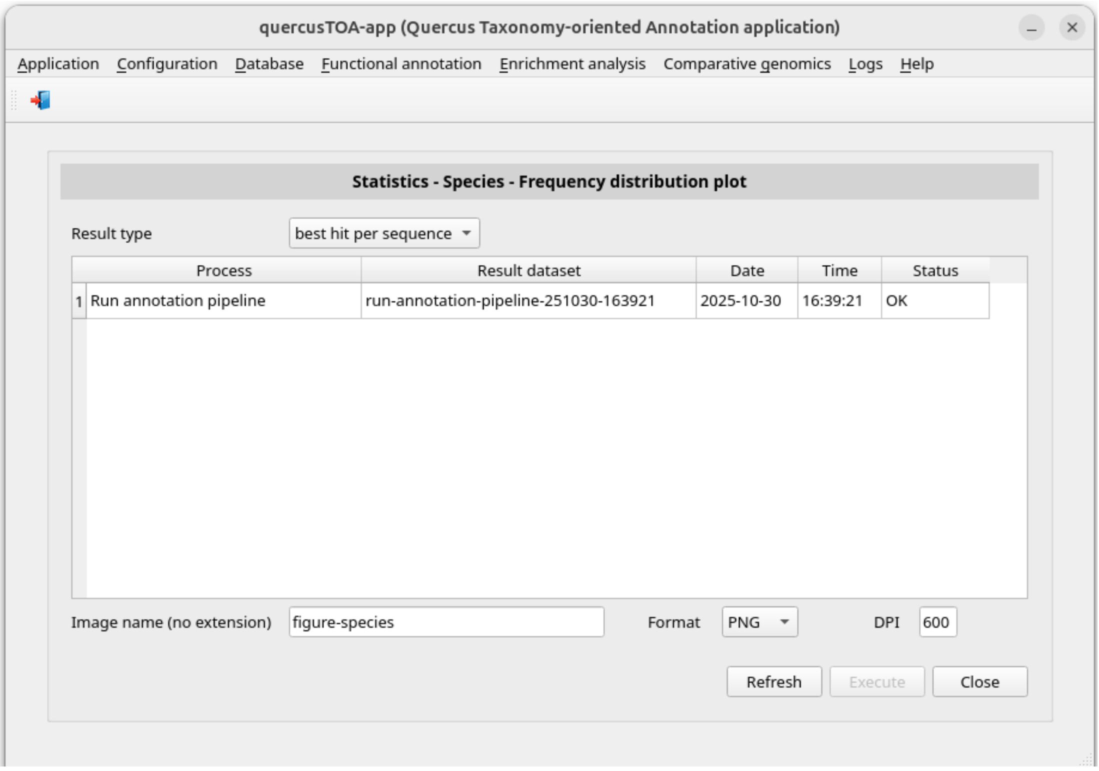


Figure 28. Window *Statistics - Species - Frequency distribution plot* showing annotation pipelines finished.

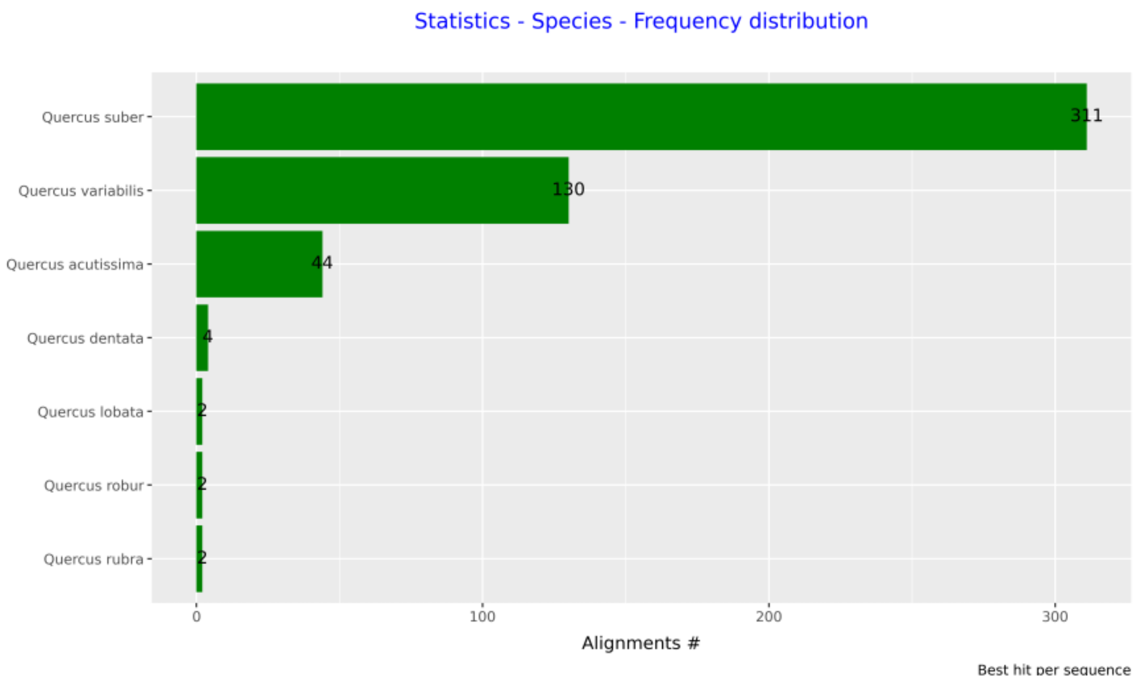


Figure 29. Plot *Statistics - Species - Frequency distribution* with the top ten of species.

The frequency distribution per GO term can be consulted in the menu item with this path:

> *Main menu > Functional annotation > Statistics > Gene Ontology > Frequency distribution per GO term data*

In the new window (Figure 30), double-click on the annotation process or select with a click on it and press the *Execute* button. A pop-up window will appear with data of frequency distribution per GO term (Figure 31).

We view the plot corresponding to the top ten data selecting the menu item:

> *Main menu > Functional annotation > Statistics > Gene Ontology > Frequency distribution per GO term plot*

In the window shown below (Figure 32), you can modify the default values of the file name and its format and resolution. After, double-click on the annotation process or select with a click on it and press the *Execute* button. A pop-up window will appear with the plot of frequency distribution per GO term (Figure 33). The corresponding file is saved in the subdirectory of *…\quercusTOA-results\run* corresponding to the script run.

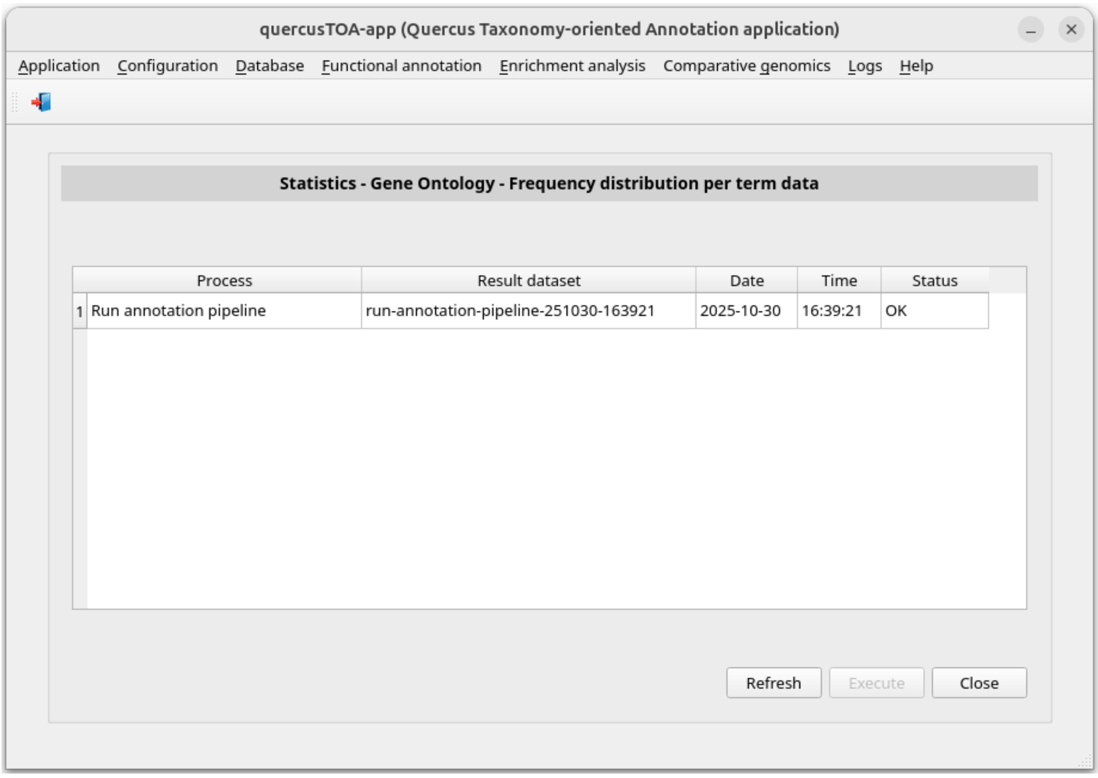
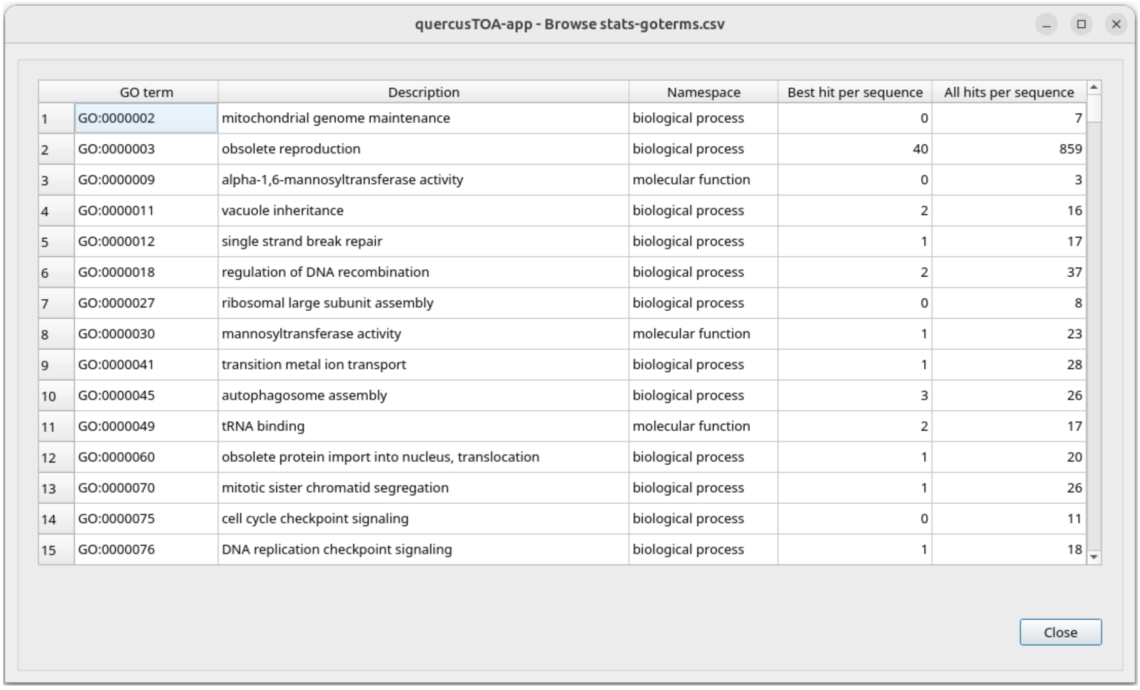


Figure 30. Window *Statistics – Gene Ontology - Frequency distribution per term data* showing annotation pipelines finished.



	GO term	Description	Namespace	Best hit per sequence	All hits per sequence
1	GO:0000002	mitochondrial genome maintenance	biological process	0	7
2	GO:0000003	obsolete reproduction	biological process	40	859
3	GO:0000009	alpha-1,6-mannosyltransferase activity	molecular function	0	3
4	GO:0000011	vacuole inheritance	biological process	2	16
5	GO:0000012	single strand break repair	biological process	1	17
6	GO:0000018	regulation of DNA recombination	biological process	2	37
7	GO:0000027	ribosomal large subunit assembly	biological process	0	8
8	GO:0000030	mannosyltransferase activity	molecular function	1	23
9	GO:0000041	transition metal ion transport	biological process	1	28
10	GO:0000045	autophagosome assembly	biological process	3	26
11	GO:0000049	tRNA binding	molecular function	2	17
12	GO:0000060	obsolete protein import into nucleus, translocation	biological process	1	20
13	GO:0000070	mitotic sister chromatid segregation	biological process	1	26
14	GO:0000075	cell cycle checkpoint signaling	biological process	0	11
15	GO:0000076	DNA replication checkpoint signaling	biological process	1	18

Figure 31. Pop-up window showing the complete list of frequency distribution performed by the process *run-annotation-pipeline-251030-163921*.

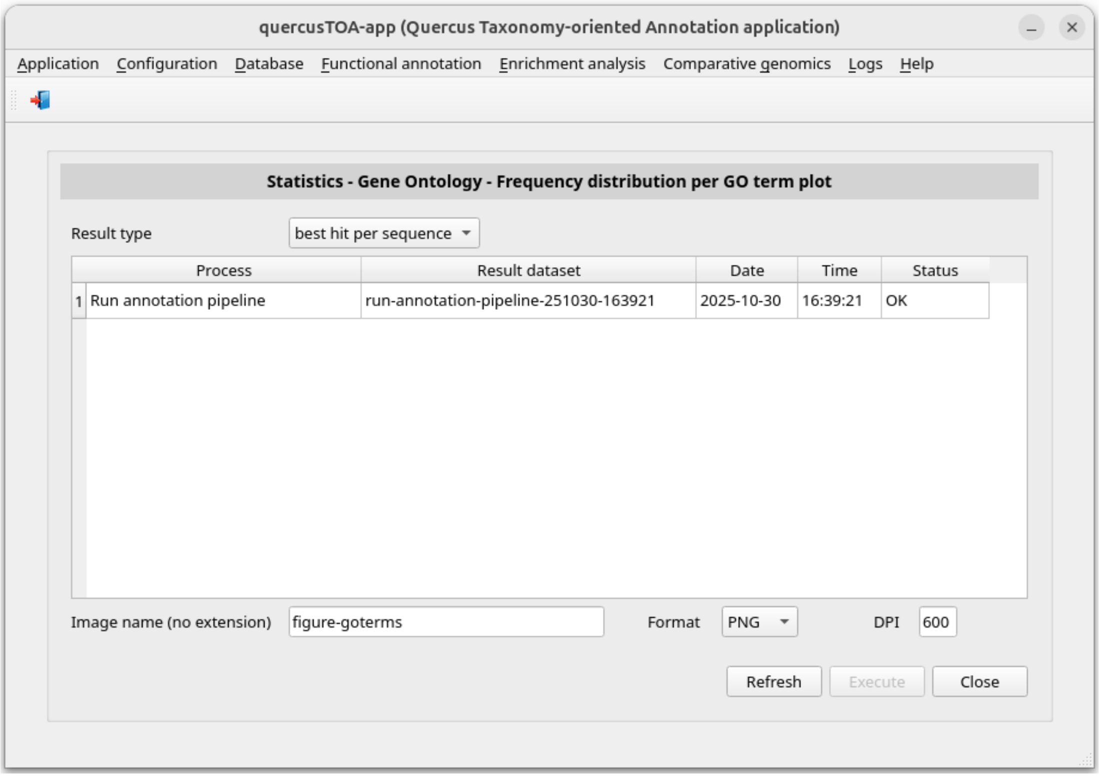


Figure 32. Window *Statistics - Gene Ontology - Frequency distribution per GO term* showing annotation pipelines finished.

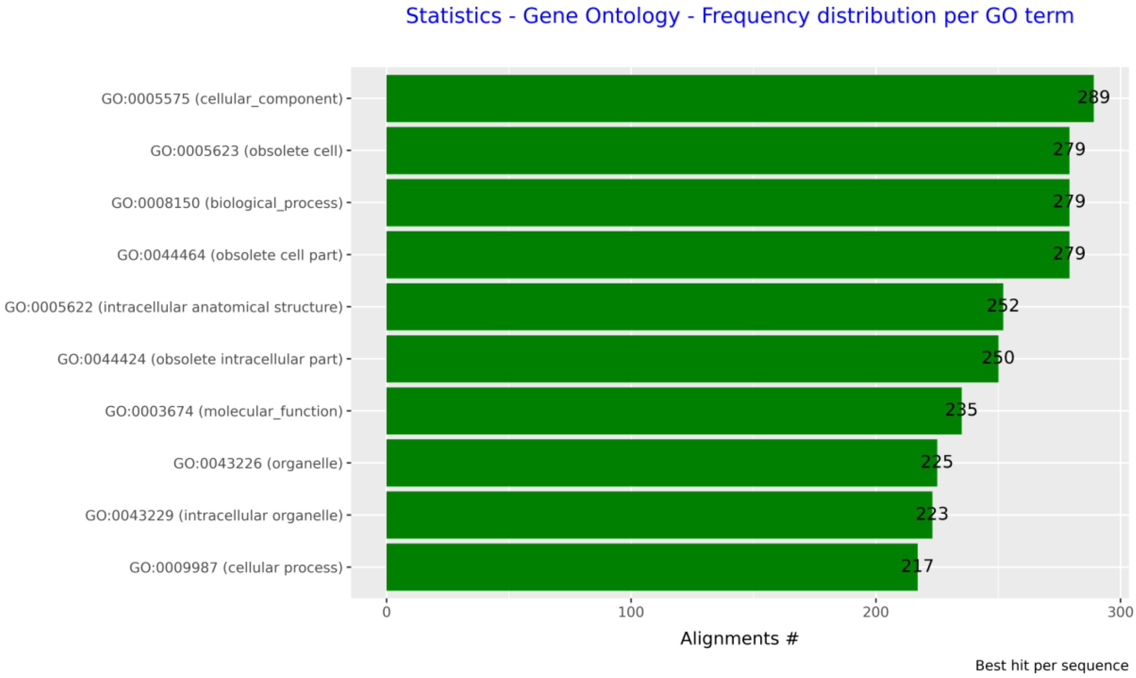


Figure 33. Plot *Statistics - Gene Ontology - Frequency distribution per GO term* with the top ten of GO terms.

To consult the frequency distribution per namespace, go to the menu item with this path:

> *Main menu > Functional annotation > Statistics > Gene Ontology > Frequency distribution per namespace data*

In the new window (Figure 34), double-click on the annotation process or select with a click on it and press the *Execute* button. A pop-up window will appear with data of frequency distribution per namespace (Figure 35).

We view the plot corresponding to this data selecting the menu item

Main menu > Functional annotation > Statistics > Gene Ontology > Frequency distribution per namespaces data

In the window shown below (Figure 36), you can modify the default values of the image name and its format and resolution. After, double-click on the annotation process or select with a click on it and press the *Execute* button. A pop-up window will appear with the plot of frequency distribution per namespace (Figure 37). The corresponding file is saved in the subdirectory of *…\quercusTOA-results\run* corresponding to the script run.

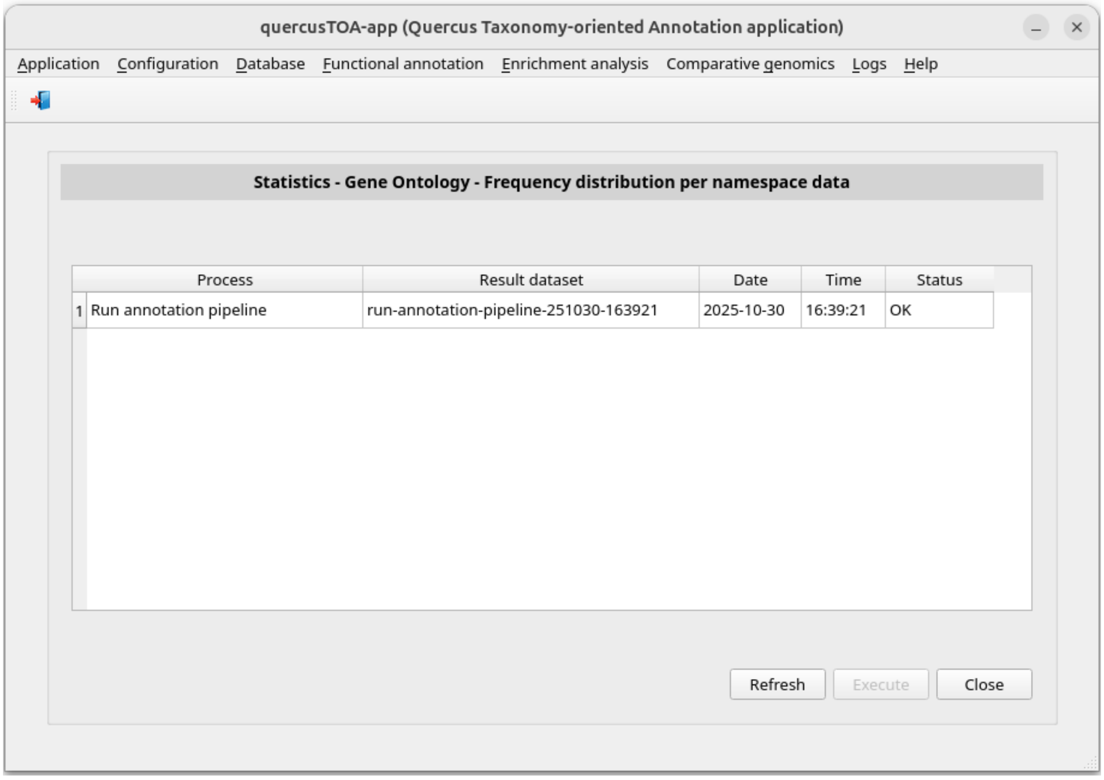


Figure 34. Window *Statistics - Gene Ontology - Frequency distribution per namespace data* showing annotation pipelines finished.

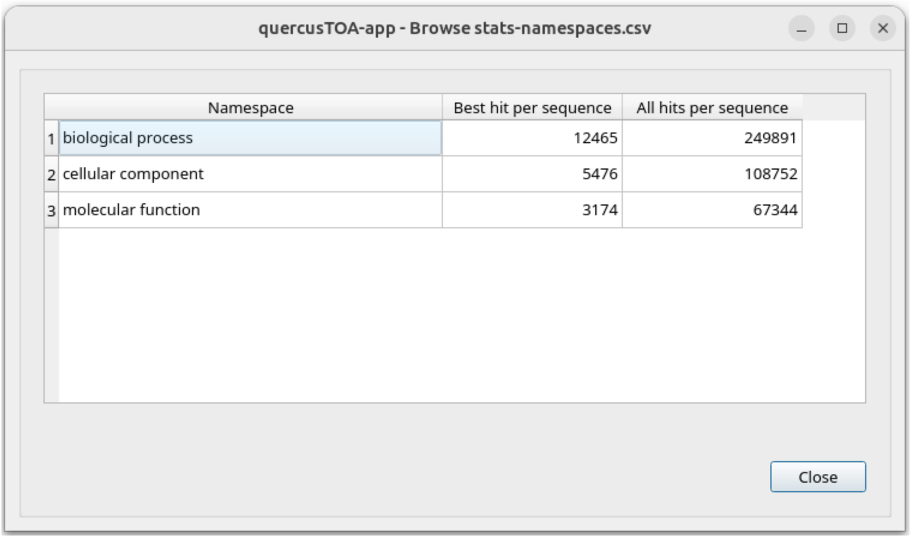


Figure 35. Pop-up window showing the complete list of frequency distribution per namespace performed by the process *run-annotation-pipeline-251030-163921*.

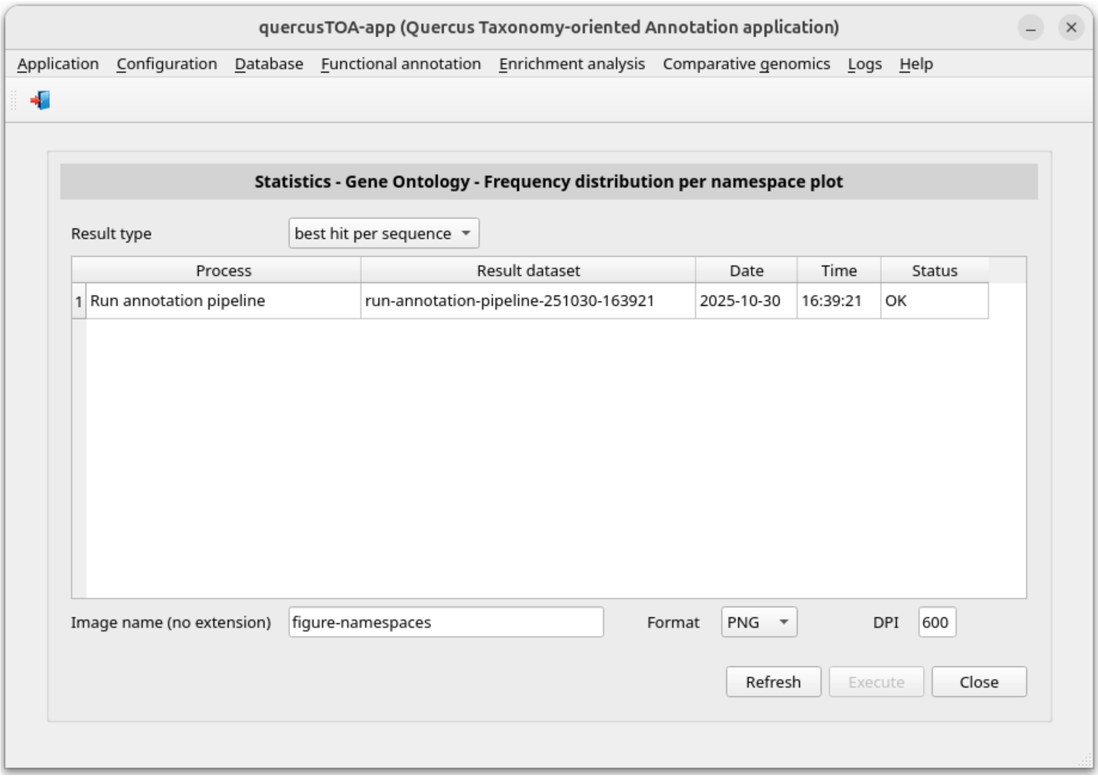


Figure 36. Window *Statistics - Gene Ontology - Frequency distribution per namespace plot* showing annotation pipelines finished.

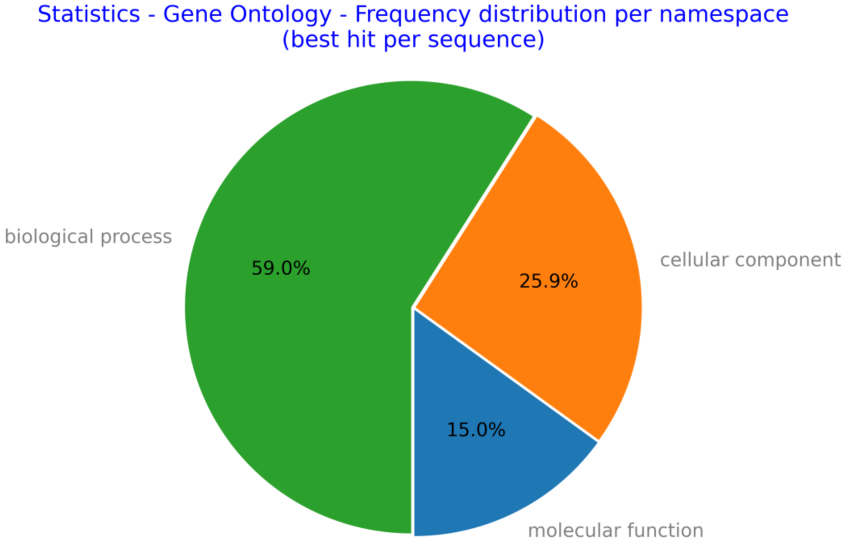


Figure 37. Plot *Statistics - Gene Ontology - Frequency distribution per namespace*.

The statistics of sequences number per GO terms number can be consulted in the menu item with this path:

> *Main menu > Functional annotation > Statistics > Gene Ontology > Sequences # per GO terms # data*

In the new window (Figure 38), double-click on the annotation process or select with a click on it and press the *Execute* button. A pop-up window will appear with data of sequences number per GO terms number (Figure 39).

We view the plot corresponding to the top ten data selecting the menu item

> *Main menu > Functional annotation > Statistics > Gene Ontology > Sequences # per GO terms # plot*

In the window shown below (Figure 40), you can modify the default values of the file name and its format and resolution. After, double-click on the annotation process or select with a click on it and press the *Execute* button. A pop-up window will appear with the plot of sequences number per GO terms number (Figure 41). The corresponding file is saved in the subdirectory of *…\quercusTOA-results\run* corresponding to the script run.

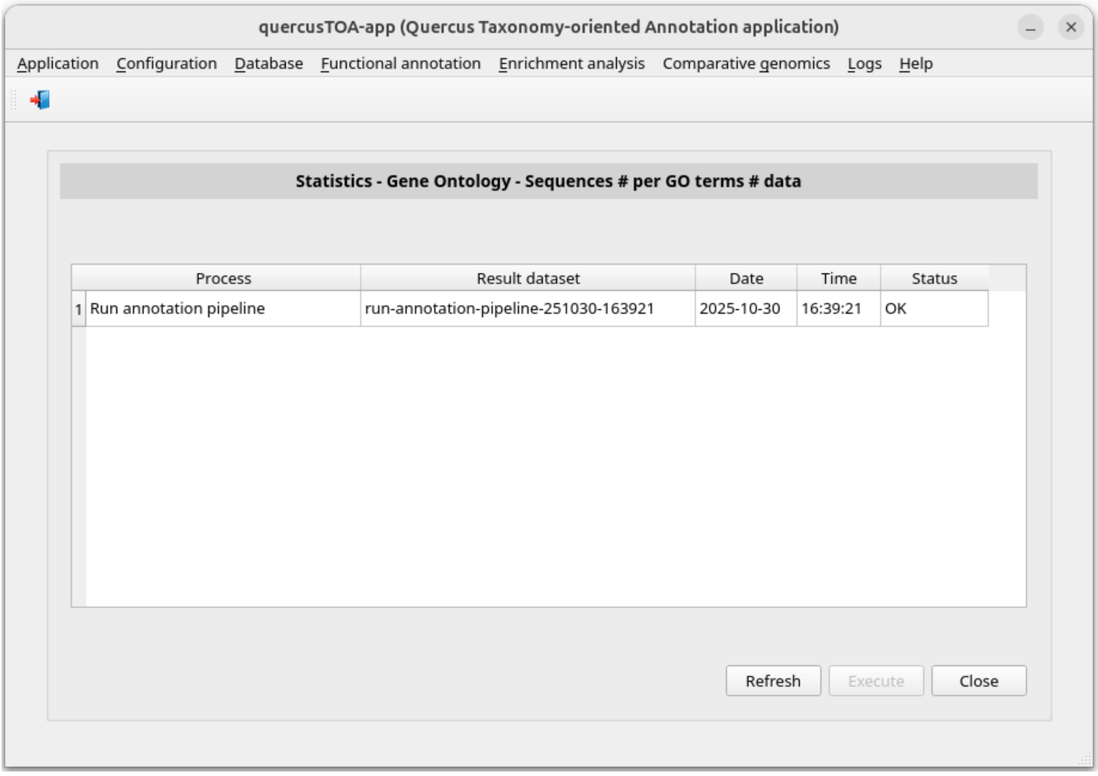


Figure 38. Window *Statistics - Gene Ontology – Sequences # per Go terms # data* showing annotation pipelines finished.

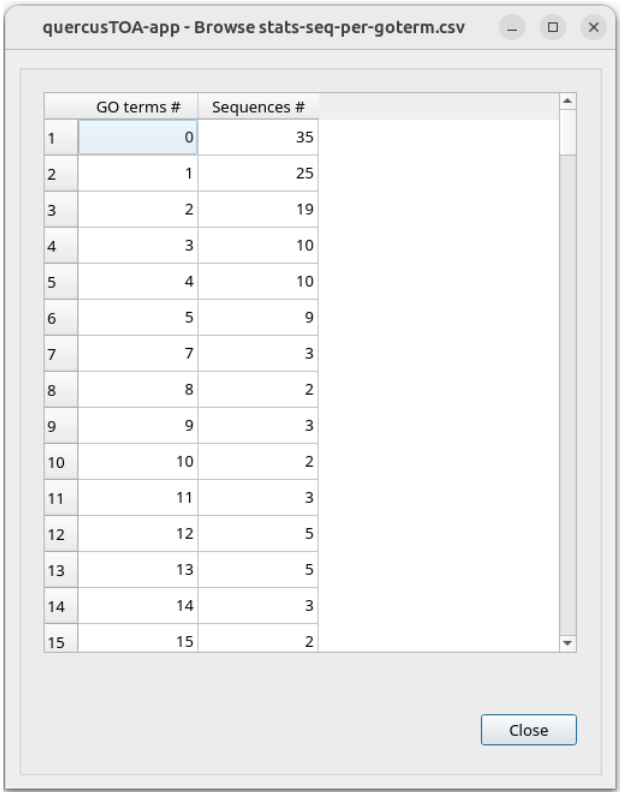


Figure 39. Pop-up window showing the complete list of GO terms number per sequences number performed by the process *run-annotation-pipeline-251030-163921*.

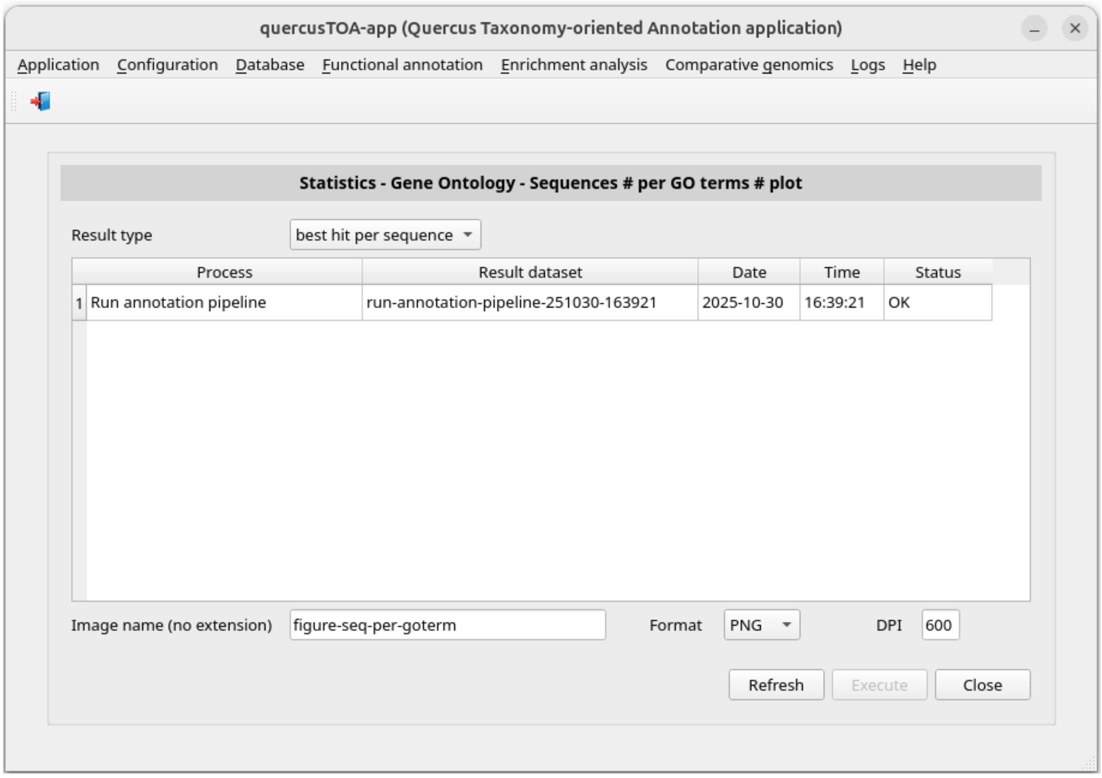


Figure 40. Window *Statistics - Gene Ontology - Sequences # per Go terms # plot* showing annotation pipelines finished.

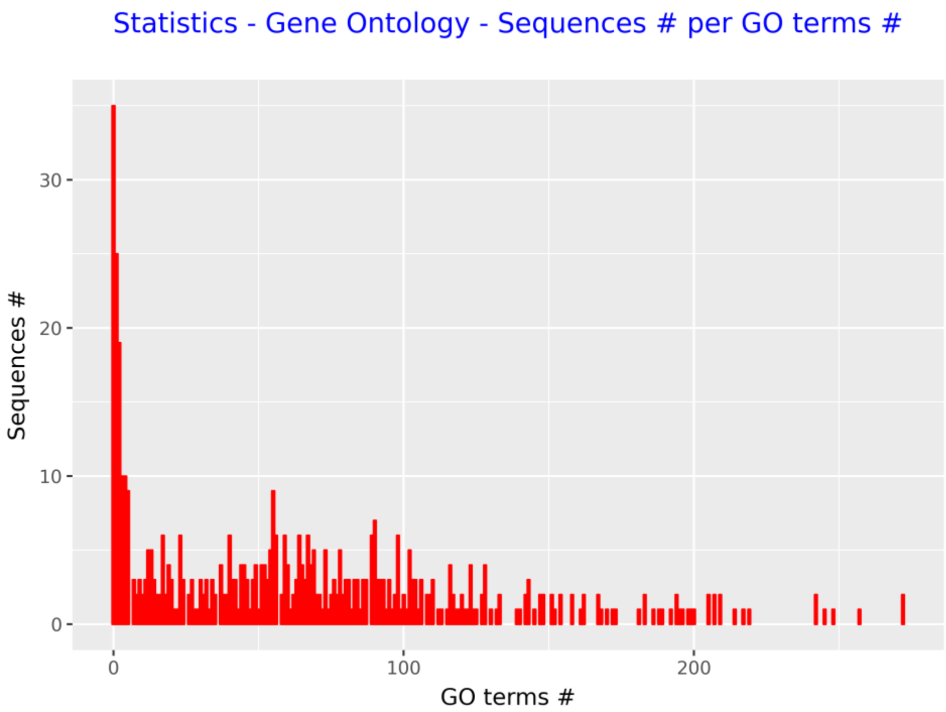


Figure 41. Plot *Statistics - Gene Ontology – Sequences # per GO terms #*.

Enrichment analysis

Run an enrichment analysis

To perform the enrichment analysis (Figure 5) with data of an annotation process we are going to select the menu item with this path:

Main menu > Enrichment analysis > Run analysis

The window *Run an enrichment analysis* appears (Figure 42). Then select the annotation pipeline to be studied, in our example *run-annotation-pipeline-251030-163921*. The window will be updated showing the transcript file of the annotation process (Figure 43). Then you can modify the default values by choosing **all species** or a specific species, changing the FDR method (**Benjamini-Hochberg** or **Benjamini-Yekutieli**), and the minimum sequence number of annotations and species to be considered. Below, click on the push button *Execute*.

You can check the process status and when it has been completed reviewing its log. So, click the following menu item:

Main menu > Logs > Result logs

Select **run** in the *Process type* combo-box. Then the window is updated (Figure 44).

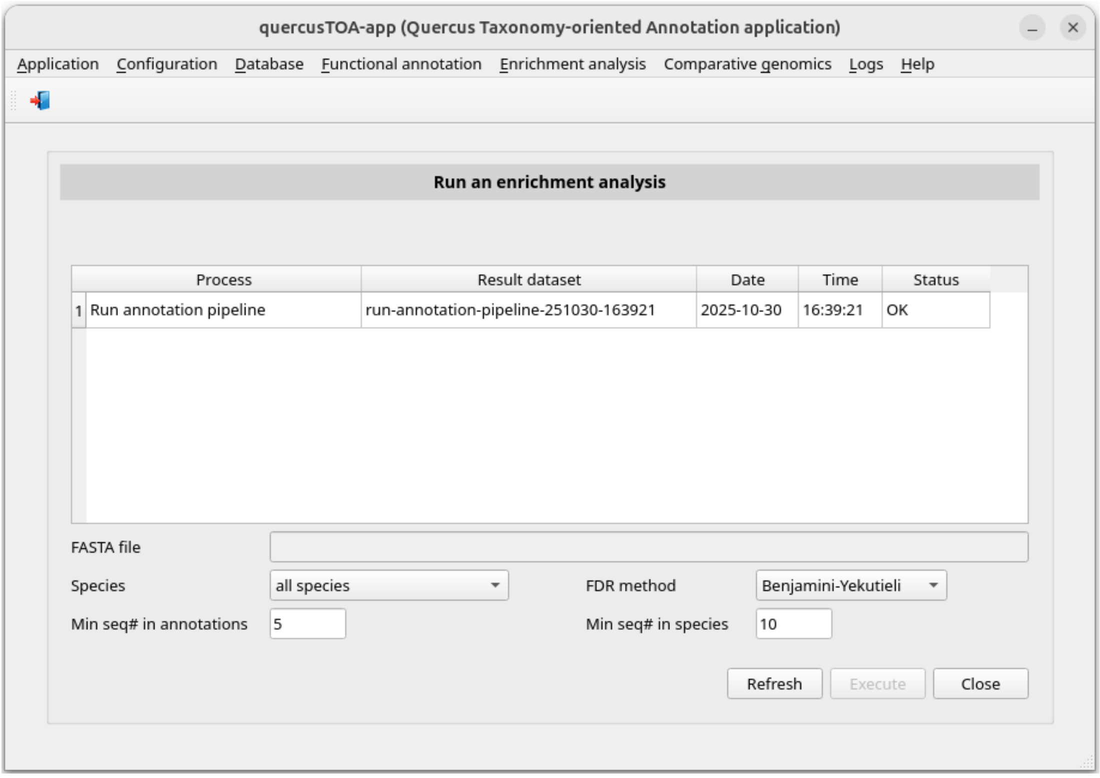


Figure 42. Initial appearance of the window *Run an enrichment analysis*.

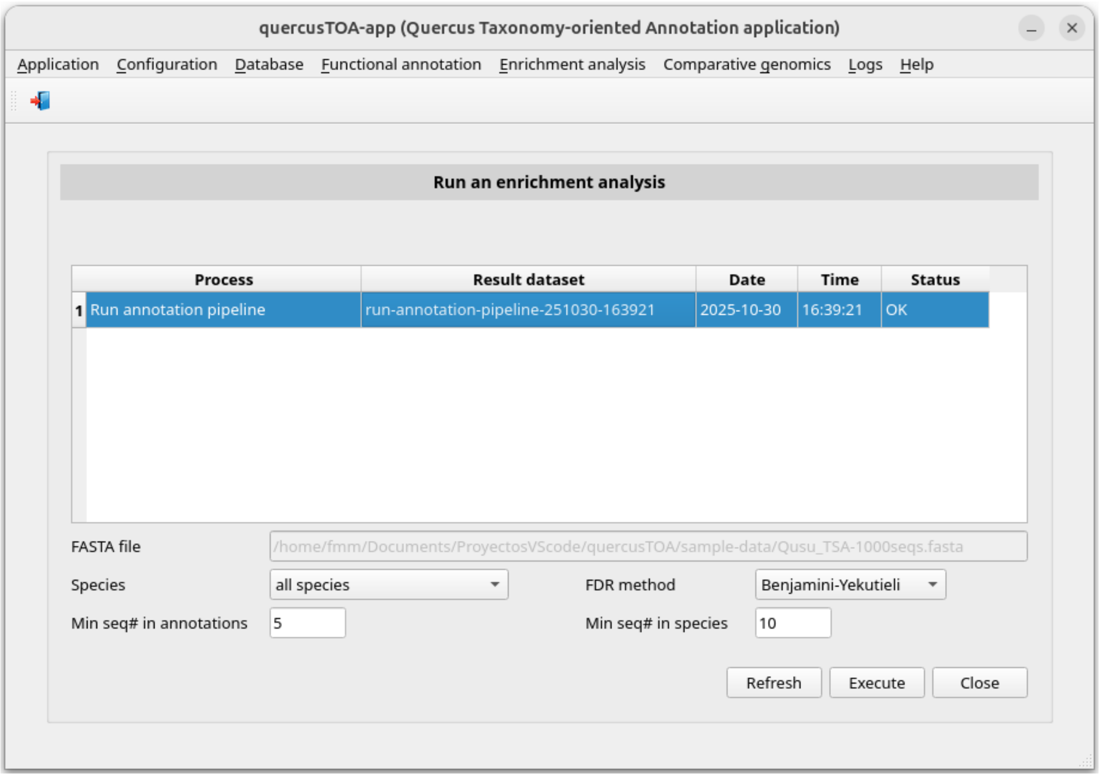


Figure 43. Window *Run an enrichment analysis* once the annotation pipeline has been selected.

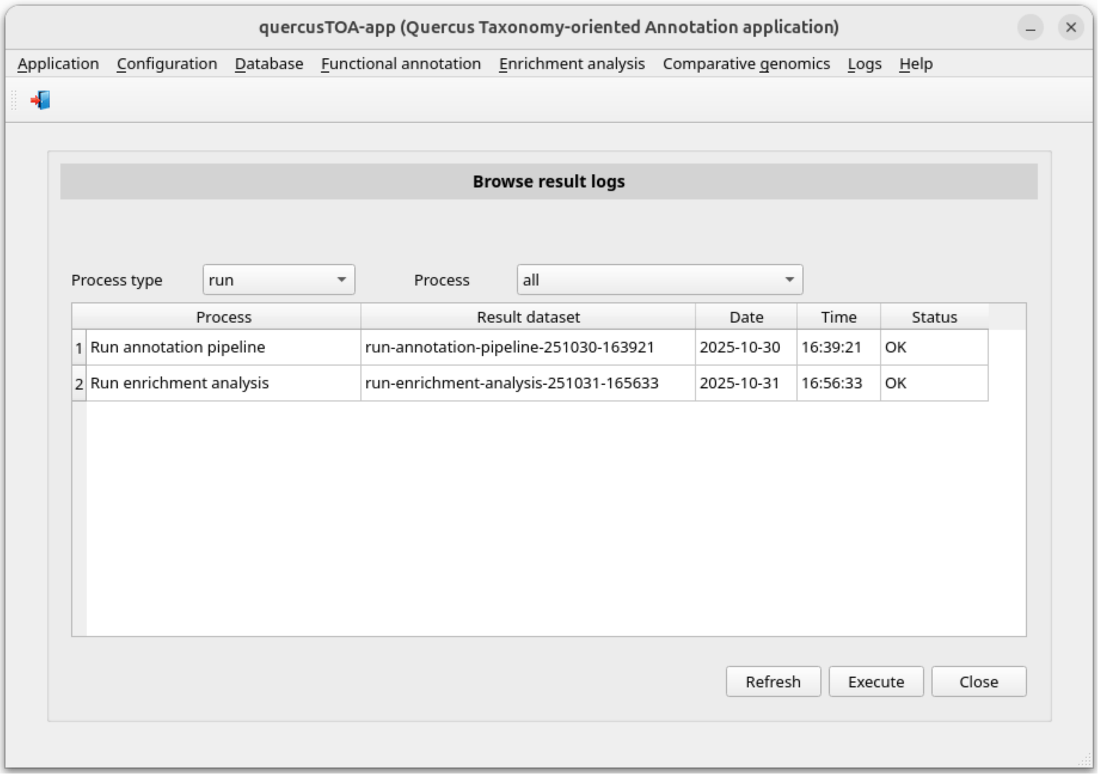


Figure 44. Window *Browse results logs* showing the annotation process and enrichment analysis finished.

When the process has finished, we could go the subdirectory of *…\quercusTOA-app-results\run* corresponding to the script run, e.g. *run-enrichment_analysis-251031-165633*, and consult the generated files (Figure 45). Some of these files are:

- Analysis using the file that contains the annotations of the subject sequence identification with best hit yielded by blast programs for every query sequence identification:
  - *besthit-goterm-enrichment-analysis.csv*: enrichment of GO terms
  - *besthit-kegg-ko-enrichment-analysis.csv*: enrichment of KEGG KOs
  - *besthit-kegg-pathway-enrichment-analysis.csv*: enrichment of KEGG pathways
  - *besthit-metacyc-pathway-enrichment-analysis.csv*: enrichment of MetaCyc pathways
- Analysis using the file that contains all annotations yielded by blast programs for every query sequence identification.
  - *complete-goterm-enrichment-analysis.csv*: enrichment of GO terms
  - *complete-kegg-ko-enrichment-analysis.csv*: enrichment of KEGG KOs
  - *complete-kegg-pathway-enrichment-analysis.csv*: enrichment of KEGG pathways
  - *complete-metacyc-pathway-enrichment-analysis.csv*: enrichment of MetaCyc pathways

Name	Modified	Size
status	31/10/25 16:58	5 items
besthit-goterm-enrichment-analysis.csv	31/10/25 16:57	90.3 kB
besthit-kegg-ko-enrichment-analysis.csv	31/10/25 16:58	215 bytes
besthit-kegg-pathway-enrichment-analysis.csv	31/10/25 16:58	4.0 kB
besthit-metacyc-pathway-enrichment-analysis.csv	31/10/25 16:58	151.3 kB
complete-goterm-enrichment-analysis.csv	31/10/25 16:58	149.7 kB
complete-kegg-ko-enrichment-analysis.csv	31/10/25 16:58	287 bytes
complete-kegg-pathway-enrichment-analysis.csv	31/10/25 16:58	6.8 kB
complete-metacyc-pathway-enrichment-analysis.csv	31/10/25 16:58	160.2 kB
log.txt	31/10/25 16:58	2.0 kB
params.txt	31/10/25 16:56	411 bytes
run-enrichment-analysis-process.sh	31/10/25 16:56	7.0 kB
run-enrichment-analysis-process-starter.sh	31/10/25 16:56	291 bytes

Figure 45. Folders and files in *…/quercusTOA-app-results/run/run-enrichment_analysis-251031-165633* yielded by the enrichment analysis.

Browse results of the enrichment analysis

If you edit the files **-enrichment-analysis*, you will see the result of the enrichment analysis. You can browse them using QUERCUSTOA-APP. For example, if you want to browse the enrichment analysis of GO terms, select the menu item:

Main menu > Enrichment analysis > Browse results > GO enrichment analysis

Below, the window (Figure 46) allows you to choose **best hit per sequence** (to see functional-annotations-besthit.csv) or **all hits per sequence** (to see functional-annotations-complete.csv) in the combo-box Result type. Then, select the enrichment analysis process that you are interested in consulting by clicking on the corresponding row, in our example *run-enrichment-analysis-251031-165633*. The parameters of the process will be shown (Figure 47). Finally, click on the push button *Execute*. A pop-up window will appear showing the enrichment analysis data (Figure 48).

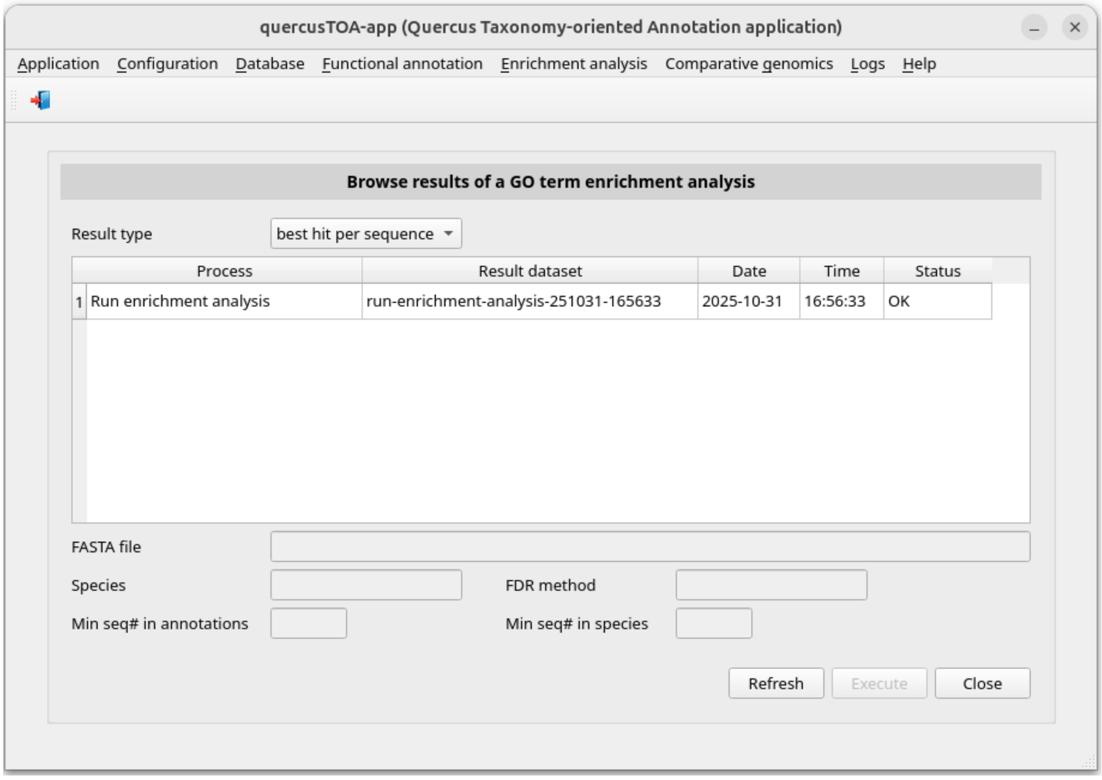


Figure 46. Window *Browse results of a GO term enrichment analysis* showing enrichment analysis finished.

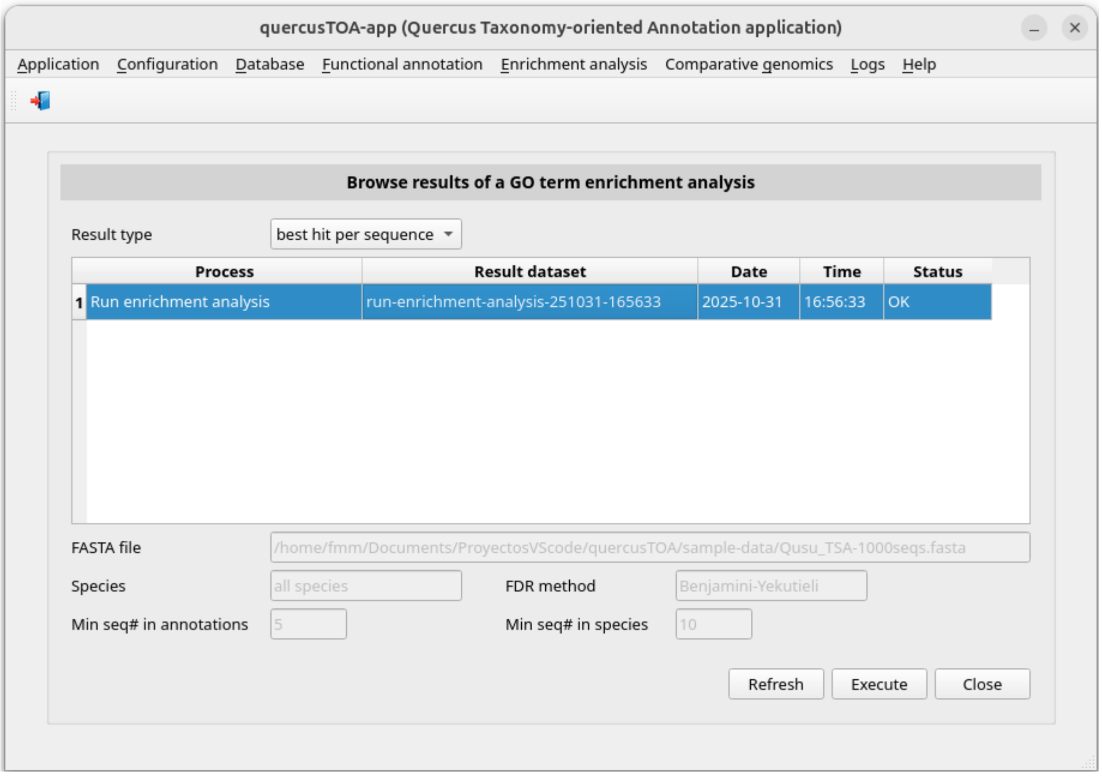
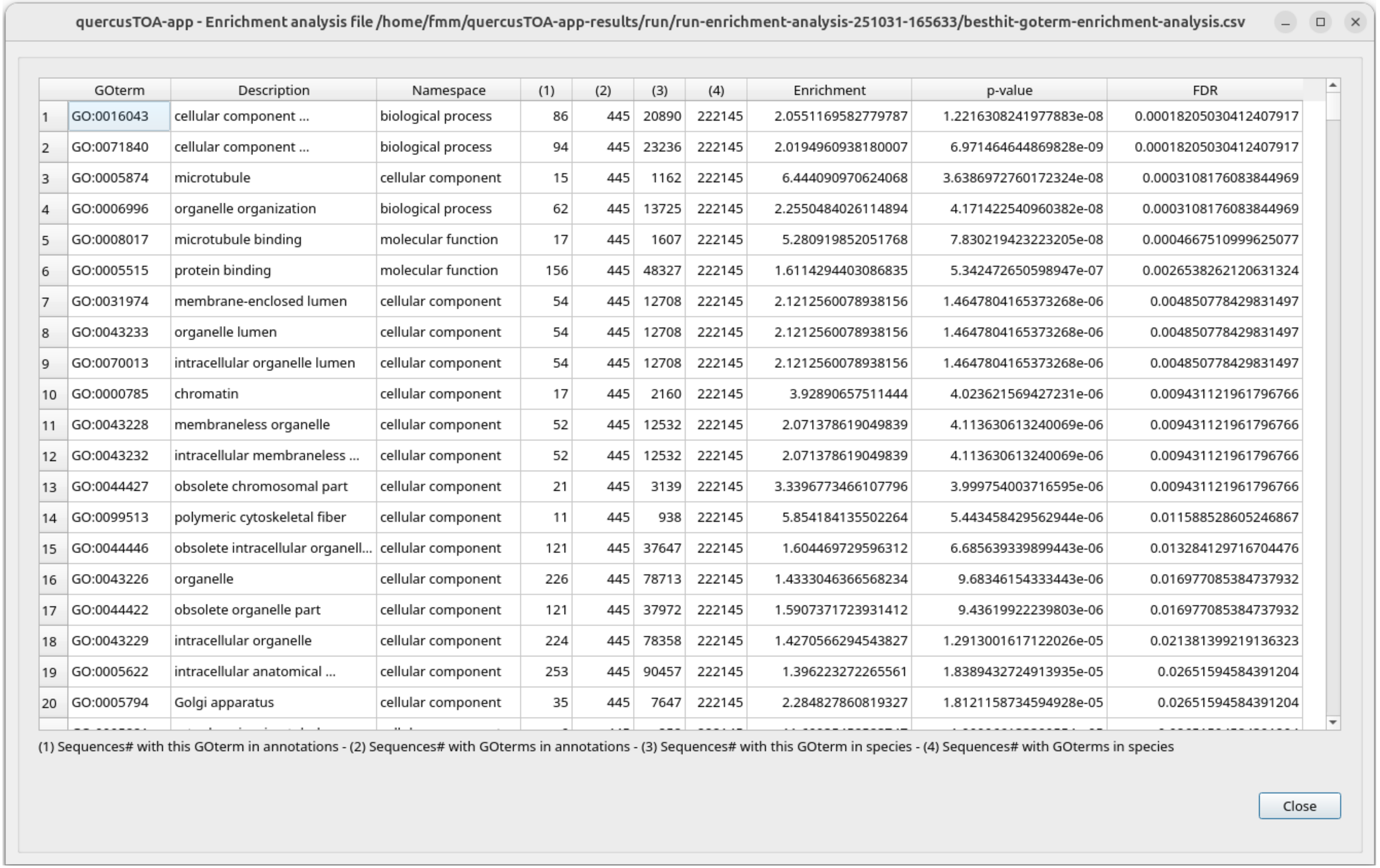


Figure 47. Window *Browse results of a GO term enrichment analysis* after *run-enrichment-analysis -251031-165633* was selected.


quercusTOA-app - Enrichment analysis file /home/fmm/quercusTOA-app-results/run/run-enrichment-analysis-251031-165633/besthit-goterm-enrichment-analysis.csv

	GOterm	Description	Namespace	(1)	(2)	(3)	(4)	Enrichment	p-value	FDR
1	GO:0016043	cellular component ...	biological process	86	445	20890	222145	2.0551169582779787	1.2216308241977883e-08	0.00018205030412407917
2	GO:0071840	cellular component ...	biological process	94	445	23236	222145	2.0194960938180007	6.971464644869828e-09	0.00018205030412407917
3	GO:0005874	microtubule	cellular component	15	445	1162	222145	6.444090970624068	3.6386972760172324e-08	0.0003108176083844969
4	GO:0006996	organelle organization	biological process	62	445	13725	222145	2.2550484026114894	4.171422540960382e-08	0.0003108176083844969
5	GO:0008017	microtubule binding	molecular function	17	445	1607	222145	5.280919852051768	7.830219423223205e-08	0.0004667510999625077
6	GO:0005515	protein binding	molecular function	156	445	48327	222145	1.6114294403086835	5.342472650598947e-07	0.0026538262120631324
7	GO:0031974	membrane-enclosed lumen	cellular component	54	445	12708	222145	2.1212560078938156	1.4647804165373268e-06	0.004850778429831497
8	GO:0043233	organelle lumen	cellular component	54	445	12708	222145	2.1212560078938156	1.4647804165373268e-06	0.004850778429831497
9	GO:0070013	intracellular organelle lumen	cellular component	54	445	12708	222145	2.1212560078938156	1.4647804165373268e-06	0.004850778429831497
10	GO:0000785	chromatin	cellular component	17	445	2160	222145	3.92890657511444	4.023621569427231e-06	0.009431121961796766
11	GO:0043228	membraneless organelle	cellular component	52	445	12532	222145	2.071378619049839	4.113630613240069e-06	0.009431121961796766
12	GO:0043232	intracellular membraneless ...	cellular component	52	445	12532	222145	2.071378619049839	4.113630613240069e-06	0.009431121961796766
13	GO:0044427	obsolete chromosomal part	cellular component	21	445	3139	222145	3.3396773466107796	3.999754003716595e-06	0.009431121961796766
14	GO:0099513	polymeric cytoskeletal fiber	cellular component	11	445	938	222145	5.854184135502264	5.443458429562944e-06	0.011588528605246867
15	GO:0044446	obsolete intracellular organell...	cellular component	121	445	37647	222145	1.604469729596312	6.685639339899443e-06	0.013284129716704476
16	GO:0043226	organelle	cellular component	226	445	78713	222145	1.4333046366568234	9.68346154333443e-06	0.016977085384737932
17	GO:0044422	obsolete organelle part	cellular component	121	445	37972	222145	1.5907371723931412	9.43619922239803e-06	0.016977085384737932
18	GO:0043229	intracellular organelle	cellular component	224	445	78358	222145	1.4270566294543827	1.2913001617122026e-05	0.021381399219136323
19	GO:0005622	intracellular anatomical ...	cellular component	253	445	90457	222145	1.396223272265561	1.8389432724913935e-05	0.02651594584391204
20	GO:0005794	Golgi apparatus	cellular component	35	445	7647	222145	2.284827860819327	1.8121158734594928e-05	0.02651594584391204

(1) Sequences# with this GOterm in annotations - (2) Sequences# with GOterms in annotations - (3) Sequences# with this GOterm in species - (4) Sequences# with GOterms in species

Close

Figure 48. Pop-up window showing the analysis performed by the process *run-enrichment-analysis -251031-165633*.

Comparative genomics

Search of sequence homology

Through this process, we can search for homology relationships of FASTA nucleotide or amino acid sequences entered directly or contained in a file. In this example, we will use a subset of 81 transcript sequences that we obtained from RAD-seq and RNA-seq experiments in a study on cork formation in *Quercus suber*, file *transcripts-81-suber-genes.fasta*, included in the subdirectory *sample-data* of QUERCUSTOA-APP.

To search homology of sequences we are going to select the menu item with this path:

Main menu > Comparative genomics > Search homology of sequences

The window Search homology sequences will appear (Figure 49). Select the FASTA type option: **TRANSCRIPTS** in our case. Then click on radio button FASTA file and type the path of the FASTA file or select it using the button Search... (in this example we use the file transcripts file *transcripts-81-suber-genes.fasta* located in the subdirectory *sample-data* of the QUERCUSTOA-APP software). Then the Execute button will be available (Figure 50).

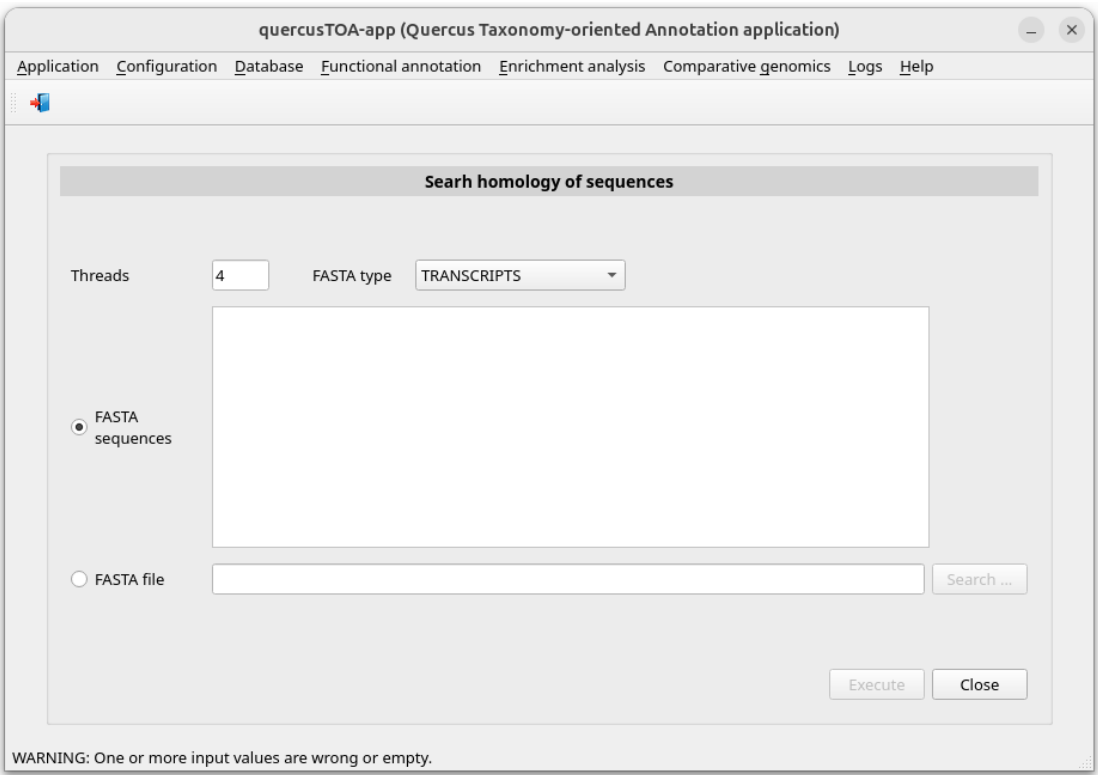


Figure 49. Initial appearance of the window *Search homology of sequences*.

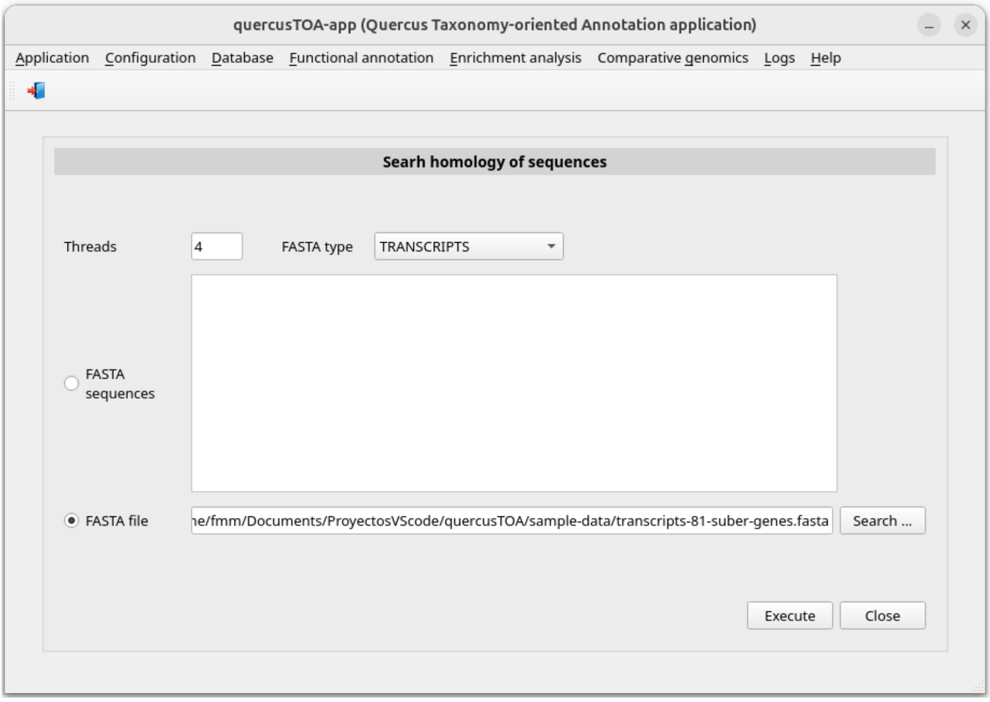


Figure 50. Window *Search homology of sequences* once the transcriptome file has been typed.

Finally, click on the Execute button. You can check the process status and when it has been completed reviewing its log. So, click the following menu item:

Main menu > Logs > Result logs

Select **run** in the *Process type* combo-box. Then the window is updated (Figure 51).

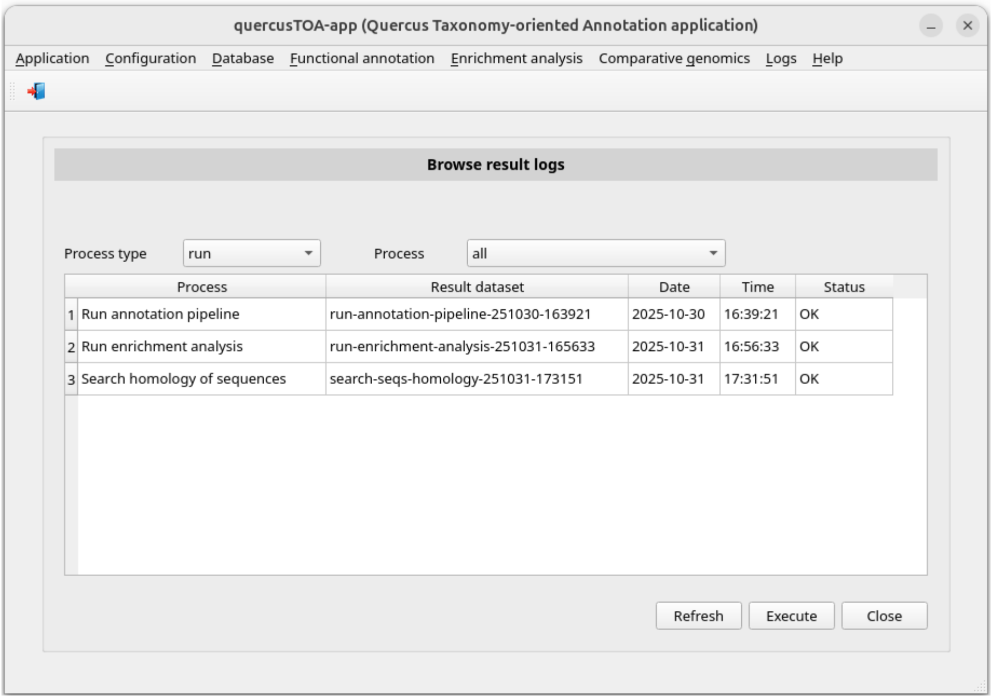


Figure 51. Window *Browse results logs* showing search homology of sequences pipeline finished.

When the process has finished, we could go the subdirectory of *…\quercusTOA-app-results\run* corresponding to the script run, e.g. *search-seqs-homology-251031-173150*, and consult the generated files (Figure 52). Some of these files are:

Figure 52. Folders and files in …/quercusTOA-app-results/run/run-annotation-pipeline-251031-173150 yielded by the annotation pipeline.

The file *homology-relationships-file.csv* contains the homology relationship of the sequences analyzed.

For each sequence, the following files are in the subdirectory *seqs-alignments* (Figure 53):

- **homologous-genes.fasta*: sequences FASTA of the homologous genes.
- **homologous-genes.fasta.aln*: alignment of sequences FASTA of the genes corresponding to homologous proteins.
- **homologous-genes.fasta.aln.pdf*: alignment plot of sequences FASTA of the genes corresponding to homologous proteins.
- **homologous-proteins.fasta*: sequences FASTA of the homologous proteins.
- **homologous-proteins.fasta.aln*: alignment of sequences FASTA of the homologous proteins.
- **homologous-proteins.fasta.aln.pdf*: alignment plot of sequences FASTA of the homologous proteins.
- **homologous-proteins.fasta.tree*: phylogenetic tree of homologous proteins.
- **homologous-proteins.fasta.tree.pdf*: phylogenetic tree plot of homologous proteins.

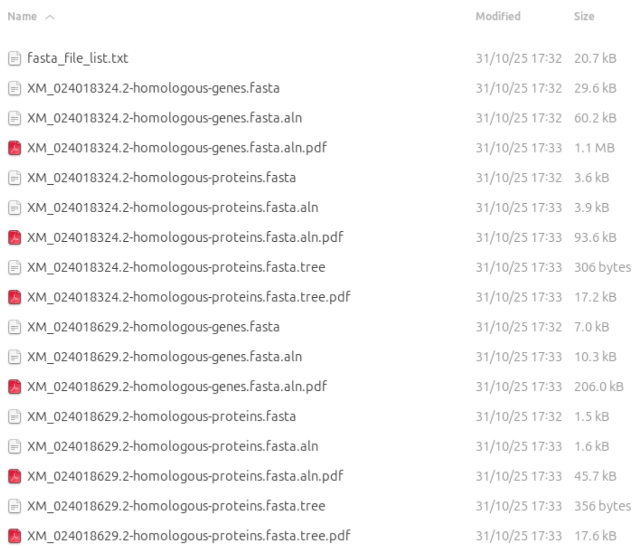


Figure 53. Files of the first sequences yielded by the process of search of sequence homology script in the subdirectory *seqs-alignments* of directory *.../quercusTOA-app-results/run/run-annotation-pipeline-251031-173150.*

Browse results of the homology search

To review the homology search process, click the following menu item:

Main menu > Comparative genomics > Browse results

Then, QUERCUSTOA-APP presents a window (Figure 54) where you must select homology search process identification clicking on its identification. In this example, the identification of the process is *search-seqs-homology-251031-173151*. When the identification is selected, the parameters used in the imputation process are shown (Figure 55).

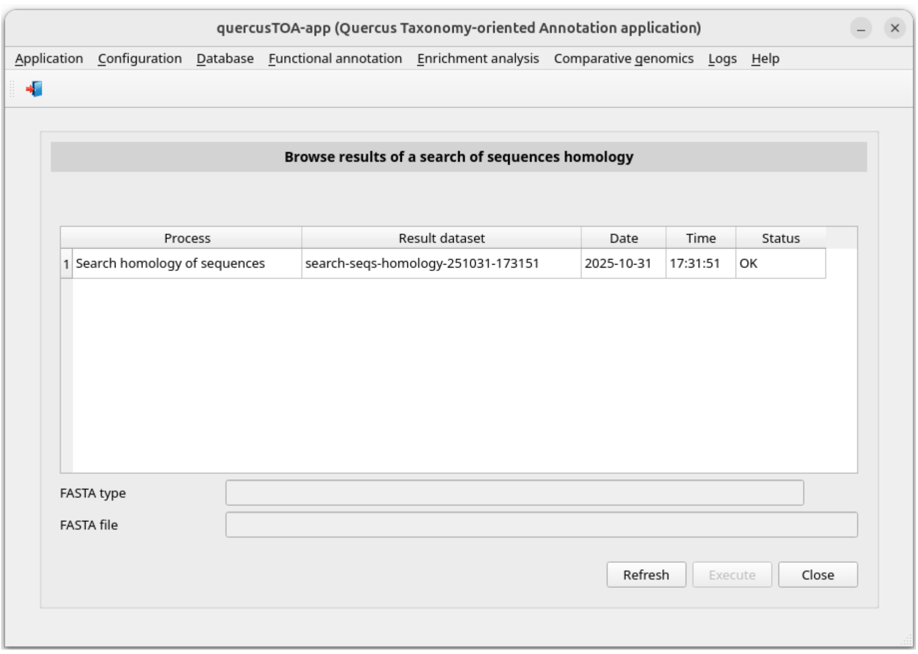


Figure 54. Initial appearance of the window *Browse results of a search of sequences homology*.

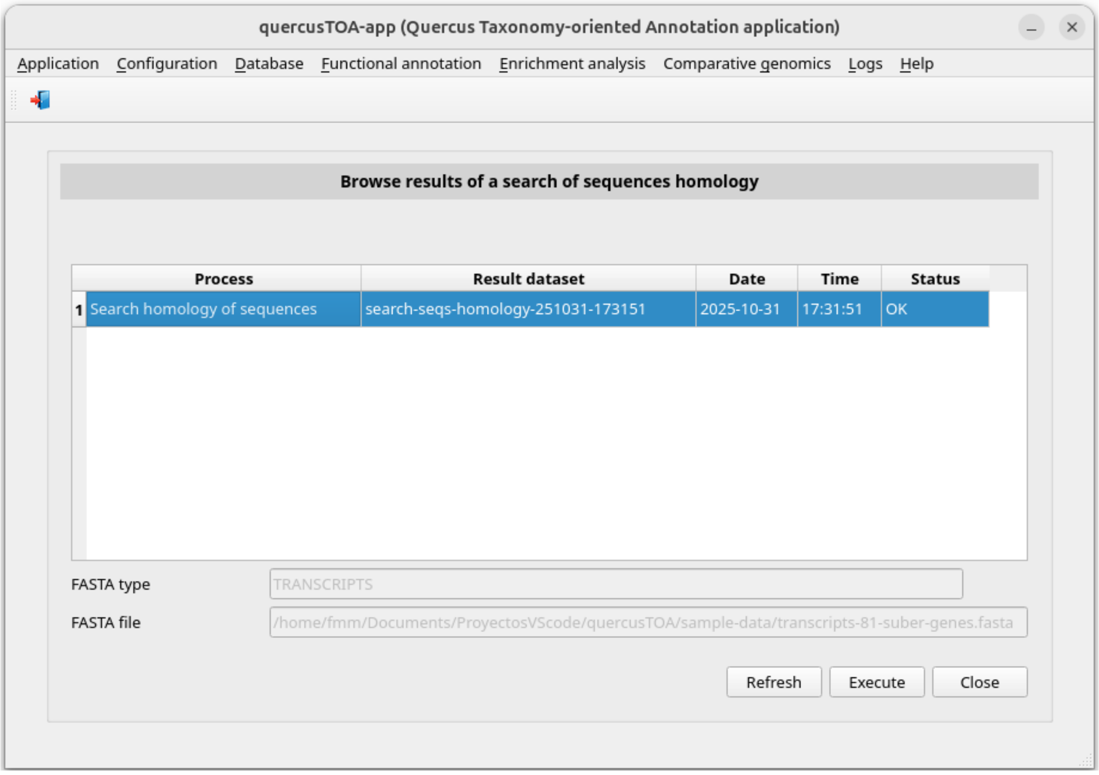


Figure 55. Window *Browse results of a search of sequences homology* once the sequences homology process has been selected.

Click the execute button. Then QUERCUSTOA-APP shows a window with the homology data for each sequence identification (Figure 56). Each row corresponds to one homologous protein isoform of a query sequence identification.

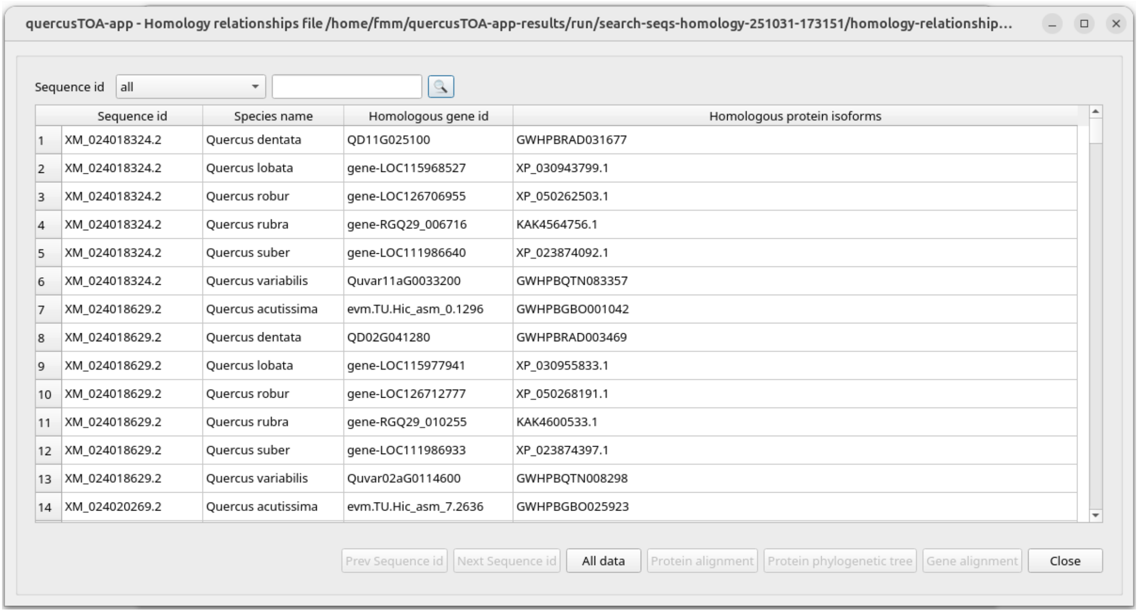


Figure 56. Window showing the homology relationships.

If you click on a query sequence identification (Figure 57), the rows of this sequence will be selected and the buttons *Prev sequence id*, *Next sequence id* will be available. These buttons allow you to browse sequence by sequence the homology relationships of each query sequence. To show all query sequences, click on the button *All data*.

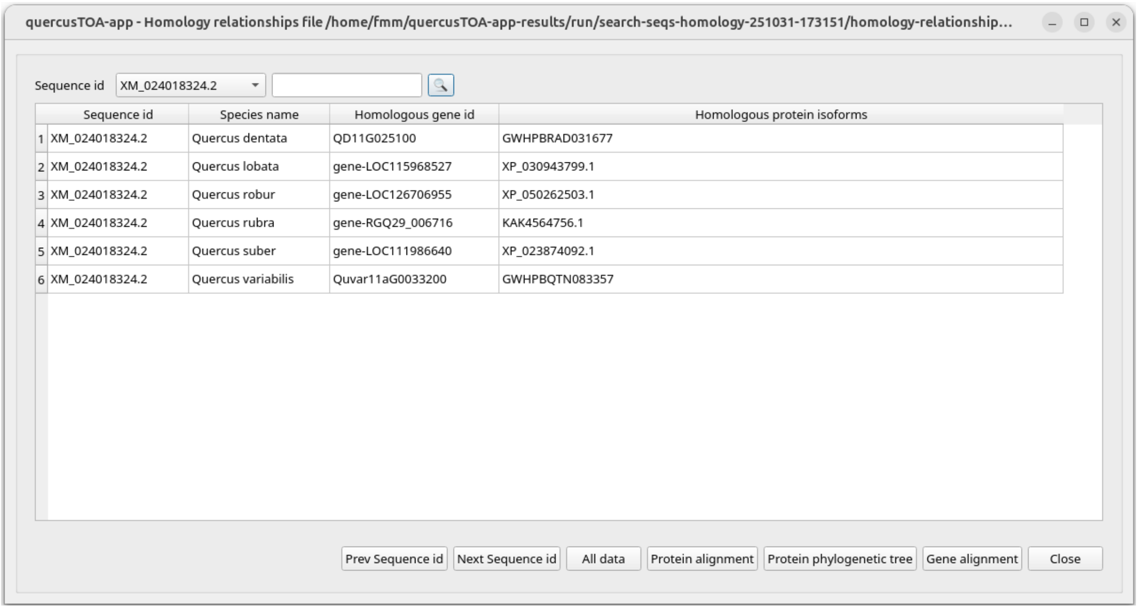


Figure 57. Window showing the homology relationships of the query sequence identification **XM_024018324.2**.

Also, the buttons *Protein alignment*, *Protein phylogenetic tree* and *Gene alignment* will be available. The button *Protein alignment* shows an alignment plot of sequences FASTA of the homologous proteins (Figure 58).

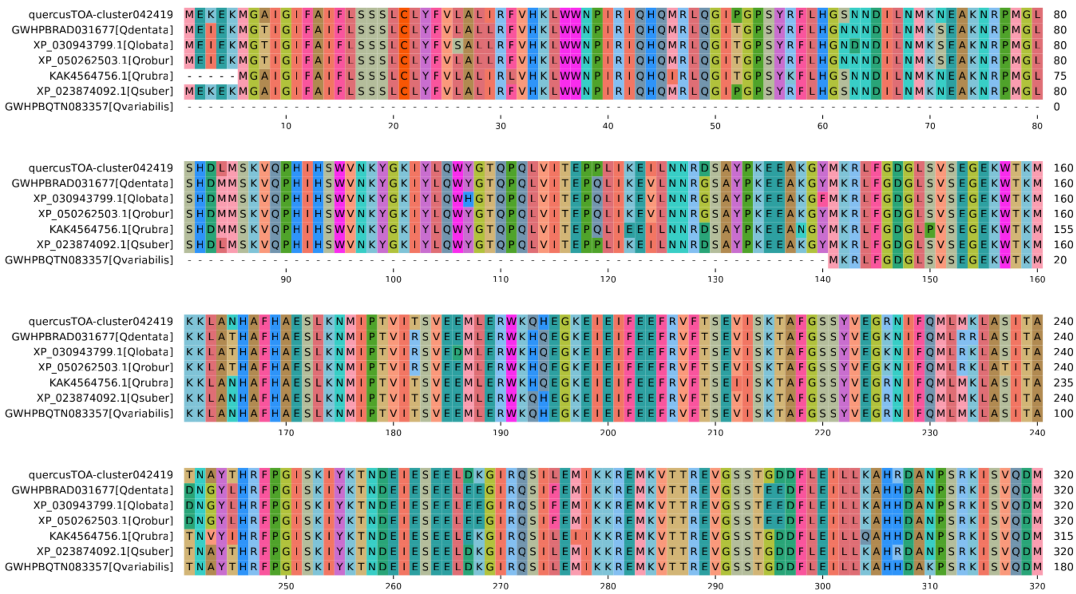


Figure 58. Plot of the alignment of the homologous protein isoform sequences of the query sequence identification **XM_024018324.2**.

The button *Protein phylogenetic tree* shows a phylogenetic tree plot of homologous proteins (Figure 59).

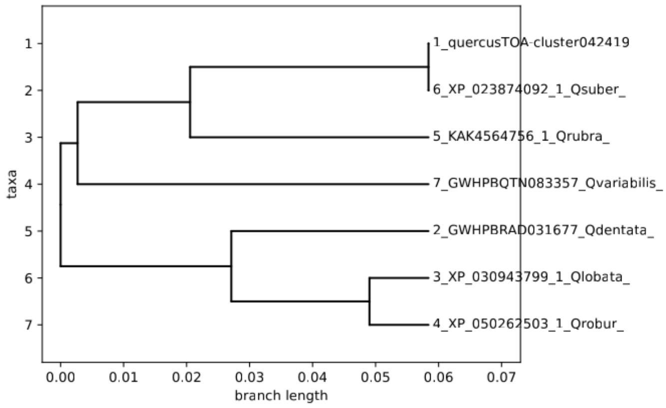


Figure 59. Plot of the phylogenetic tree of the homologous protein isoform of the query sequence identification **XM_024018324.2**.

And finally, the button *Gene alignment* shows an alignment plot of sequences FASTA of the genes corresponding to homologous proteins (Figure 60).

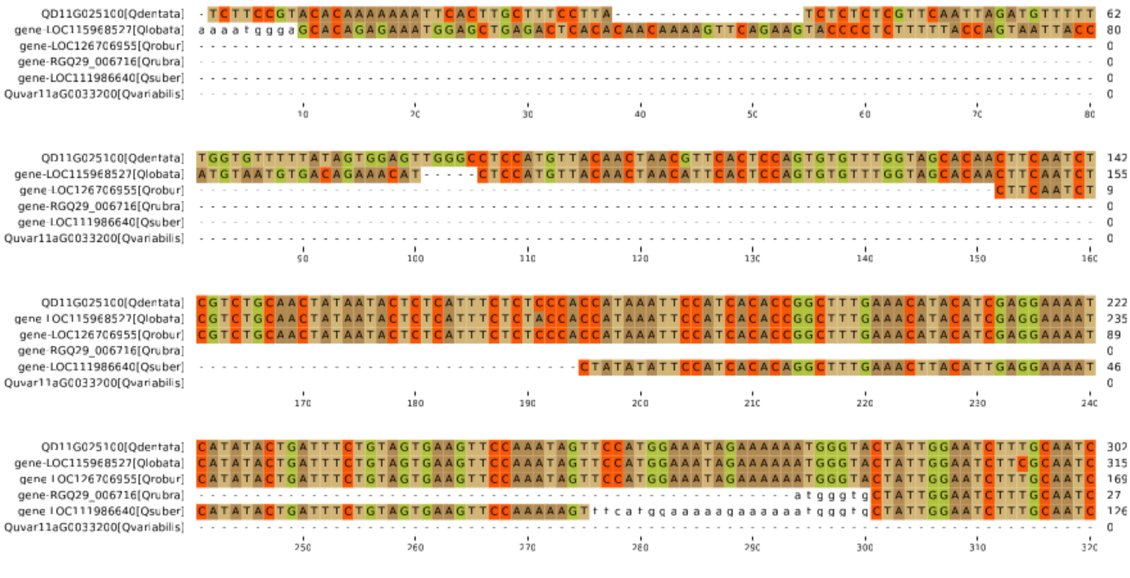


Figure 60. Plot of the alignment of the gene sequences corresponding to the homologous protein isoforms of the query sequence identification **XM_024018324.2**.

Functional annotation, enrichment analysis and homology search on Linux servers

In addition to the GUI version, QUERCUSTOA-APP includes three bash scripts to run the functional annotation, enrichment analysis and homology search processes from other Bash scripts on a Linux server without having to enter the QUERCUSTOA-APP graphical environment.

Below are the steps for executing these processes in an Ubuntu Server using some locations ($HOME/Apps/Miniforge3, $HOME/Apps/quercusTOA-app, $HOME/BioData/quercusTOA-db) that must be adapted to right locations on the server where the processes will be run. Also, the environments names (quercus, quercus-blast, quercus-codan, quercus-diamond, quercus-liftoff and quercus-mafft) could be adapted. Notice that these locations and environments names are in red typeface.

QUERCUSTOA-APP installation

The unzip command is used to decompress some ZIP files. To install this command, if necessary, type the following command in a terminal:

```
$ sudo apt install unzip
```

Now, we will download the QUERCUSTOA-APP software in the proper location, e.g. $HOME/Apps/quercusTOA-app, and decompress the ZIP file using the following command:

```
$ mkdir -p $HOME/Apps

$ cd $HOME/Apps

$ wget --output-document main.zip https://github.com/GGFHF/quercusTOA-app/archive/refs/heads/main.zip

$ unzip main.zip

$ mv quercusTOA-app-main quercusTOA-app

$ rm main.zip
```

We will have the directory quercusTOA-app in $HOME/Apps. Then, the execution permissions of the programs must be set by using these commands:

```
$ find $HOME/Apps/quercusTOA-app -type f \( -name "*.py" -o -name "*.sh" \) -exec chmod +x {} \;
```

QUERCUSTOA database download

To download and decompress in the directory $HOME/BioData/quercusTOA-db, we will type the following commands:

```
$ mkdir -p $HOME/BioData/quercusTOA-db

$ cd $HOME/BioData

$ wget --output-document quercusTOA-db.zip https://drive.upm.es/public.php/dav/files/oZL6TnvNkM7paOC/quercusTOA-db.zip

$ unzip -o quercusTOA-db.zip

$ rm quercusTOA-db.zip
```

Conda and bioinformatics software installation

The following are the commands for the installation of **Miniforge3** in the right location, e.g. $HOME/Apps/Miniforge3, if there is no previous installation:

```
$ mkdir -p $HOME/Apps

$ cd $HOME/Apps

$ wget "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-$(uname)-$(uname -m).sh"

$ bash Miniforge3-$(uname)-$(uname -m).sh -b -p $HOME/Apps/Miniforge3

$ rm Miniforge3-$(uname)-$(uname -m).sh

$ $HOME/Apps/Miniforge3/condabin/conda init bash
```

Close the terminal and open a new terminal. The **Conda** environment base is now active. Then type the following commands to configure the download environment:

```
$ conda config --add channels bioconda

$ conda config --add channels conda-forge

$ conda config --set channel_priority strict
```

Now we will create the *quercustoa* environment used to run QUERCUSTOA-APP programs typing the following command:

```
$ conda env create -f $HOME/Apps/quercusTOA-app/yml/quercustoa.yml
```

Finally, we will create the bioinformatics software environments corresponding to **BLAST+**, **CodAn**, **DIAMOND**, **Liftoff**, **LiftoffTools** and **MAFFT**. We will use three environments to avoid dependencies problems. Use the following commands to create the environment corresponding to each software:

- **BLAST+** environment (quercustoa-blast):

```
$ conda create --yes --name quercustoa-blast blast
```

- **CodAn** environment (quercustoa-codan)

```
$ conda create --yes --name quercustoa-codan codan

$ conda activate quercustoa-codan

$ MODELS_DIR=`echo $CONDA_PREFIX`/models

$ mkdir -p $MODELS_DIR

$ wget --output-document $MODELS_DIR/PLANTS_full.zip https://github.com/pedronachtigall/CodAn/raw/master/models/PLANTS_full.zip

$ unzip -o -d $MODELS_DIR $MODELS_DIR/PLANTS_full.zip

$ rm $MODELS_DIR/PLANTS_full.zip

$ wget --output-document $MODELS_DIR/PLANTS_partial.zip https://github.com/pedronachtigall/CodAn/raw/master/models/PLANTS_partial.zip

$ unzip -o -d $MODELS_DIR $MODELS_DIR/PLANTS_partial.zip

$ rm $MODELS_DIR/PLANTS_partial.zip

$ conda deactivate
```

- **DIAMOND** environment (quercustoa-diamond):

```
$ conda create --yes --name quercustoa-diamond diamond
```

- **LIftoff** environment (quercustoa-liftoff):

```
$ conda create --yes --name quercustoa-liftoff liftoff
```

- **MAFFT** environment (quercustoa-mafft):

```
$ conda env create -f $HOME/Apps/quercusTOA-app/yml/quercustoa-
mafft.yml
```

Running functional annotation and enrichment analysis processes

Two bash scripts are included in the folder pipelines of the QUERCUSTOA-APP software package: `run-annotation-pipeline-process.sh` and `run-enrichment-analysis-process.sh`. These scripts can be used to run functional annotation and enrichment analysis processes from other Bash scripts without having to enter the QUERCUSTOA-APP graphical environment once the above steps have been carried out. There is another script in the folder, `test-quercustoa-processes.sh`, which is an example of running the annotation and enrichment processes from another Bash script.

The script `run-annotation-pipeline-process.sh` needs 12 input parameters detailed in the following table:

Table: **run-annotation-pipeline-process.sh** parameters

Parameter	Comment
quercustoa_app_dir	path of the QUERCUSTOA-APP directory
quercustoa_db_dir	path of the QUERCUSTOA database directory
fasta_type	FASTA file type: "TRANSCRIPTS" or "PROTEINS"
fasta_file	transcripts file path
model	CodAn model: "PLANTS_full" or "PLANTS_partial"
aligner	alignment software: "BLAST+" or "DIAMOND"
ev	evalue (BLAST+ and DIAMOND parameter)
mts	max_target_seqs (BLAST+ and DIAMOND parameter)
mh	max_hsps (BLAST+ and DIAMOND parameter)
qhp	qcov_hsp_perc (BLAST+ parameter) or 0 (if DIAMOND)
threads	threads number
annotation_dir	path of the annotation output directory

If `run-annotation-pipeline-process.sh` is called with an incorrect number of 11 parameters, a warning message is displayed (Figure 61):

```
*** ERROR: The following 12 parameters are required:

    quercustoa_app_dir <- path of the quercusTOA-app directory.
    quercustoa_db_dir <- path of the quercusTOA-db directory.
    fasta_type <- FASTA file type: "TRANSCRIPTS" or "PROTEINS".
    fasta_file <- FASTA file path.
    model <- CodAn model: "PLANTS_full" or "PLANTS_partial".
    aligner <- alignment software: "BLAST+" or "DIAMOND".
    ev <- evalue (BLAST+ and DIAMOND parameter).
    mts <- max_target_seqs (BLAST+ and DIAMOND parameter).
    mh <- max_hsps (BLAST+ and DIAMOND parameter).
    qhp <- qcov_hsp_perc (BLAST+ parameter) or 0 (if DIAMOND).
    threads <- threads number.
    annotation_dir <- path of the annotation output directory.

Use: run-annotation-pipeline-process.sh quercustoa_app_dir quercustoa_db_dir fasta_type fasta_file model aligner ev mts mh qhp threads annotation_dir
```

Figure 61. Warning message displayed when the parameters are not correctly passed to the script `run-annotation-pipeline-process.sh`.

The script `run-enrichment-analysis-process.sh` needs 8 input parameters detailed in the following table:

Table: `run-enrichment-analysis-process.sh` parameters

Parameter	Comment
quercustoa_app_dir	path of the QUERCUSTOA-APP directory
quercustoa_db_dir	path of the QUERCUSTOA database directory
annotation_dir	path of the annotation output directory
species	"all_species" or specific spcecies name
method	FDR method: "bh" (Benjamini-Hochberg) or "by" (Benjamini-Yekutieli)
msqannot	minimum sequences number in annotations
msqspec	minimum sequences number in species
enrichment_dir	path of the enrichment output directory

If `run-enrichment-analysis-process.sh` is called with an incorrect number of 11 parameters, a warning message is displayed (Figure 62).

```
*** ERROR: The following 8 parameters are required:

    quercustoa_app_dir <- path of the quercusTOA-app directory.
    quercustoa_db_dir <- path of the quercusTOA-db directory.
    annotation_dir <- path of the annotation input directory.
    species <- "all_species" or specific spcecies name.
    method <- FDR method: "bh" (Benjamini-Hochberg) or "by" (Benjamini-Yekutieli).
    msqannot <- minimum sequences number in annotations.
    msqspec <- minimum sequences number in species.
    enrichment_dir <- path of the enrichment output directory.

Use: run-enrichment-analysis-process.sh quercustoa_app_dir quercustoa_db_dir annotation_dir species method msqannot msqspec enrichment_dir
```

Figure 62. Warning message displayed when the parameters are not correctly passed to the script `run-enrichment-analysis-process.sh`.

No changes need to be made to the scripts `run-annotation-pipeline-process.sh` and `run-enrichment-analysis-process.sh` unless you have chosen environment names other than *quercustoa*, *quercustoa-blast*, *quercustoa-codan*, *quercustoa-diamond* and *quercustoa-liftoff*.

Running the scripts `run-annotation-pipeline-process.sh` and `run-enrichment-analysis-process.sh` can be made directly from the command line by passing the values of the required parameters, although a more convenient way is to call them from another script. The script `test-quercustoa-processes.sh` is an example of calling functional annotation and enrichment analysis scripts from other bash scripts. You can run this script as a test without any changes as long as the installation has been performed in the indicated locations. If not, simply modify the path of the affected directories. The following command will run `test-quercustoa-processes.sh`:

```
$ $HOME/Apps/quercusTOA-app/pipelines/test-quercustoa-processes.sh
```

In this test the value passed to *annotation_dir* parameter is *$HOME/results/annotation-test*. Figure 63 shows the folders and files yielded in this directory when `run-annotation-pipeline-process.sh` is ended.

```
(base) fmm@fmmpc72:~$ ls -la $HOME/results/annotation-test
total 58164
drwxrwxr-x 4 fmm fmm     4096 Nov  3 12:19 .
drwxrwxr-x 5 fmm fmm     4096 Nov  3 12:20 ..
-rw-rw-r-- 1 fmm fmm   832884 Nov  3 12:19 agrigo-input-file.txt
drwxrwxr-x 2 fmm fmm     4096 Nov  3 12:00 codan_output
-rw-rw-r-- 1 fmm fmm  2655853 Nov  3 12:19 functional-annotations-besthit.csv
-rw-rw-r-- 1 fmm fmm 55647172 Nov  3 12:19 functional-annotations-complete.csv
-rw-rw-r-- 1 fmm fmm    55986 Nov  3 12:19 revigo-input-file.txt
-rw-rw-r-- 1 fmm fmm   324013 Nov  3 12:19 stats-goterms.csv
-rw-rw-r-- 1 fmm fmm      133 Nov  3 12:19 stats-namespaces.csv
-rw-rw-r-- 1 fmm fmm     1564 Nov  3 12:19 stats-seq-per-goterm.csv
-rw-rw-r-- 1 fmm fmm      210 Nov  3 12:19 stats-species.csv
drwxrwxr-x 3 fmm fmm     4096 Nov  3 12:19 temp
```

Figure 63. Folders and files yielded by `run-annotation-pipeline-process.sh`.

These files are:

- Annotation:
  - *functional-annotations-complete.csv*: It contains all annotations yielded by blast programs for every query sequence identification.
  - *functional-annotations-besthit.csv*: It contains the annotations of the subject sequence identification with best hit yielded by blast programs for every query sequence identification.
- Statistics:
  - *stats-goterms.csv*: It contains the frequency distribution per GO term.
  - *stats-namespaces.csv*: It contains the frequency distribution per namespace.
  - *stats-seq-per-goterm.csv*: It contains the sequence number per GO term number.
  - *stats-species.csv*: It contains the frequency distribution of species.
- Other applications input:
  - *agrigo-input-file.txt*: It contains data to be used as agriGO input.
  - *revigo-input-file.txt*: It contains data to be used as REVIGO input.

In this test the value passed to *enrichment_dir* parameter is *$HOME/results/enrichment-test*. Figure 64 shows the folders and files yielded in this directory when `run-enrichment-analysis-process.sh` is ended.

```
(base) fmm@fmmpc72:~$ ls -la $HOME/results/enrichment-test
total 576
drwxrwxr-x 2 fmm fmm   4096 Nov  3 12:20 .
drwxrwxr-x 5 fmm fmm   4096 Nov  3 12:20 ..
-rw-rw-r-- 1 fmm fmm  90282 Nov  3 12:19 besthit-goterm-enrichment-analysis.csv
-rw-rw-r-- 1 fmm fmm    215 Nov  3 12:20 besthit-kegg-ko-enrichment-analysis.csv
-rw-rw-r-- 1 fmm fmm   3993 Nov  3 12:20 besthit-kegg-pathway-enrichment-analysis.csv
-rw-rw-r-- 1 fmm fmm 151288 Nov  3 12:20 besthit-metacyc-pathway-enrichment-analysis.csv
-rw-rw-r-- 1 fmm fmm 149726 Nov  3 12:20 complete-goterm-enrichment-analysis.csv
-rw-rw-r-- 1 fmm fmm    287 Nov  3 12:20 complete-kegg-ko-enrichment-analysis.csv
-rw-rw-r-- 1 fmm fmm   6831 Nov  3 12:20 complete-kegg-pathway-enrichment-analysis.csv
-rw-rw-r-- 1 fmm fmm 160189 Nov  3 12:20 complete-metacyc-pathway-enrichment-analysis.csv
```

Figure 64. Files yielded by the script `run-enrichment-analysis-process.sh`.

These files are:

- Analysis using the file that contains the annotations of the subject sequence identification with best hit yielded by blast programs for every query sequence identification:
  - *besthit-goterm-enrichment-analysis.csv*: enrichment of GO terms
  - *besthit-kegg-ko-enrichment-analysis.csv*: enrichment of KEGG KOs
  - *besthit-kegg-pathway-enrichment-analysis.csv*: enrichment of KEGG pathways
  - *besthit-metacyc-pathway-enrichment-analysis.csv*: enrichment of MetaCyc pathways
- Analysis using the file that contains all annotations yielded by blast programs for every query sequence identification.
  - *complete-goterm-enrichment-analysis.csv*: enrichment of GO terms
  - *complete-kegg-ko-enrichment-analysis.csv*: enrichment of KEGG KOs
  - *complete-kegg-pathway-enrichment-analysis.csv*: enrichment of KEGG pathways
  - *complete-metacyc-pathway-enrichment-analysis.csv*: enrichment of MetaCyc pathways

Homology search process

The homology search is a process integrated into QUERCUSTOA-APP, but it can be run without having to enter the QUERCUSTOA-APP graphical environment using the Bash script `test-quercustoa-processes.sh`. Just like `run-annotation-pipeline-process.sh` and `run-enrichment-analysis-process.sh`, `run-homology-search-process.sh` is in the folder pipelines; the script `test-quercustoa-processes.sh` has an example of running the homology search process from another Bash script.

The script `run-homology-search-process.sh` needs 7 input parameters detailed in the following table:

Table: **run-homology-search-process.sh** parameters

Parameter	Comment
quercustoa_app_dir	path of the QUERCUSTOA-APP directory
quercustoa_db_dir	path of the QUERCUSTOA database directory
analysis_type	FASTA file type: "TRANSCRIPTS" or "PROTEINS"
analysis_file	FASTA file with analysis protein sequences
model	CodAn model: "PLANTS_full" or "PLANTS_partial"
threads	threads numbe
homology_dir	path of the homology search output directory

If `run-homology-search-process.sh` is called with an incorrect number of parameters, the script displays a warning message (Figure 65).

```
*** ERROR: The following 7 parameters are required:

    quercustoa_app_dir <- path of the quercusTOA-app directory.
    quercustoa_db_dir <- path of the quercusTOA-db directory.
    analysis_type <- FASTA file type: "TRANSCRIPTS" or "PROTEINS".
    analysis_file <- FASTA file with analysis protein sequences.
    model <- CodAn model: "PLANTS_full" or "PLANTS_partial".
    threads <- threads number.
    homology_dir <- path of the homology search output directory.

Use: run-homology-search-process.sh quercustoa_app_dir quercustoa_db_dir analysis_type analysis_file model threads homology_dir
```

Figure 65. Warning message displayed when the parameters are not correctly passed to the script `run-homology-search-process.sh`.

In this test the value passed to *homology_dir* parameter is *$HOME/results/homology-test*. Figure 66 shows the folders and files yielded in this directory when `run-homology-search-process.sh` is ended. The file *homology-relationships-file.csv* contains the homology relationship of the sequences analyzed.

```
(base) fmm@fmmpc72:~$ ls -la $HOME/results/homology-test
total 112
drwxrwxr-x 5 fmm fmm  4096 Nov  3 16:05 .
drwxrwxr-x 5 fmm fmm  4096 Nov  3 14:46 ..
drwxrwxr-x 2 fmm fmm  4096 Nov  3 14:46 codan_output
-rw-rw-r-- 1 fmm fmm 31399 Nov  3 14:47 homology-relationships-file.csv
drwxrwxr-x 2 fmm fmm 61440 Nov  3 15:50 seqs-alignments
drwxrwxr-x 2 fmm fmm  4096 Nov  3 14:46 temp
```

Figure 66. Files yielded by the script **run-homology-search-process.sh** in the directory **homology-test**.

For each query sequence, the following files are in the subdirectory *seqs-alignments* (Figure 67):

- **homologous-genes.fasta*: sequences FASTA of the homologous genes.
- **homologous-genes.fasta.aln*: alignment of sequences FASTA of the genes corresponding to homologous proteins.

- **homologous-genes.fasta.aln.pdf*: alignment plot of sequences FASTA of the genes corresponding to homologous proteins.
- **homologous-proteins.fasta*: sequences FASTA of the homologous proteins.
- **homologous-proteins.fasta.aln*: alignment of sequences FASTA of the homologous proteins.
- **homologous-proteins.fasta.aln.pdf*: alignment plot of sequences FASTA of the homologous proteins.
- **homologous-proteins.fasta.tree*: phylogenetic tree of homologous proteins.
- **homologous-proteins.fasta.tree.pdf*: phylogenetic tree plot of homologous proteins.

```
(base) fmm@fmmpc72:~$ ls -la $HOME/results/homology-test/seqs-alignments
total 103120
drwxrwxr-x 2 fmm fmm   61440 Nov  3 15:50 .
drwxrwxr-x 5 fmm fmm    4096 Nov  3 16:05 ..
-rw-rw-r-- 1 fmm fmm   14175 Nov  3 14:47 fasta_file_list.txt
-rw-rw-r-- 1 fmm fmm   29574 Nov  3 14:47 XM_024018324.2-homologous-genes.fasta
-rw-rw-r-- 1 fmm fmm   60163 Nov  3 14:47 XM_024018324.2-homologous-genes.fasta.aln
-rw-rw-r-- 1 fmm fmm 1122796 Nov  3 14:48 XM_024018324.2-homologous-genes.fasta.aln.pdf
-rw-rw-r-- 1 fmm fmm    4124 Nov  3 14:47 XM_024018324.2-homologous-proteins.fasta
-rw-rw-r-- 1 fmm fmm    4440 Nov  3 14:48 XM_024018324.2-homologous-proteins.fasta.aln
-rw-rw-r-- 1 fmm fmm  103801 Nov  3 14:48 XM_024018324.2-homologous-proteins.fasta.aln.pdf
-rw-rw-r-- 1 fmm fmm     343 Nov  3 14:48 XM_024018324.2-homologous-proteins.fasta.tree
-rw-rw-r-- 1 fmm fmm   17483 Nov  3 14:48 XM_024018324.2-homologous-proteins.fasta.tree.pdf
-rw-rw-r-- 1 fmm fmm    7027 Nov  3 14:47 XM_024018629.2-homologous-genes.fasta
-rw-rw-r-- 1 fmm fmm   10349 Nov  3 14:48 XM_024018629.2-homologous-genes.fasta.aln
-rw-rw-r-- 1 fmm fmm  205964 Nov  3 14:48 XM_024018629.2-homologous-genes.fasta.aln.pdf
-rw-rw-r-- 1 fmm fmm    1718 Nov  3 14:47 XM_024018629.2-homologous-proteins.fasta
-rw-rw-r-- 1 fmm fmm    1778 Nov  3 14:48 XM_024018629.2-homologous-proteins.fasta.aln
-rw-rw-r-- 1 fmm fmm   49572 Nov  3 14:48 XM_024018629.2-homologous-proteins.fasta.aln.pdf
-rw-rw-r-- 1 fmm fmm     393 Nov  3 14:48 XM_024018629.2-homologous-proteins.fasta.tree
-rw-rw-r-- 1 fmm fmm   17717 Nov  3 14:48 XM_024018629.2-homologous-proteins.fasta.tree.pdf
-rw-rw-r-- 1 fmm fmm   23523 Nov  3 14:47 XM_024020269.2-homologous-genes.fasta
-rw-rw-r-- 1 fmm fmm   25861 Nov  3 14:48 XM_024020269.2-homologous-genes.fasta.aln
-rw-rw-r-- 1 fmm fmm  507899 Nov  3 14:48 XM_024020269.2-homologous-genes.fasta.aln.pdf
-rw-rw-r-- 1 fmm fmm    1743 Nov  3 14:47 XM_024020269.2-homologous-proteins.fasta
-rw-rw-r-- 1 fmm fmm    1769 Nov  3 14:48 XM_024020269.2-homologous-proteins.fasta.aln
-rw-rw-r-- 1 fmm fmm   48916 Nov  3 14:48 XM_024020269.2-homologous-proteins.fasta.aln.pdf
-rw-rw-r-- 1 fmm fmm     393 Nov  3 14:48 XM_024020269.2-homologous-proteins.fasta.tree
-rw-rw-r-- 1 fmm fmm   17685 Nov  3 14:48 XM_024020269.2-homologous-proteins.fasta.tree.pdf
-rw-rw-r-- 1 fmm fmm   94634 Nov  3 14:47 XM_024021998.2-homologous-genes.fasta
-rw-rw-r-- 1 fmm fmm  104397 Nov  3 14:48 XM_024021998.2-homologous-genes.fasta.aln
-rw-rw-r-- 1 fmm fmm 1793432 Nov  3 14:49 XM_024021998.2-homologous-genes.fasta.aln.pdf
-rw-rw-r-- 1 fmm fmm    8853 Nov  3 14:47 XM_024021998.2-homologous-proteins.fasta
-rw-rw-r-- 1 fmm fmm    9340 Nov  3 14:49 XM_024021998.2-homologous-proteins.fasta.aln
-rw-rw-r-- 1 fmm fmm  175525 Nov  3 14:50 XM_024021998.2-homologous-proteins.fasta.aln.pdf
-rw-rw-r-- 1 fmm fmm    1310 Nov  3 14:50 XM_024021998.2-homologous-proteins.fasta.tree
-rw-rw-r-- 1 fmm fmm   18840 Nov  3 14:50 XM_024021998.2-homologous-proteins.fasta.tree.pdf
```

Figure 67. Files of the first sequences yielded by the script `run-homology-search-process.sh` in the subdirectory *seqs-alignments* of directory *homology-test*.

Appendix A: Description of QUERCUSTOA tables

This appendix contains the description of the SQLite database of the QUERCUSTOA database. It is composed of three files: *functional-annotation.db*, *comparative-genomics.db* and *sequences.db*.

The file *functional-annotation.db* contains the following tables:

Table *mmseq2_protein_clusters*

Column	Type	Index	Comment
cluster_id	TEXT	1	identification of protein cluster
seq_id	TEXT	2	NCBI protein sequence identification
description	TEXT		description from the NCBI protein sequence
species	TEXT		species from the NCBI protein sequence

Table *interproscan_annotations*

Column	Type	Index	Comment
cluster_id	TEXT	1	cluster identification
interpro_goterms	TEXT		concatenated list of GO terms from InterPro
panther_goterms	TEXT		concatenated list of GO terms from Panther
x_goterms	TEXT		concatenated list of GO terms from other sources
metacyc_pathways	TEXT		concatenated list of pathway identifications from MetaCyc
reactome_pathways	TEXT		concatenated list of pathway identifications from Reactome
x_pathways	TEXT		concatenated list of pathway identifications from other sources

Table *emapper_annotations*

Column	Type	Index	Comment
cluster_id	TEXT	1	cluster identification
ortholog_seq_id	TEXT		ortholog sequence identification from eggNOG
ortholog_species	TEXT		species from eggNOG
eggnog_ogs	TEXT		OGs (Orthologous Groups) of proteins from eggNOG
cog_category	TEXT		COG (Cluster of Orthologous Genes) from eggNOG
description	TEXT		description from eggNOG
goterms	TEXT		concatenated list of GO terms from eggNOG
ec	TEXT		concatenated list of EC (Enzyme Commission) numbers
kegg_kos	TEXT		concatenated list of KO from KEGG
kegg_pathways	TEXT		concatenated list of pathway identifications from KEGG

kegg_modules	TEXT		concatenated list of module identifications from KEGG
kegg_reactions	TEXT		concatenated list of chemical reactions identifications from KEGG
kegg_rclasses	TEXT		concatenated list of reactions classification identifications from KEGG
brite	TEXT		functional hierarchy of OGs assigned to the sequence
kegg_tc	TEXT		T cell receptor (TCR) signaling pathway
cazy	TEXT		concatenated list of Carbohydrate-Active Enzymes (CAZymes)
pfams	TEXT		concatenated list of protein families from Pfam

Table *tair10_info*

Column	Type	Index	Comment
tair10_peptide_id	TEXT	1	*A. thaliana* peptide identification
description	TEXT		Description of the *A. thalina* peptide

Table *tair10_orthologs*

Column	Type	Index	Comment
cluster_id	TEXT	1	cluster identification
ortholog_seq_id	TEXT		ortholog sequence identification of *A. thalina*

Table *go_ontology*

Column	Type	Index	Comment
go_id	TEXT	1	GO term identification
go_name	TEXT		GO term description
namespace	TEXT		Molecular function, biological process or cellular component

The file *comparative-genomics.db* contains the following tables:

Table *liftoff_gff_cds_data*

Column	Type	Index	Comment
reference_species_id	TEXT	1,2	identification of the reference species
target_species_id	TEXT	1	identification of the target species
reference_protein_id	TEXT	2	protein identification of the reference species
target_seq_id	TEXT	1	sequence identification of the reference species
target_start	INT		start position in the reference species genome
target_end	INT		end position in the reference species genome
target_strand	TEXT		strand in the reference species genome

Table *liftoff_gff_gene_data*

Column	Type	Index	Comment
reference_species_id	TEXT	1	identification of the reference species
target_species_id	TEXT	1	identification of the target species
reference_gene_id	TEXT	2	gene identification of the reference species
target_seq_id	TEXT	1	sequence identification of the reference species
target_start	INT		start position in the reference species genome
target_end	INT		end position in the reference species genome
target_strand	TEXT		strand in the reference species genome
reference_protein_id_list	TEXT		protein identification list of the reference species

Table *liftoff_homologous_proteins*

Column	Type	Index	Comment
reference_species_id	TEXT	1	identification of the reference species
target_species_id	TEXT	1	identification of the target species
reference_protein_id	TEXT	1,2	protein identification of the reference species
target_protein_id	TEXT		protein identification of the target species

Table *liftoff_unmapped_genes*

Column	Type	Index	Comment
reference_species_id	TEXT	1	identification of the reference species
target_species_id	TEXT	1	identification of the target species
reference_gene_id	TEXT	1,2	gene identification of the reference species

Table *mmseqs2_concatenated_cds_clusers*

Column	Type	Index	Comment
species_id	TEXT	1,2,3,4	species identification
cluster_id	TEXT	1	identification of concatenated CDS cluster
seq_id	TEXT	2	sequence identification
gene_id	TEXT	4	gene identification
protein_id	TEXT	3	protein identification

The file *sequences.db* contains the following tables:

Table *species_protein_seqs*

Column	Type	Index	Comment
protein_id	TEXT	1	protein identification
species_id	TEXT		species identification
seq	TEXT		sequence of the protein

Table *species_gene_seqs*

Column	Type	Index	Comment
gene_id	TEXT	1	gene identification
species_id	TEXT		species identification
seq	TEXT		sequence of the protein